

Structural Principle of Entanglement (PSE)

Foundations of Emergence:
Physical quantities as images of a fundamental pregeometric structure

Axiomatic Mathematical Formalization



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Abstract

The Structural Principle of Entanglement (PSE) is an axiomatic mathematical theory that describes the emergence of physical structures from a completely pregeometric fundamental domain. The formalization introduces a rigorous ontological hierarchy composed of five structural levels: the fundamental domain S , the fundamental distance $D_S^{(f)}$, the fundamental correlation C_f , the emergent distance $D_S^{(e)}$, and the emergent correlation C_e . These levels constitute the complete ontological structure of the PSE and are defined by axioms ensuring coherence, continuity, compressivity, and informational saturation.

In the fundamental domain, consisting of a discrete set without geometry or physical quantities, a structural distance $D_S^{(f)}$ and a selective binary correlation C_f are defined, determined by a non-physical structural threshold Δ . The two functions are linked by a non-invertible constraint that enforces informational rigidity at the fundamental level. Through monotone families of regularization operators $\{F_\lambda\}$ and $\{G_\lambda\}$, the emergent domain is constructed as a continuous and compressed limit of the fundamental domain. At this level, the emergent distance $D_S^{(e)}$ and the emergent correlation C_e arise as continuous, regular, saturating functions, rigidly related through a decreasing monotonicity.

The structural projection

$$\Phi: \left(X, D_S^{(e)}, C_e \right)_{\text{reg}} \rightarrow \mathcal{D}_{\text{phys}}$$

does not belong to the ontological hierarchy: it is a mathematical transformation that becomes defined only when the emergent structure reaches a sufficient level of complexity to generate physical representations. The projection is monotone, compressive, and non-invertible: the physical domain contains less information than the emergent structure and cannot reconstruct it. The physical structures in $\mathcal{D}_{\text{phys}}$ are effective and compressed images of the emergent structure, determined solely by the structural properties of $D_S^{(e)}$ and C_e .

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1. Mathematical Formalization

The mathematical formalization of the Structural Principle of Entanglement (PSE) is constructed as a rigorous ontological chain linking a discrete pregeometric structure to an emergent physical domain. The starting point is a fundamental discrete set S , devoid of metric, causality, and dynamics, on which a discrete structural distance $D_S^{(f)}$ and a fundamental correlation C_f are defined, subject to conditions of regularity and coherence. These objects are neither physical nor observable: they constitute the primary ontological structure of the model.

The formation of the emergent domain is described through monotonic families of regularization operators $\{F_\lambda\}$ and $\{G_\lambda\}$, which generate the emergent structural distance $D_S^{(e)}$ and the emergent correlation C_e as continuous limits of the fundamental quantities. The emergent domain

$$X = \lim_{\lambda \rightarrow \infty} S_\lambda$$

is continuous, regular, and stable, yet still entirely pregeometric: it contains no physical metric, causality, or dynamics. The structural conditions of continuity, stability, regularity, and correlational monotonicity ensure that the emergent structure is mathematically well-defined and suitable for physical representation.

The projection map

$$\Phi: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow \mathcal{D}_{\text{phys}}$$

formalizes the structural compression that allows the emergent domain to be represented in physical terms. Φ is a structural, non-ontological and non-physical operator, monotonic and non-invertible, which associates the emergent structure with a set of observable physical quantities. The physical domain $\mathcal{D}_{\text{phys}}$ is a compressive representation of the emergent structure: it necessarily contains less information than $(X, D_S^{(e)}, C_e)_{\text{reg}}$ and cannot reconstruct it. The physical quantities present in $\mathcal{D}_{\text{phys}}$ are determined exclusively by the structural properties of the emergent distance and the emergent correlation.

The entire formalization is constructed axiomatically, ensuring that every step of the ontological chain is mathematically determined, non-arbitrary, and coherent. This structure places the PSE within the context of emergentist theories of spacetime, providing a rigorous foundation for the interpretation of non-local correlations and for comparison with Bell's results and with contemporary ontological approaches based on informational structures.

1. Fundamental Domain

The fundamental domain constitutes the primary ontological level of the PSE. In this section we introduce the primitive objects of the theory and specify their formal properties. Everything that follows is completely pregeometric: no metric, no causality, and no physical quantity is defined at this level.

The fundamental structure S reorganizes according to a fundamental reorganization $\mathfrak{D}$, composed of the structural index t , the coherence bound c , and the $E + C$ principle described in the conceptual framework.

S is discrete, bounded, and regulated by a structural threshold Δ , which determines the selectivity of the fundamental correlations C_f . Only the correlations that exceed Δ contribute to the construction of the emergent distance $D_S^{(e)}$ and the emergent correlation C_e .

Axiom F1 — Fundamental Set

There exists a discrete set

$$S = \{s_i \mid i \in I\},$$

where:

- I is a discrete index set, not necessarily countable;
- each element s_i has no internal structure;
- no physical interpretation exists for the elements of S ;
- no geometry, coordinate system, or metric is associated with S .

Structural Properties of S

- **Discrete:** no internal continuity.
- **Pregeometric:** no notion of space, physical time, causality, or metric.
- **Non-observable:** no physical quantity can be defined or measured on S .
- **Non-physical dynamics:** no physical law acts on S .

Axiom F2 — Structural Index t

There exists a discrete parameter

$$t \in \mathbb{N},$$

called the **Structural Index**, which **orders** the fundamental configurations of the structure:

$$S = \{S_t \mid t \in \mathbb{N}\}.$$

The index t does **not** define any evolution, nor does it introduce a fundamental dynamics: no ontological law of the form

$$S_t \rightarrow S_{t+1}$$

is admitted, since such notation would imply a causal or temporal succession that is not allowed in the fundamental domain of the PSE. The index t is **purely ordinal**: it labels the discrete configurations without generating them.

Properties of the Structural Index t

- **Non-physical:** it does not represent duration, coordinate, causality, or propagation.
- **Ordinal:** it orders the fundamental configurations without implying transformations or dynamics.
- **Bounded by c :** there exists a structural parameter $c > 0$ that limits the admissible variation between adjacent configurations:

$$\forall (t, t') \in \mathbb{N}^2, t' = \text{is the immediate successor of } t \Rightarrow |D_S^{(f)}(t') - D_S^{(f)}(t)| \leq c.$$

This constraint is not dynamical: it is a **structural coherence condition** preventing pathological discontinuities in the fundamental structure.

Ontological Role

The Structural Index t :

- enables the **discrete reorganization** of the fundamental structure,
- guarantees **emergent stability**,
- ensures the **continuity of the limit** $D_S^{(e)}$,
- enforces the **non-retroactivity** of the ontological chain,
- preserves the **non-physical nature** of the fundamental domain.

Axiom F3 — Fundamental Structural Distance $D_S^{(f)}$

There exists a fundamental function

$$D_S^{(f)}: S \times S \rightarrow \mathbb{R}^+,$$

assigning a positive real value to each pair (s_i, s_j) .

Properties of $D_S^{(f)}$

- **Non-metric:** no symmetry, triangle inequality, or identity of indiscernibles.
- **Non-geometric:** does not represent any physical distance.
- **Non-observable:** no physical procedure can measure it.
- **Structural:** an internal constraint of the prespatial structure.
- **Discrete:** defined on discrete pairs.
- **Bounded:** there exists

$$D_{\max} = \sup_{i,j} D_S^{(f)}(i, j) < +\infty.$$

Boundedness is the root of future emergent saturation.

Axiom F4 — Structural Threshold Δ

There exists a positive real value

$$\Delta \in (0, D_{\max}),$$

called the **coherence threshold**, determining the selectivity of fundamental correlations.

Properties of Δ

- bounded: $0 < \Delta < D_{\max}$;
- non-physical;
- non-geometric;

- non-dynamic;
- non-observable;
- structural.

Δ determines the binary coherence of the fundamental level.

Axiom F5 — Fundamental Binary Correlation C_f

There exists a fundamental function

$$C_f: S \times S \rightarrow \{0,1\},$$

defined through Δ :

$$C_f(i, j) = 1 \text{ if } D_S^{(f)}(i, j) \leq \Delta,$$

$$C_f(i, j) = 0 \text{ if } D_S^{(f)}(i, j) > \Delta.$$

Properties of C_f

- binary;
- selective;
- compressive;
- non-invertible;
- non-physical;
- discrete.

The boundedness of C_f allows the emergent correlation to be compressive and saturating.

Axiom F6 — Structural Constraint among $D_S^{(f)}$, C_f , Δ

The fundamental structural distance, the fundamental correlation, and the structural threshold are linked by an internal relation:

$$F(D_S^{(f)}, C_f, \Delta) = 0,$$

where F is a non-arbitrary constraint imposing coherence among the three quantities.

Properties of the Constraint

- non-physical;
- non-geometric;
- non-dynamic;
- non-observable;
- non-invertible;
- non-arbitrary;
- compressive.

Axiom F7 — Total Absence of Physics in the Fundamental Domain

In the fundamental domain, none of the following exist:

- space,
- physical time,
- metric,
- causality,
- energy,
- mass,
- particles,
- fields,
- systems,
- physical dynamics.

The fundamental domain is entirely prespatial.

Axiom F8 — Structural Bound c

There exists a structural parameter

$$c > 0,$$

called the **coherence bound**, which constrains the admissible variation between adjacent fundamental configurations:

$$t' > t(\text{structural adjacency}) \Rightarrow |D_S^{(f)}(t') - D_S^{(f)}(t)| \leq c.$$

The index t does not represent time nor dynamics: it is a non-physical structural ordering parameter, as established in Axiom F2.

Properties of the bound c

- non-physical;
- non-geometric;
- non-dynamical;
- non-observable;
- structural;
- ensures continuity and regularity of the emergent limit $D_S^{(e)}$;
- prevents pathological discontinuities in the fundamental domain.

Ontological role

The bound c establishes the **structural coherence** of the fundamental domain, ensuring that the fundamental reorganization does not produce variations incompatible with emergent stability.

Axiom F9 — Structural Principle $E + C$

There exists an internal structural principle within the fundamental domain, called the **$E + C$ principle**, which stabilizes the admissible fundamental configurations.

The principle consists of two conditions:

- **E (Elimination)**: reduction of internal contradictions among fundamental relations;
- **C (Compression)**: reduction of local tensions among nearby correlations.

Properties of the $E + C$ principle

- non-physical;
- non-geometric;
- non-dynamical;
- non-observable;
- structural;
- selective: determines which fundamental configurations are stable;
- compressive: prepares emergent saturation.

Ontological role

The $E + C$ principle is an **ontological stabilization rule** acting on $D_S^{(f)}$ and C_f , determining which fundamental configurations persist under the fundamental reorganization $\mathfrak{D}$. It guarantees emergent stability and the continuity of the ontological chain.

Minimal Example: Fundamental Structure S

1. Fundamental Set

Let

$$S = \{s_1, s_2, \dots, s_N\}$$

with finite N . The elements of S have no internal structure and no physical interpretation, in accordance with Axiom F1.

2. Fundamental Structural Distance $D_S^{(f)}$

Define

$$D_S^{(f)}: S \times S \rightarrow \mathbb{R}^+$$

via a block-structured matrix. Let

$$S = S_A \cup S_B, S_A \cap S_B = \emptyset.$$

Set

$$D_S(s_i, s_j) = \begin{cases} 0, & i = j, \\ a, & s_i, s_j \in S_A \text{ or } s_i, s_j \in S_B, \\ b, & s_i \in S_A, s_j \in S_B \text{ or vice versa,} \end{cases} \quad 0 < a < b.$$

The maximum value is

$$D_{\max} = b.$$

3. Structural Threshold Δ

Choose

$$\Delta \in (0, D_{\max}), \Delta = \frac{a+b}{2}.$$

4. Fundamental Binary Correlation C_f

Define

$$C_f: S \times S \rightarrow \{0, 1\}$$

$$C_f(i, j) = \begin{cases} 1, & D_S^{(f)}(i, j) \leq \Delta, \\ 0, & D_S^{(f)}(i, j) > \Delta. \end{cases}$$

Elements in the same block are correlated (1), elements in different blocks are not (0).

5. Structural Constraint

$$F(D_S^{(f)}, C_f, \Delta) = \sum_{i,j} [C_f(i, j) - \chi_{[0, \Delta]}(D_S^{(f)}(i, j))] = 0.$$

This ensures that the binary correlation is exactly the structural compression of the fundamental distance, as required by Axiom F6.

6. Structural Index t and Coherence Limit c

Introduce a family of fundamental configurations

$$S_t, D_S^{(f)}(t), C_f^{(t)}, t \in \mathbb{N}.$$

For every $t' > t$ (structural adjacency):

$$|D_S^{(f)}(t')(i, j) - D_S^{(f)}(t)(i, j)| \leq c \forall i, j.$$

A minimal example consists in allowing the parameters $a(t)$ and $b(t)$ to vary within the coherence limit between structurally adjacent indices:

$$t' > t \Rightarrow |a(t') - a(t)| \leq c, |b(t') - b(t)| \leq c.$$

The index t does not represent time nor dynamics: it is a non-physical structural ordering parameter, as established in Axiom F2.

This construction satisfies Axiom F8 and ensures that the fundamental structure remains coherent and free from pathological discontinuities.

7. Structural Principle $E + C$

The E+C principle acts on $D_S^{(f)}(t)$ and $C_f^{(t)}$:

- **E (Elimination):** reduction of internal contradictions between distance and correlation.
- **C (Compression):** reduction of local tensions between nearby correlations.

Stability is ensured by:

- the binary correlation rigidly determined by Δ ;
- variations of $D_S^{(f)}(t)$ bounded by c ;
- the constraint

$$F(D_S^{(f)}, C_f, \Delta) = 0$$

which prevents incoherent configurations.

Summary

This minimal example shows that:

- the fundamental structure can be fully discrete, non-physical, and non-geometric;
- the fundamental distance can be explicitly defined and non-metric;
- the binary correlation is a structural compression of the distance;
- the internal constraint ensures coherence;
- the structural index t and the limit c guarantee stability;
- the $E + C$ principle selects admissible configurations.

This example provides a concrete model satisfying Axioms F1–F9 and prepares the construction of the emergent domain through the regularization operators.

2. Non-Observable Emergent Domain

The non-observable emergent domain constitutes the intermediate level of the PSE ontological chain. It is continuous, coherent, and structural, yet still entirely pregeometric: no metric, no causality, and no physical quantity is defined at this level. The emergent domain is an informational compression of the prespatial structure S , of the fundamental structural distance $D_S^{(f)}$, and of the fundamental correlation C_f , regulated by the structural threshold Δ and the non-physical structural time t .

Axiom E1 — Family of Approximations of the Fundamental Domain

There exists a family of sets

$$\{S_\lambda\}_{\lambda \in \Lambda}$$

such that:

- each S_λ is a continuous approximation of the prespatial structure S ;
- Λ is a set of scale parameters with $\lambda \rightarrow \infty$, corresponding to increasingly fine regularizations.

The family $\{S_\lambda\}$ is **compressive**: each approximation reduces the information contained in S , $D_S^{(f)}$, and C_f in a regular and bounded way, preparing the emergence of continuous structures.

Axiom E2 — Definition of the Emergent Domain

The emergent domain is defined as the structural limit of the family S_λ :

$$X = \lim_{\lambda \rightarrow \infty} S_\lambda.$$

Properties of X

- **continuity:** X lacks the discreteness of the fundamental domain;
- **pregeometric nature:** X possesses no physical metric, coordinates, causality, or geometric structure;
- **non-observability:** no physical quantity can be defined or measured on X ;
- **emergent compressivity:** X contains less information than the prespatial structure S ;
- **emergent rigidity:** microscopic variations in S produce bounded variations in X .

Axiom E3 — Regularization Operators

There exist two families of regularization operators

$$F_\lambda: D_S^{(f)} \rightarrow D_\lambda, \quad G_\lambda: C_f \rightarrow C_\lambda,$$

where D_λ and C_λ are spaces of regularized functions, such that:

(1) Structural Monotonicity

For every $\lambda_2 > \lambda_1$:

$$\begin{aligned} F_{\lambda_2}(D_S^{(f)}) &\text{ is more regular than } F_{\lambda_1}(D_S^{(f)}), \\ G_{\lambda_2}(C_f) &\text{ is more regular than } G_{\lambda_1}(C_f). \end{aligned}$$

Regularity increases with the approximation scale.

(2) Compressivity

The operators reduce the information contained in $D_S^{(f)}$ and C_f :

$$\begin{aligned} \text{Info}(F_\lambda(D_S^{(f)})) &< \text{Info}(D_S^{(f)}), \\ \text{Info}(G_\lambda(C_f)) &< \text{Info}(C_f). \end{aligned}$$

(3) Structural Compatibility

The operators do not introduce any new structure not present in the fundamental domain.

- **Compatibility with the fundamental distance:**

$$D_S^{(f)}(x, y) \leq D_S^{(f)}(x', y') \Rightarrow D_\lambda(x, y) \leq D_\lambda(x', y').$$

- **Compatibility with the fundamental correlation:**

$$C_f(x, y) = 1 \Rightarrow C_\lambda(x, y) = 1.$$

(4) Stability

For every pair of immediate successors t, t' such that

$$|D_S^{(f)}(t') - D_S^{(f)}(t)| \leq c,$$

the variations induced in D_λ and C_λ are controlled and not amplified:

$$|D_\lambda(t') - D_\lambda(t)| \leq k_\lambda,$$

$$|C_\lambda(t') - C_\lambda(t)| \leq h_\lambda,$$

with k_λ and h_λ bounded.

(5) Continuity of the Limit

The limits

$$\lim_{\lambda \rightarrow \infty} F_\lambda(D_S^{(f)}), \quad \lim_{\lambda \rightarrow \infty} G_\lambda(C_f)$$

exist and are continuous functions.

These limits define the emergent distance $D_S^{(e)}$ and the emergent correlation C_e .

(6) Hierarchical Idempotence (Weak Non-Invertibility)

For every λ_1, λ_2 :

$$F_{\lambda_2} \circ F_{\lambda_1} = F_{\max(\lambda_1, \lambda_2)},$$

$$G_{\lambda_2} \circ G_{\lambda_1} = G_{\max(\lambda_1, \lambda_2)}.$$

The operators do not allow any reconstruction of compressed information.

(7) Non-Physicality

The operators do not represent any physical dynamics and do not introduce coordinates, metrics, causality, or geometric structure.

Axiom E4 — Emergent Structural Distance $D_S^{(e)}$

The emergent distance is defined as the limit:

$$D_S^{(e)} = \lim_{\lambda \rightarrow \infty} F_\lambda(D_S^{(f)}),$$

with domain:

$$D_S^{(e)}: X \times X \rightarrow \mathbb{R}^+.$$

Properties of $D_S^{(e)}$

- **continuity** on $X \times X$;
- **regularity**: absence of discontinuities and singularities;
- **pregeometric nature**: not a physical metric;
- **non-observability**: no physical procedure can measure it;
- **saturation**: the boundedness of $D_S^{(f)}$ and the threshold Δ imply

$$\sup_{x, y \in X} D_S^{(e)}(x, y) < +\infty.$$

E4.1 — Domain and Codomain of the Emergent Structural Distance

The emergent structural distance is a continuous function

$$D_S^{(e)}: X \times X \rightarrow \mathbb{R}^+$$

defined on the emergent domain X , which is devoid of geometric structure.

E4.2 — Structural Aggregation

There exists a structural aggregation operator

$$A: D_S^{(f)} \rightarrow D_S^{(e)}$$

such that

$$D_S^{(e)}(x, y) = A \left(D_S^{(f)} \right)_{\lambda \rightarrow \infty}.$$

E4.3 — Strong Emergent Continuity

For every $x, y \in X$:

$$\lim_{\lambda \rightarrow \infty} F_\lambda \left(D_S^{(f)} \right) (x, y)$$

is continuous and free of discontinuities.

E4.4 — Informational Compressivity

For every pair (x, y) :

$$D_S^{(e)}(x, y) \leq \sup D_S^{(f)}.$$

E4.5 — Emergent Stability

There exists a structural parameter c such that

$$| D_S^{(f)}(t') - D_S^{(f)}(t) | \leq c$$

implies

$$| D_S^{(e)}(x', y') - D_S^{(e)}(x, y) | \leq C.$$

E4.6 — Emergent Regularity

The function $D_S^{(e)}$ is free of singularities, divergences, and discontinuities.

E4.7 — Non-Physicality

$$\exists g_{\mu\nu}, \exists d_{\text{fisico}}, \exists t_{\text{fisico}}.$$

E4.8 — Compatibility with Regularization Operators

For every λ :

$$F_\lambda \left(D_S^{(f)} \right) \rightarrow D_S^{(e)}$$

in a monotone and compressive manner.

E4.9 — Compatibility with Metric Complexity

There exists a metric complexity function C_M such that

$$C_M(x, y) = H \left(D_S^{(e)}(x, y) \right).$$

E4.10 — Synthesis of the Emergent Structural Distance

The emergent structural distance is continuous, compressive, stable, regular, and non-physical.

Axiom E5 — Emergent Correlation C_e

The emergent correlation is defined as the limit:

$$C_e = \lim_{\lambda \rightarrow \infty} G_\lambda(C_f),$$

with domain:

$$C_e: X \times X \rightarrow \mathbb{R}.$$

Properties of C_e

- **continuity** on $X \times X$;
- **pregeometric nature**: does not represent any physical correlation;
- **structural nonlocality**: not constrained by any metric;
- **compressivity**: C_e contains less information than $D_S^{(e)}$;
- **saturation**: the binarity of C_f and the threshold Δ imply

$$\sup_{x,y \in X} |C_e(x,y)| < +\infty.$$

E5.1 — Functional Class of the Emergent Correlation

There exists a class of monotonically decreasing functions

$$\mathcal{F}_{reg} = \{F: \mathbb{R}^+ \rightarrow \mathbb{R}\}$$

such that

$$C_e(x,y) = F\left(D_S^{(e)}(x,y)\right).$$

E5.2 — Strong Correlative Monotonicity

For every $x, y \in X$:

$$D_1 < D_2 \Rightarrow F(D_1) > F(D_2).$$

E5.3 — Emergent Continuity

The function

$$C_e: X \times X \rightarrow \mathbb{R}$$

is continuous.

E5.4 — Informational Compressivity

$$|C_e(x,y)| \leq \sup |C_f|.$$

E5.5 — Emergent Stability

There exists a structural parameter c such that local variations in C_f produce bounded variations in C_e .

E5.6 — Compatibility with Regularization Operators

$$G_\lambda(C_f) \rightarrow C_e$$

in a monotone and compressive manner.

E5.7 — Non-Physicality

The emergent correlation does not represent any physical correlation.

E5.8 — Compatibility with Correlational Complexity

There exists a correlational complexity function C_c such that

$$C_c(x, y) = K(C_e(x, y)).$$

E5.9 — Compatibility with the Projection Φ

The emergent correlation is a structural input of the projection:

$$\Phi(D_S^{(e)}, C_e) \rightarrow \text{physical domains.}$$

E5.10 — Synthesis of the Emergent Correlation

The emergent correlation is continuous, compressive, monotone, stable, and non-physical.

Axiom E6 — Structural Conditions for Emergence

The emergence of $D_S^{(e)}$ and C_e requires:

1. **emergent continuity:** the limits $F_\lambda(D_S^{(f)})$ and $G_\lambda(C_f)$ are continuous on X ;
2. **emergent stability:** the limit is stable with respect to the structural variations regulated by the structural index t and the parameter c ;
3. **emergent regularity:** no discontinuity or singularity is allowed;
4. **correlational monotonicity:** there exists a decreasing monotonic function

$$C_e(x, y) = F(D_S^{(e)}(x, y)), \quad F' < 0,$$

derived from the binary selectivity imposed by Δ ;

5. **global coherence:** there exists an internal constraint

$$G(D_S^{(e)}, C_e) = 0.$$

Axiom E7 — Total Absence of Geometry in the Emergent Domain

In the emergent domain, none of the following exist:

$$\nexists g_{\mu\nu}, \quad \nexists d_{phys}, \quad \nexists t_{phys}.$$

No metric, physical distance, causality, or dynamics is defined. The emergent domain is entirely pregeometric.

Axiom E8 — Emergent Stability

For all $x, y \in X$:

$$D_S^{(e)}(x, y)$$

is stable with respect to local variations of $D_S^{(f)}$. This stability is structural, not physical, and is necessary for the future construction of regular physical structures.

Axiom E9 — Emergent Coherence

For all $x, y, z \in X$:

$$C_e(x, y) \text{ and } C_e(y, z)$$

are constrained by structural coherence relations. Emergent coherence is compressive: it reduces the fundamental information to a regular and bounded correlational structure.

Axiom E10 — Structural Relation between $D_S^{(e)}$ and C_e

There exists an internal constraint:

$$G(D_S^{(e)}, C_e) = 0,$$

which is not physical, geometric, dynamic, observable, invertible, or arbitrary. The emergent correlation contains no more information than the emergent distance.

Axiom E11 — Minimal Examples of Emergence

$$X = \lim_{\lambda \rightarrow \infty} S_\lambda, \quad D_S^{(e)} = \lim_{\lambda \rightarrow \infty} F_\lambda(D_S^{(f)}), \quad C_e = \lim_{\lambda \rightarrow \infty} G_\lambda(C_f).$$

Each limit is an informational compression of the prespatial structure S .

Axiom E12 — Synthesis of the Emergent Domain

The non-observable emergent domain is the continuous structure obtained from the compression of the prespatial network S . It is characterized by:

- the **emergent structural distance** $D_S^{(e)}$,
- the **emergent correlation** $C^{(e)}$,
- **continuity, stability, and regularity**,
- **correlative monotonicity**,
- **compressivity and saturation**,
- the **complete absence of geometry and physics**.

The quantities $D_S^{(e)}$ and $C^{(e)}$ are continuous and compressed images of the prespatial structure S , of the fundamental distance $D_S^{(f)}$, and of the binary fundamental correlation C_f , determined by the threshold Δ and by the structural index t .

Minimal Example: Definition of the Regularization Operators F_λ and G_λ

In this minimal example we explicitly construct the regularization operators F_λ and G_λ , which generate the emergent distance $D_S^{(e)}$ and the emergent correlation C_e from the discrete fundamental structure $(S, D_S^{(f)}, C_f)$.

1. Emergent Domain and Immersion of the Fundamental Structure

Let the emergent domain be

$$X = [0, 1].$$

Define a discrete immersion

$$\varphi: S \rightarrow X$$

such that:

- the elements of S_A are immersed in $\left[0, \frac{1}{2}\right]$,
- the elements of S_B are immersed in $\left[\frac{1}{2}, 1\right]$.

If

$$S_A = \{s_1, \dots, s_{n_A}\}, \quad S_B = \{s_{n_A+1}, \dots, s_N\},$$

we set

$$\begin{aligned} \varphi(s_i) &= \frac{i-1}{2n_A} \text{ for } s_i \in S_A, \\ \varphi(s_j) &= \frac{1}{2} + \frac{j-n_A-1}{2n_B} \text{ for } s_j \in S_B, \end{aligned}$$

with $n_B = N - n_A$.

2. Regularization Operator F_λ on the Fundamental Distance $D_S^{(f)}$

For every $\lambda > 0$, define the regularization operator acting on the fundamental distance:

$$F_\lambda \left(D_S^{(f)} \right) (x, y) = \frac{1}{Z_\lambda(x, y)} \sum_{i,j} D_S^{(f)}(s_i, s_j) \chi_{B_\lambda(x)}(\varphi(s_i)) \chi_{B_\lambda(y)}(\varphi(s_j)),$$

where:

- $B_\lambda(x) = (x, -\lambda x + \lambda) \cap [0, 1]$,
- $\chi_{B_\lambda(x)}$ is the characteristic function of $B_\lambda(x)$,
- $Z_\lambda(x, y)$ is the number of pairs (i, j) such that $\varphi(s_i) \in B_\lambda(x)$ and $\varphi(s_j) \in B_\lambda(y)$.

The operator F_λ :

- is monotonic with respect to the scale λ ,
- is compressive (it averages the discrete values of $D_S^{(f)}$),
- is regularizing (it removes local discontinuities),
- is non-physical and non-dynamical, as required by Axioms E3–E6.

3. Emergent Limit of the Distance $D_S^{(e)}$

In the limit $\lambda \rightarrow 0$, the intervals $B_\lambda(x)$ select only the elements of S immersed in infinitesimal neighborhoods around x and y . For the chosen block structure, we obtain:

$$D_S^{(e)}(x, y) = \begin{cases} a & \text{if } x, y \in \left[0, \frac{1}{2}\right] \text{ or } x, y \in \left[\frac{1}{2}, 1\right], \\ b & \text{if } x \in \left[0, \frac{1}{2}\right], y \in \left[\frac{1}{2}, 1\right] \text{ or vice versa,} \end{cases} \quad 0 < a < b.$$

The function $D_S^{(e)}$:

- is piecewise continuous,
- is bounded,
- is pregeometric and non-physical,
- is a structural compression of the fundamental distance $D_S^{(f)}$.

4. Regularization Operator G_λ on the Fundamental Correlation C_f

Define analogously the regularization operator acting on the fundamental correlation:

$$G_\lambda(C_f)(x, y) = \frac{1}{Z_\lambda(x, y)} \sum_{i, j} C_f(i, j) \chi_{B_\lambda(x)}(\varphi(s_i)) \chi_{B_\lambda(y)}(\varphi(s_j)).$$

In the limit $\lambda \rightarrow 0$, we obtain the emergent correlation:

$$C_e(x, y) = \begin{cases} 1 & \text{if } x, y \in \left[0, \frac{1}{2}\right] \text{ or } x, y \in \left[\frac{1}{2}, 1\right], \\ 0 & \text{if } x \in \left[0, \frac{1}{2}\right], y \in \left[\frac{1}{2}, 1\right] \text{ or vice versa.} \end{cases}$$

5. Structural Relation Between $D_S^{(e)}$ and C_e

In this explicit example, the emergent correlation is a monotonically decreasing function of the emergent distance:

$$C_e(x, y) = \begin{cases} 1 & \text{if } D_S^{(e)}(x, y) = a, \\ 0 & \text{if } D_S^{(e)}(x, y) = b. \end{cases}$$

This realizes concretely:

- correlational monotonicity,
- compressivity,
- saturation,
- informational rigidity,

in perfect agreement with Axioms E3–E6.

3. Structural Regularization Operator R

The raw emergent structure

$$(X, D_S^{(e)}, C_e)$$

is continuous, coherent, and pregeometric (Axiom E4), but it is not yet compatible with the physical compression performed by the projection map Φ . An internal structural operator is required to

increase the *compressive regularity* of the emergent structure without introducing any physical quantity.

This operator is the **Structural Regularization Operator R** .

R is purely ontological: it is not physical, not dynamical, not geometrical, and not invertible. Its role is to prepare the emergent structure for the projection Φ .

R1 — Domain of Definition

$$R: (X, D_S^{(e)}, C_e) \rightarrow (X, D_{S \text{ reg}}^{(e)}, C_{e \text{ reg}})$$

where:

- X is the continuous emergent domain (image of Ψ);
- $D_S^{(e)}$ is the emergent structural distance;
- C_e is the emergent correlation;
- $D_{S \text{ reg}}^{(e)}$ and $C_{e \text{ reg}}$ are the compressively regularized versions of the emergent structures.

R does not modify the domain X . X is already continuous, isotropic, coherent, and saturated by construction (Axiom E4). The regularization acts exclusively on the emergent structures.

R2 — Definition of the Operator

There exists a structural operator:

$$R = (R_D, R_C)$$

such that:

$$D_{S \text{ reg}}^{(e)} = R_D(D_S^{(e)}), \quad C_{e \text{ reg}} = R_C(C_e)$$

R is defined solely through structural constraints internal to the emergent structure.

R3 — Structural Monotonicity

For all $x, y, z \in X$:

$$D_S^{(e)}(x, y) < D_S^{(e)}(x, z) \implies D_{S \text{ reg}}^{(e)}(x, y) < D_{S \text{ reg}}^{(e)}(x, z)$$

and analogously for C_e .

R preserves the structural order of the emergent domain.

R4 — Non-Invertibility

No inverse map exists:

$$\nexists R^{-1}: (X, D_{S \text{ reg}}^{(e)}, C_{e \text{ reg}}) \rightarrow (X, D_S^{(e)}, C_e)$$

Structural reasons:

- R is compressive;
- it eliminates residual oscillations;

- it reduces emergent information;
- it increases compressive regularity.

R5 — Non-Physicality

R does not represent any physical quantity:

R is non-dynamical, non-geometrical, non-measurable.

R is internal to the ontological hierarchy and does not introduce any physical structure.

R6 — Compressive Regularity

The regularization induced by R increases the regularity of the emergent structure:

- **compressive continuity:**

$D_{S \text{ reg}}^{(e)}$ is more regular than $D_S^{(e)}$

- **correlative stability:**

$C_{e \text{ reg}}$ is more stable than C_e

- **structural coherence:**

$$G(D_S^{(e)}, C_e) = 0 \Rightarrow G(D_{S \text{ reg}}^{(e)}, C_{e \text{ reg}}) = 0$$

R does not create new structures: it reinforces the existing ones.

R7 — Non-Arbitrariness

R must respect:

- structural constraints of the emergent domain;
- continuity;
- stability;
- coherence;
- correlative monotonicity.

No freedom is allowed beyond what is derived from $D_S^{(e)}$ and C_e .

R8 — Non-Linearity

$$R(\alpha D_S^{(e)} + \beta C_e) \neq \alpha R(D_S^{(e)}) + \beta R(C_e)$$

The non-linearity reflects the non-linear nature of the emergent structure.

R9 — Informational Compressivity

$(X, D_{S \text{ reg}}^{(e)}, C_{e \text{ reg}})$ contains less information than $(X, D_S^{(e)}, C_e)$

Consequences:

- structural rigidity;

- elimination of residual oscillations;
- compressive saturation;
- global regularity.

R10 — Preparation for Projection

The regularized structure is the only one compatible with the projection Φ :

$$\Phi \text{ is well-defined only on } \left(X, D_{S \text{ reg}}^{(e)}, C_{e \text{ reg}} \right)$$

R is the ontological bridge between:

- the pregeometric emergent structure, and
- the observable physical domain.

Summary

The structural regularization operator R :

- is internal to the ontological hierarchy;
- does not introduce any physical quantity;
- increases the *compressive* regularity of the emergent structure;
- eliminates residual oscillations and local amplifications;
- stabilizes the emergent correlation;
- reduces structural information;
- preserves monotonicity, continuity, and coherence;
- prepares the emergent structure for the physical projection Φ ;
- replaces the intuitive role of the Laplacian without violating pregeometricity;
- makes the application of the projection map Φ natural and well-defined.

Note on the notation of the regularized structure

In the document we use the compact notation

$$\left(X, D_S^{(e)}, C_e \right)_{\text{reg}}$$

to denote the regularized emergent structure. This notation **does not** imply that the domain X is modified: X remains unchanged, continuous, and coherent by construction.

The compact form

$$\left(X, D_S^{(e)}, C_e \right)_{\text{reg}}$$

is therefore equivalent to the explicit form:

$$\left(X, D_{S \text{ reg}}^{(e)}, C_{e \text{ reg}} \right)$$

where the regularization acts **only** on the emergent structures $D_S^{(e)}$ and C_e , while the domain X remains unchanged.

4. Structural Complexity Thresholds and Projection Classes Φ

The regularized emergent structure $(X, D_S^{(e)}, C_e)_{\text{reg}}$ possesses sufficient continuity, stability, and coherence to be analyzed through a global structural complexity index. The emergence of the projection classes Φ is not arbitrary: each class requires the structural complexity to exceed a specific threshold. Complexity thresholds determine the hierarchical succession of the projection maps Φ .

Axiom C1 — Emergent Complexity Function

There exists a global structural complexity function:

$$\mathcal{C}: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow \mathbb{R}^+$$

such that:

- $\mathcal{C}$ is continuous,
- $\mathcal{C}$ is monotonic with respect to structural regularity,
- $\mathcal{C}$ is compressive: it contains no more information than $(D_S^{(e)}, C_e)_{\text{reg}}$,
- $\mathcal{C}$ is non-physical, non-measurable, and non-dynamical.

$\mathcal{C}$ measures the global structural complexity of the regularized emergent domain.

Axiom C2 — Complexity Thresholds

There exists an ordered family of structural thresholds:

$$\Theta_1 < \Theta_2 < \Theta_3 < \Theta_4 < \Theta_5 < \Theta_6$$

such that each threshold Θ_k corresponds to the projection class Φ_k .

The thresholds are internal properties of the emergent structure and do not depend on any physical quantity.

Axiom C3 — Emergence Condition for Projection Φ_k

The projection Φ_k is definable if and only if:

$$\mathcal{C}(X, D_S^{(e)}, C_e)_{\text{reg}} \geq \Theta_k.$$

If the complexity does not reach the threshold Θ_k , the projection Φ_k cannot emerge.

Axiom C4 — Hierarchical Succession of Projections Φ

If:

$$\mathcal{C} \in [\Theta_k, \Theta_{k+1})$$

then the active projections are:

$$\Phi_1, \Phi_2, \dots, \Phi_k,$$

while Φ_{k+1} is not yet definable.

The succession of projection classes is determined by the growth of emergent complexity.

Axiom C5 — Non-Arbitrariness of Thresholds

The thresholds θ_k depend exclusively on structural properties of the regularized emergent domain:

- rigidity of $D_S^{(e)}$,
- correlational saturation,
- global coherence,
- correlational monotonicity,
- continuity and stability induced by R .

No freedom is allowed beyond what is determined by the emergent structure.

Axiom C6 — Non-Physical Nature of Complexity

The function $\mathcal{C}$ and the thresholds θ_k :

- do not represent any physical quantity,
- are not measurable,
- are not dynamical,
- are not geometric.

They are structural indices internal to the ontological hierarchy of the PSE.

Axiom C7 — Compatibility with the Operator R

Complexity is well-defined only on the regularized structure:

$$\mathcal{C} \text{ is well-defined only on } \left(X, D_S^{(e)}, \mathcal{C}_e \right)_{\text{reg}}.$$

The operator R is therefore necessary to ensure that $\mathcal{C}$ is continuous, monotonic, and non-arbitrary.

Axiom C8 — Activation of Projection Φ

The active projection Φ is determined by the threshold reached:

$$\Phi_k: \left(X, D_S^{(e)}, \mathcal{C}_e \right)_{\text{reg}} \rightarrow D_{\text{fisico}}^{(k)}$$

if and only if:

$$\mathcal{C} \geq \theta_k.$$

Each projection class Φ is therefore a structural compression compatible with the emergent complexity.

Axiom C9 — Stability of Projection Class Φ_k

If the complexity remains within:

$$\mathcal{C} \in [\theta_k, \theta_{k+1})$$

then the projection Φ_k is stable: no transition to Φ_{k+1} is possible.

Transitions between classes are discrete and governed by the thresholds.

Axiom C10 — Synthesis of Complexity Thresholds

Complexity thresholds:

- determine the succession of projection classes Φ ,
- guarantee the non-arbitrariness of the projection process,
- ensure that physical quantities emerge only when the structure is sufficiently complex,
- are internal properties of the regularized emergent domain,
- constitute the bridge between the operator R and the projection maps Φ .

Summary of Chapter

This chapter establishes that:

- the regularized emergent structure possesses a global complexity index,
- structural thresholds regulate the emergence of projection classes Φ ,
- each Φ_k requires a minimum complexity,
- the active projection Φ is determined by the threshold reached,
- the succession of classes is hierarchical, non-arbitrary, and non-physical.

The complete operative chain becomes:

$$S \rightarrow D_S^{(f)} \rightarrow C_f \rightarrow D_S^{(e)} \rightarrow C_e \rightarrow R \rightarrow \Theta_k \rightarrow \Phi_k \rightarrow \mathcal{O}_{\text{phys}}.$$

5. Projection Map Φ

The projection map Φ is the structural operator that formalizes the transition from the regularized emergent domain

$$E_{\text{reg}} = (X, D_S^{(e)}, C_{e \text{ reg}})$$

to the observable physical domain of class k ,

$$\Phi: (X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)} \rightarrow \mathcal{D}_{\text{physical}}^{(k)}$$

Φ represents the only point in the ontological chain where physical quantities appear. The projection is monotonic, compressive, non-linear, and non-invertible: the physical domain necessarily contains less information than the emergent structure.

Axiom $\Phi 0$ — Domain of Definition

The projection map operates on the regularized emergent domain:

$$E_{\text{reg}} = R(X, D_S^{(e)}, C_e),$$

where:

- $X = \lim_{\lambda \rightarrow \infty} S_\lambda$ is the continuous emergent domain;
- $D_S^{(e)} = \lim_{\lambda \rightarrow \infty} F_\lambda(D_S^{(f)})$ is the emergent structural distance;
- $C_e = \lim_{\lambda \rightarrow \infty} G_\lambda(C_f)$ is the emergent correlation;
- R is the structural regularization operator ensuring continuity, stability, and coherence.

The emergent complexity of the regularized domain is defined as:

$$C_{\text{reg}} = C(E_{\text{reg}}).$$

The projection class Φ_k is activated if and only if:

$$C_{\text{reg}} \geq \Theta_k.$$

The structural properties of continuity, stability, saturation, and correlational monotonicity:

$$C_e(x, y) = F(D_S^{(e)}(x, y)), \quad F' < 0,$$

constitute the conditions that make the projection well-defined and non-arbitrary.

Axiom $\Phi 1$ — Definition of the Projection

There exists a family of projection maps:

$$\Phi_k: E_{\text{reg}} \rightarrow D_{\text{phys}}^{(k)},$$

where $D_{\text{phys}}^{(k)}$ is a physical observable domain containing geometric, dynamical, statistical, field-like, or discrete structures, depending on the class activated by the threshold Θ_k .

Formally:

$$\Phi_k(X, D_S^{(e)}, C_e)_{\text{reg}} = \mathcal{O}_{\text{phys}}^{(k)},$$

where $\mathcal{O}_{\text{phys}}^{(k)}$ denotes the set of physical quantities emerging in class k .

Axiom $\Phi 2$ — Structural Monotonicity

For all $x, y, z \in X$:

$$D_S^{(e)}(x, y) < D_S^{(e)}(x, z) \implies \Phi_k(x, y) < \Phi_k(x, z),$$

where $<$ is a structural order induced in the physical domain (e.g., intensity, proximity, probability, correlation).

The projection preserves the structural order induced by $D_S^{(e)}$, without requiring linearity or isometry.

Axiom $\Phi 3$ — Non-Invertibility

No inverse map exists:

$$\nexists \Phi_k^{-1}: D_{\text{phys}}^{(k)} \rightarrow E_{\text{reg}}.$$

Structural reasons:

- the physical domain is a compressive representation of the emergent structure;
- the emergent correlation C_e contains less information than the emergent distance $D_S^{(e)}$;
- the emergent structure contains less information than the fundamental structure;
- the threshold Δ rigidly limits the fundamental correlation C_f ;
- the compressivity of the operator R further reduces the available information.

Non-invertibility is a consequence of informational compressivity.

Axiom Φ_4 — Non-Physical Nature of the Projection

The map Φ_k is not a physical operator:

Φ_k is non-dynamical, non-interactive, non-measurable.

It is a structural operator internal to the ontological hierarchy.

Axiom Φ_5 — Induced Regularity

Emergent regularity constrains the resulting physical structures:

- continuity of $D_S^{(e)} \Rightarrow$ continuity of physical quantities;
- stability of $D_S^{(e)} \Rightarrow$ stability of physical structures;
- coherence of $C_e \Rightarrow$ coherence of physical correlations.

Induced regularity is a joint consequence of R and Φ_k .

Axiom Φ_6 — Non-Arbitrariness

The map Φ_k must respect:

- structural constraints of the emergent domain,
- coherence constraints,
- stability constraints,
- monotonicity constraints,
- the threshold Θ_k .

No physical freedom is allowed beyond what derives from $D_S^{(e)}$, C_e , R, and $\mathcal{C}$.

Axiom Φ_7 — Non-Linearity

The map Φ_k is non-linear:

$$\Phi_k \left(\alpha D_S^{(e)} + \beta C_e \right) \neq \alpha \Phi_k \left(D_S^{(e)} \right) + \beta \Phi_k (C_e).$$

Non-linearity reflects the non-linear nature of the emergent structure.

Axiom Φ_8 — Informational Compressivity

The map Φ_k is compressive: it translates a finite amount of structural information

$$(X, D_S^{(e)}, C_e)_{\text{reg}}$$

into a regular physical representation $\mathcal{O}_{\text{phys}}^{(k)}$.

Consequences:

Physical rigidity Physical quantities cannot indefinitely amplify emergent information.

Physical saturation Saturation arises exclusively from the compressivity of:

- the operators F_λ and G_λ ,
- the emergent distance $D_S^{(e)}$,
- the emergent correlation C_e ,
- the regularization operator R ,
- the projection Φ_k .

Structural limitation The physical domain contains less information than the regularized emergent domain.

Global regularity Compressivity prevents uncontrolled physical divergences: emergent physical quantities are regular and finite across the entire physical domain.

Minimal Example of Structural Projection Φ

Emergent domain

The emergent domain is the pregeometric continuum:

$$X = [0, 1].$$

Regularized emergent structure

The regularized emergent structure is:

$$E_{\text{reg}} = (X, D_S^{(e)}, C_e)_{\text{reg}}$$

In this minimal example:

$$D_S^{(e)}(x, y) = \begin{cases} a & x, y \in [0, 1/2] \text{ or } x, y \in [1, 2/2], \\ b & x \in [0, 1/2], y \in [1, 2/2] \text{ or vice versa,} \end{cases} \quad 0 < a < b,$$

$$C_e(x, y) = \begin{cases} 1 & x, y \in [0, 1/2] \text{ or } x, y \in [1, 2/2], \\ 0 & x \in [0, 1/2], y \in [1, 2/2] \text{ or vice versa.} \end{cases}$$

The emergent correlation is monotonically decreasing with respect to the emergent distance:

$$C_e(x, y) = F(D_S^{(e)}(x, y)), \quad F' < 0.$$

Projection Φ

Define:

$$\Phi_1: E_{\text{reg}} \rightarrow D_{\text{phys}}^{(1)}.$$

The projection acts on the emergent structure by transforming the emergent distance into an effective metric:

$$\Phi_1(D_S^{(e)}, C_e) = g(x, y),$$

with:

$$g(x, y) = \begin{cases} \alpha & D_S^{(e)}(x, y) = a, \\ \beta & D_S^{(e)}(x, y) = b, \end{cases} \alpha < \beta.$$

Structural interpretation

The map Φ_1 :

- **preserves the structural order** induced by $D_S^{(e)}$ (Axiom $\Phi 2$);
- **is compressive** (Axiom $\Phi 8$): $\{a, b\} \rightarrow \{\alpha, \beta\}$;
- **is non-invertible** (Axiom $\Phi 3$): $g(x, y)$ does not allow reconstruction of $D_S^{(e)}$ or C_e ;
- **is non-physical** (Axiom $\Phi 4$): Φ does not represent dynamics or interactions;
- **respects joint dependence**: the physical metric depends simultaneously on $D_S^{(e)}$ and C_e .

Summary of Chapter Φ

In this chapter:

- the regularized emergent domain is the starting point of the projection;
- the emergent complexity $\mathcal{C}$ and the thresholds Θ_k determine the projection class Φ_k ;
- Φ_k is monotonic, compressive, non-linear, and non-invertible;
- Φ_k does not introduce physics, but generates regular physical quantities;
- the projection is the only point in the ontological chain where physical quantities appear.

The full operative chain is:

$$S \rightarrow D_S^{(f)} \rightarrow C_f \rightarrow D_S^{(e)} \rightarrow C_e \rightarrow R \rightarrow \Theta_k \rightarrow \Phi_k \rightarrow \mathcal{O}_{\text{phys}}.$$

6. Emergence of Physical Quantities

The previous section describes the transition from the regularized pregeometric emergent structure

$$E_{\text{reg}} = (X, D_S^{(e)}, C_e)_{\text{reg}}$$

to observable physical quantities. In this neutral formulation, no physical quantity is assumed to be fundamental: geometry, dynamics, statistical distributions, fields, discrete structures, and quantum structures are all compressed images of the same structural information, obtained through a projection class Φ_k .

Emergent physical quantities are treated as particular cases of monotonic, compressive, non-linear, and non-invertible projections, activated by the complexity threshold Θ_k .

Axiom P1 — Physical Domain as the Image of Φ_k

The physical domain of class k is defined as:

$$D_{\text{phys}}^{(k)} = \Phi_k(E_{\text{reg}}),$$

where the regularized emergent domain is:

$$E_{\text{reg}} = R(X, D_S^{(e)}, C_e),$$

and:

- $X = \lim_{\lambda \rightarrow \infty} S_\lambda$ is the continuous pregeometric emergent domain;
- $D_S^{(e)} = \lim_{\lambda \rightarrow \infty} F_\lambda(D_S^{(f)})$ is the emergent structural distance;
- $C_e = \lim_{\lambda \rightarrow \infty} G_\lambda(C_f)$ is the emergent correlation;
- R is the structural regularization operator.

The emergent complexity of the regularized domain is:

$$C_{\text{reg}} = \mathcal{C}(E_{\text{reg}}).$$

The projection class Φ_k is active if and only if:

$$C_{\text{reg}} \geq \Theta_k.$$

Properties

- **Observability:** unlike S , $D_S^{(f)}$, $D_S^{(e)}$, and C_e , the physical domain contains measurable quantities.
- **Non-fundamentality:** $D_{\text{phys}}^{(k)}$ is not ontologically primary; it is a compressed representation of the emergent structure.
- **Structural dependence:** every physical quantity is determined solely by E_{reg} and the threshold Θ_k .

Axiom P2 — Emergent Physical Quantities

Every physical quantity Q is defined as the image of a monotonic projection belonging to class Φ_k :

$$Q = \Phi_{k,Q}(E_{\text{reg}}),$$

where $\Phi_{k,Q}$ is a map internal to class Φ_k .

Properties

Monotonicity

$$D_S^{(e)}(x, y) < D_S^{(e)}(x, z) \Rightarrow \Phi_{k,Q}(x, y) < \Phi_{k,Q}(x, z),$$

where $<$ is an induced physical order.

Non-linearity

$$\Phi_{k,Q}(\alpha D_S^{(e)} + \beta C_e) \neq \alpha \Phi_{k,Q}(D_S^{(e)}) + \beta \Phi_{k,Q}(C_e).$$

Non-invertibility

$$\exists \Psi_Q: Q \rightarrow E_{\text{reg}}.$$

Compressivity

$$\text{Info}(Q) < \text{Info}\left(D_S^{(e)}, C_e\right)_{\text{reg}}.$$

Axiom P3 — Classes of Emergence Maps

The projection classes are:

$$\Phi = \{\Phi_1, \Phi_2, \Phi_3, \Phi_4, \Phi_5, \Phi_6\},$$

activated respectively by the thresholds:

$$\theta_1 < \theta_2 < \theta_3 < \theta_4 < \theta_5 < \theta_6.$$

Each class Φ_k generates physical quantities consistent with the emergent structure, including:

- geometric structures (metrics, physical distances, connections),
- dynamical structures (equations of motion, evolutionary constraints),
- statistical structures (distributions, probabilities),
- field structures (intensities, potentials, configurations),
- discrete structures (graphs, networks, finite operators),
- quantum structures (state spaces, correlational constraints, coherence thresholds).

Properties

- each class is monotonic and compressive;
- no class is ontologically privileged;
- each class produces physical quantities consistent with E_{reg} .

Axiom P4 — Physical Saturation

Every emergent physical quantity Q is saturating: there exists a maximum value Q_{max} such that:

$$Q(x, y) \leq Q_{\text{max}}, \forall x, y \in X.$$

Structural Motivation

Saturation arises from the compressivity of:

- the operators F_λ and G_λ ,
- the emergent distance $D_S^{(e)}$,
- the emergent correlation C_e ,
- the regularization operator R ,
- the projection Φ_k .

Thus:

$$Q_{\max} = \sup_{x,y \in X} \Phi_{k,Q} \left(D_S^{(e)}, (x,y) C_e(x,y) \right) < +\infty.$$

Axiom P5 — Physical Rigidity

Every emergent physical quantity is rigidly constrained by the emergent structure:

$$| \delta Q | \leq K | \delta D_S^{(e)} |,$$

where K is a structural constant depending on class Φ_k .

Consequences

- no physical quantity can indefinitely amplify microscopic variations;
- physical structures are regular, finite, and stable;
- rigidity is a structural, not dynamical, property.

Axiom P6 — Non-Retroactivity

The physical domain cannot influence the emergent structure:

$$\frac{\partial \left(D_S^{(e)}, C_e \right)}{\partial Q} = 0.$$

Formally:

$$\exists \Psi: D_{\text{phys}}^{(k)} \rightarrow E_{\text{reg}}.$$

Consequences

- no physical quantity can modify the pregeometric structure;
- the ontological hierarchy is unidirectional;
- non-invertibility is guaranteed.

Axiom P7 — Emergent Observability

A quantity is observable only if it belongs to $D_{\text{phys}}^{(k)}$.

The structural quantities:

$$\{S, D_S^{(f)}, C_f, D_S^{(e)}, C_e\}$$

are real but not observable.

Formally:

$$\mathcal{R} = \{S, D_S^{(f)}, C_f, D_S^{(e)}, C_e\}, \quad \mathcal{O} = D_{\text{fisico}}^{(k)}, \quad \mathcal{R} \cap \mathcal{O} = \emptyset.$$

7. Ontological Structure and Functional Spaces of the PSE

In this section we formally define the normed spaces used in Theorems 19–30. These spaces constitute the mathematical backbone of the operative chain of the PSE:

$$F \rightarrow E \rightarrow E_{\text{reg}} \rightarrow \Theta_k \rightarrow \Phi_k \rightarrow D_{\text{phys}}(k).$$

7.1 Fundamental Space F

The pregeometric emergent set is X . The fundamental structure is a function:

$$F: X \times X \rightarrow \mathbb{R}.$$

We define:

$$\mathcal{F} := C^0(X \times X)$$

as the space of continuous functions on $X \times X$, equipped with the sup norm:

$$\|F\|_{\mathcal{F}} := \sup_{(x,y) \in X \times X} |F(x,y)|.$$

7.2 Emergent Space E

The emergent domain is the pair:

$$E = (D_S^{(e)}, C_e),$$

where both components are continuous functions on $X \times X$.

We define:

$$E := C^0(X \times X) \times C^0(X \times X),$$

with norm:

$$\|E\|_E := \max(\|D_S^{(e)}\|_{\infty}, \|C_e\|_{\infty}).$$

7.3 Regularized Emergent Space E_{reg}

The regularity operator:

$$R: E \rightarrow E$$

produces the regularized emergent domain:

$$E_{\text{reg}} = R(E).$$

Regularization ensures:

- structural continuity,
- boundedness,
- absence of divergences,
- compatibility with the complexity thresholds Θ_k .

7.4 Space of Emergent Physical Metrics $\mathcal{G}$

An emergent physical metric is a symmetric rank-2 tensor:

$$g_{\mu\nu}: X \rightarrow \mathbb{R}.$$

We define:

$$\mathcal{G} := C^0(X; \text{Sym}^2),$$

the space of continuous metric tensors, with norm:

$$\|g_{\mu\nu}\|_{\mathcal{G}} := \sup_{x \in X} |g_{\mu\nu}(x)|.$$

7.5 Space of Emergent Physical Fields $\mathcal{F}_{\text{phys}}$

The emergent physical fields generated by the projections Φ_k are:

$$\mathcal{F}_{\text{phys}} = \{\phi, A_\mu, T_{\mu\nu}, J_\mu, \rho, \sigma, P, F_\mu, \dots\}.$$

Formally:

$$\mathcal{F}_{\text{phys}} := \prod_i C^0(X; V_i),$$

where each V_i is the appropriate vector space (scalars, vectors, tensors).

Norm:

$$\|\mathcal{F}_{\text{phys}}\| := \max_i \|F_i\|_\infty.$$

7.6 Space of Emergent Physical Domains $\mathcal{P}$

Each projection class Φ_k generates a distinct emergent physical domain:

$$D_{\text{phys}}(k) = \Phi_k(E_{\text{reg}}).$$

The full physical domain is:

$$P = (g_{\mu\nu}, \mathcal{F}_{\text{phys}}).$$

We define:

$$\mathcal{P} := \mathcal{G} \times \mathcal{F}_{\text{phys}},$$

with norm:

$$\|P\|_{\mathcal{P}} := \max(\|g_{\mu\nu}\|_{\mathcal{G}}, \|\mathcal{F}_{\text{phys}}\|).$$

7.7 Space of Observable Physical Quantities $\mathcal{Q}_{\text{phys}}$

Observable emergent physical quantities are:

$$\mathcal{Q}_{\text{phys}} = \{Q_1, Q_2, \dots\}.$$

We define:

$$\mathcal{Q}_{\text{phys}} := \prod_j C^0(X; W_j),$$

where each W_j is the vector space associated with the corresponding physical quantity.

Norm:

$$\|Q\|_{\mathcal{Q}_{\text{phys}}} := \max_j \|Q_j\|_\infty.$$

7.8 Space of Emergent Physical Dynamics $\mathcal{D}$

Emergent dynamics are defined as:

$$\mathcal{D}_{\text{phys}} = \Delta(P),$$

where Δ produces equations of motion, evolution operators, and physical laws.

We define:

$$\mathcal{D} = C^0(X; U),$$

where U is the space of dynamical structures.

Norm:

$$\| \mathcal{D}_{\text{phys}} \|_{\mathcal{D}} = \sup_{x \in X} | \Delta(P)(x) |.$$

7.9 Definition of a Compressive Map

A map between normed spaces:

$$T: A \rightarrow B$$

is **compressive** if there exists $K < 1$ such that:

$$\| T(u) - T(v) \|_B \leq K \| u - v \|_A.$$

Role in the PSE

All fundamental maps are compressive:

$$R: E \rightarrow E_{\text{reg}}, \Phi_k: E_{\text{reg}} \rightarrow D_{\text{phys}}(k).$$

Compressivity guarantees:

- emergent saturation,
- physical stability,
- global regularity,
- non-invertibility,
- rigidity of physical quantities.

7.10 Fundamental Domain Isomorphisms

Isomorphisms encode the absence of physical structure in the fundamental domain.

Fundamental Isomorphisms

Let:

$$\sigma: X \rightarrow X$$

be a bijection of the emergent set.

A fundamental isomorphism satisfies:

$$F(\sigma(x)\sigma(y)) = F(x, y).$$

Interpretation:

- preserves fundamental relations;

- introduces no geometry or physics;
- expresses that labels of elements of X are purely conventional.

Action of Isomorphisms on the Emergent Domain

Given:

$$E = (D_S^{(e)}, C_e),$$

we define:

$$E_\sigma = (D_{S,\sigma}^{(e)}, C_{e,\sigma}),$$

with:

$$\begin{aligned} D_{S,\sigma}^{(e)}(x, y) &= D_S^{(e)}(\sigma(x)\sigma(y)), \\ C_{e,\sigma}(x, y) &= C_e(\sigma(x)\sigma(y)). \end{aligned}$$

Emergent Isotropy

$$E_\sigma = E.$$

Action of Isomorphisms on the Physical Domain

The emergent physical domain is:

$$P = (g_{\mu\nu}, \mathcal{F}_{\text{phys}}).$$

We define:

$$P_\sigma = (g_{\mu\nu,\sigma}, \mathcal{F}_{\text{phys},\sigma}),$$

with:

$$g_{\mu\nu,\sigma}(x) = g_{\mu\nu}(\sigma(x)).$$

Physical Isotropy

$$P_\sigma = P.$$

Compatibility with the Ontological Chain

The maps of the PSE satisfy:

$$\begin{aligned} R(E_\sigma) &= R(E)_\sigma, \\ \Phi_k(E_{\text{reg},\sigma}) &= \Phi_k(E_{\text{reg}}). \end{aligned}$$

Thus the entire chain:

$$F \rightarrow E \rightarrow E_{\text{reg}} \rightarrow \Theta_k \rightarrow \Phi_k \rightarrow D_{\text{phys}}(k)$$

is invariant under isomorphisms.

7.11 Summary

An isomorphism σ :

- is a bijection of the emergent set X ;
- preserves the fundamental structure;
- acts as a relabeling;
- leaves the emergent domain invariant;
- leaves the physical domain invariant;
- is compatible with R and all projections Φ_k ;
- expresses that labels have no physical meaning.

8. Physical Domains of the Projection Φ_k

The projection Φ_k formalizes the transition from the regularized pregeometric emergent domain

$$E_{\text{reg}} = R\left(X, D_S^{(e)}, C_e\right)$$

to the observable physical domain $D_{\text{phys}}^{(k)}$. In this section we formalize the classes of emergent physical quantities, each obtained as a compressive image of the regularized emergent structure. No physical quantity is fundamental: all are effective representations of the same structural information.

Physical saturation arises exclusively from the compressivity of the regularization operators F_λ, G_λ , of the operator R , and of the projection Φ_k . The threshold Δ guarantees structural limitation in the fundamental domain.

Axiom Φ_9 — Classes of Physical Projection

There exist six classes of physical projections, activated by complexity thresholds:

$$\theta_1 < \theta_2 < \theta_3 < \theta_4 < \theta_5 < \theta_6.$$

The classes are:

$$\Phi = \{\Phi_1, \Phi_2, \Phi_3, \Phi_4, \Phi_5, \Phi_6\},$$

which respectively generate:

- **discrete structures:** graphs, networks, finite operators;
- **geometric structures:** physical distance, metrics, connections, curvature;
- **dynamical structures:** equations of motion, evolutionary constraints, physical temporal structures;
- **statistical structures:** densities, distributions, probabilities, statistical correlations;
- **field structures:** potentials, intensities, field configurations;
- **quantum structures:** state spaces, correlational constraints, coherence thresholds.

Each class Φ_k is monotonic, compressive, nonlinear, and non-invertible.

Axiom Φ_{10} — Joint Dependence

Every emergent physical quantity depends jointly on the emergent distance and the emergent correlation of the regularized domain:

$$Q = \Phi_{k,Q}(E_{\text{reg}}) = \Phi_{k,Q}\left(X, D_S^{(e)}, C_e\right)_{\text{reg}}.$$

The dependence is total: no physical quantity can depend on fundamental physical magnitudes, because such magnitudes do not exist in the PSE.

8.1 Discrete Domain

The projection class Φ_1 , activated by the threshold Θ_1 , is the first physical class in the PSE hierarchy. It projects the regularized emergent domain

$$E_{\text{reg}} = \left(X, D_S^{(e)}, C_e\right)_{\text{reg}}$$

into emergent discrete physical structures:

$$\Phi_1: E_{\text{reg}} \rightarrow G = (V, E).$$

Definition Φ_9 — Vertices and Edges of the Graph

$$V = \Phi_1(X), E = \Phi_1\left(D_S^{(e)}, C_e\right).$$

Properties of the Emergent Graph

- **Discrete monotonicity:** structural variations produce monotonic variations in connectivity.
- **Discrete regularity:** the graph is finite and free of divergences.
- **Discrete saturation:**

$$|E| \leq E_{\text{max}}.$$

Definition Φ_{10} — Discrete Operators $\mathcal{O}_{\text{disc}}$

$$\mathcal{O}_{\text{disc}} = \Phi_1\left(X, D_S^{(e)}, C_e\right)_{\text{reg}}.$$

Properties of Discrete Operators

- **Monotonicity:** structural variations produce monotonic variations of the operator.
- **Regularity:** operators are finite and stable.
- **Saturation:**

$$|\mathcal{O}_{\text{disc}}| \leq \mathcal{O}_{\text{max}}.$$

Discrete Rigidity and Continuous Projection

The family of stable discrete configurations

$$\Gamma = \{\gamma\}$$

formalized in the Discrete Domain represents the set of coherent structures generated by the emergent correlation C_e .

For each stable discrete configuration $\gamma \in \Gamma$, the emergent rigidity associated with the correlation C_e defines a continuous parameter through the projection:

$$\gamma_{\text{cont}} = \Phi_{1, \text{rig}}(\gamma).$$

The metric parameter γ used in emergent deviations,

$$\epsilon(r) = \frac{\beta \alpha r}{1 + \gamma r},$$

is therefore the continuous projection of the discrete rigidity of the configuration γ .

The projection $\Phi_{1,\text{rig}}$ is consistent with Axioms $\Phi 1$ – $\Phi 10$: it is compressive, non-invertible, and monotonic with respect to structural variations of $D_S^{(e)}$ and C_e .

Emergent Discrete Physical Domain

The discrete domain describes the effective physical representation of the discrete configurations emerging from the regularized domain E_{reg} . Discrete quantities are not fundamental: they are compressed images of the emergent structure, obtained through the projection Φ_1 .

1. Emergent Discrete Configurations

The emergent structure $(D_S^{(e)}, C_e)$ exhibits coherent and incoherent regions. We define an emergent discrete configuration as a partition of X induced by the emergent correlation:

$$\Gamma = \{X_\alpha, X_\beta\},$$

where:

$$X_\alpha = [0, 1/2], \quad X_\beta = [1, 21].$$

This partition is generated by the emergent structure:

$$C_e(x, y) = \begin{cases} 1 & x, y \in X_\alpha \text{ or } x, y \in X_\beta, \\ 0 & x \in X_\alpha, y \in X_\beta \text{ or vice versa.} \end{cases}$$

2. Discrete Projection Φ_1

We define the discrete projection as:

$$\Phi_1: E_{\text{reg}} \rightarrow \mathcal{N},$$

where $\mathcal{N}$ is a set of emergent discrete physical quantities.

The projection is defined as:

$$\mathcal{N}(x) = \Phi_1(D_S^{(e)}, C_e)(x).$$

In the minimal case, the discrete projection assigns a discrete value to each emergent region:

$$\mathcal{N}(x) = \begin{cases} n_\alpha & x \in X_\alpha, \\ n_\beta & x \in X_\beta, \end{cases} \quad n_\alpha \neq n_\beta.$$

Interpretation

- each emergent region generates a discrete quantity;
- discrete quantities are non-invertible compressions of the emergent structure.

3. Emergent Discrete Quantities

The emergent discrete physical quantities are:

$$\mathcal{N} = \{n_\alpha, n_\beta\}.$$

They represent:

- discrete states,
- physical labels,
- quantized quantities,
- configuration classes.

In the minimal example:

- n_α represents the discrete quantity associated with region X_α ;
- n_β represents the discrete quantity associated with region X_β .

4. Physical Discrete Configurations

A physical discrete configuration is defined as:

$$\Gamma_{\text{phys}} = \Phi_1(\Gamma) = \{n_\alpha, n_\beta\}.$$

The physical configuration is therefore a compression of the emergent partition.

5. Properties of the Discrete Domain

The emergent discrete domain satisfies:

- **non-invertibility:** from $\mathcal{N}$ it is not possible to recover E_{reg} ;
- **compressivity:** discrete quantities contain less information than the emergent structure;
- **monotonicity:** monotonic structural variations generate monotonic discrete variations;
- **joint dependence:** discrete quantities depend simultaneously on $D_S^{(e)}$ and C_e ;
- **non-fundamentality:** discrete quantities do not exist in the fundamental domain.

6. Synthesis of the Discrete Domain

In the minimal example:

- discrete quantities are compressed images of the emergent structure;
- the emergent partition generates two discrete quantities;
- the discrete projection is non-invertible and non-fundamental;
- the discrete domain completes the first physical phase of the PSE ontological chain.

8.2 Geometric Domain

The geometric domain describes how the regularized emergent structure

$$E_{\text{reg}} = (X, D_S^{(e)}, C_e)_{\text{reg}}$$

is projected into observable physical geometric structures. The geometric projection is a compressive and non-invertible map belonging to class Φ_2 :

$$\Phi_2(E_{\text{reg}}) \rightarrow (M, d, g_{\mu\nu}),$$

where:

- M is the geometric physical domain,
- d is the physical distance,
- $g_{\mu\nu}$ is the emergent physical metric.

Geometric saturation arises from the compressivity of the operators F_λ, G_λ , of the operator R , and of the projection Φ_2 .

Definition Φ_1 — Physical Distance d

Physical distance is defined as the image of the regularized emergent distance:

$$d = \Phi_2 \left(D_S^{(e)} \right), \quad d: M \times M \rightarrow \mathbb{R}^+.$$

Properties of Physical Distance

- represents measurable separations between points or events in M ;
- satisfies symmetry, identity of indiscernibles, and the triangle inequality;
- is observable through physical procedures;
- is locally defined on M ;
- is bounded:

$$d(p, q) \leq d_{\max}.$$

Physical distance is therefore a compressive representation of the regularized emergent distance.

Definition Φ_2 — Physical Metric $g_{\mu\nu}$

The physical metric is defined as the joint image of the regularized emergent distance and correlation:

$$g_{\mu\nu} = \Phi_2 \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}.$$

Properties of the Physical Metric

- $g_{\mu\nu} \in C^\infty(M)$;
- regular and free of uncontrolled singularities;
- defines the causal structure of the physical domain;
- enables the formulation of field equations and dynamical laws;
- the curvature of $g_{\mu\nu}$ encodes gravity as a compressive image of the emergent structure;
- indirectly observable through physical phenomena;
- geometric saturation:

$$|g_{\mu\nu}(p)| \leq g_{\max}.$$

The physical metric is therefore a compressive and regular image of the regularized emergent structure, consistent with the ontological hierarchy of the PSE.

Emergent Rigidity

Geometric rigidity is defined as a compressive image of the discrete rigidity of stable configurations $\gamma \in \Gamma$ formalized in the Discrete Domain. The geometric projection contains a monotonic sub-map:

$$\Phi_{2,\text{rig}} \subset \Phi_2,$$

which transports discrete rigidity into a continuous parameter:

$$\gamma_{\text{cont}} = \Phi_{2,\text{rig}}(\gamma).$$

The parameter γ thus obtained controls metric saturation, for example in the form:

$$\epsilon(r) = \frac{\beta ar}{1 + \gamma r}.$$

Properties of Emergent Rigidity

- monotonic with respect to variations of $D_S^{(e)}$ and C_e ;
- informationally compressive;
- non-invertible;
- ensures geometric coherence: γ guarantees boundedness and regularity of the emergent geometry.

Emergent rigidity is therefore a geometric property derived from the regularized pregeometric structure, consistent with Axioms $\Phi 1$ – $\Phi 10$.

Emergent Geometric Physical Domain

In this section we define the emergent geometric physical domain generated by the projection Φ_2 applied to the regularized emergent domain. The goal is to show how a discrete and compressive pregeometric structure can produce an effective physical metric, consistent with Axioms $\Phi 1$ – $\Phi 10$.

1. Emergent Distance and Emergent Correlation

From the fundamental structure and the regularization operators F_λ, G_λ , we obtain:

$$D_S^{(e)}(x, y) = \begin{cases} a & x, y \in [0, 1/2] \text{ or } x, y \in [1, 2/3], \\ b & x \in [0, 1/2], y \in [1, 2/3] \text{ or vice versa,} \end{cases} \quad 0 < a < b,$$

$$C_e(x, y) = \begin{cases} 1 & x, y \in [0, 1/2] \text{ or } x, y \in [1, 2/3], \\ 0 & x \in [0, 1/2], y \in [1, 2/3] \text{ or vice versa.} \end{cases}$$

Emergent correlation is monotonically decreasing with respect to emergent distance:

$$C_e(x, y) = F\left(D_S^{(e)}(x, y)\right), \quad F' < 0.$$

2. Geometric Projection Φ_2

We define the projection map:

$$\Phi_2: \left(X, D_S^{(e)}, C_e\right)_{\text{reg}} \rightarrow D_{\text{geom}},$$

where:

$$D_{\text{geom}} = (X, g).$$

The projection is defined as a structural transformation of the emergent distance:

$$\Phi_2(D_S^{(e)}, C_e) = g(x, y),$$

with:

$$g(x, y) = \begin{cases} \alpha & D_S^{(e)}(x, y) = a, \\ \beta & D_S^{(e)}(x, y) = b, \end{cases} \alpha < \beta.$$

3. Geometric Interpretation

The metric g defines an effective physical geometry with two regions:

- **Region 1:** $x \in [0, 1/2]$
- **Region 2:** $x \in [1, 21]$

Within each region:

- physical distance is α ;
- physical correlation is maximal;
- the structure is locally homogeneous.

Between the two regions:

- physical distance is β ;
- physical correlation is zero;
- the structure is discontinuous but coherent.

The resulting geometry is a 1D manifold with two distinct metric domains and a compressive structural transition.

4. Properties of the Projection Φ_2

The map Φ_2 satisfies:

- **Structural monotonicity ($\Phi 2$)**

$$D_S^{(e)}(x, y) < D_S^{(e)}(x, z) \Rightarrow g(x, y) < g(x, z).$$

- **Compressivity ($\Phi 8$)** $\{a, b\} \rightarrow \{\alpha, \beta\}$.
- **Non-invertibility ($\Phi 3$)** From $g(x, y)$ it is not possible to reconstruct $D_S^{(e)}$ or C_e .
- **Non-physicality ($\Phi 4$)** Φ_2 does not represent dynamics or interactions.
- **Joint dependence ($\Phi 10$)** g depends simultaneously on $D_S^{(e)}$ and C_e .

5. Synthesis of the Geometric Domain

The emergent geometric domain shows that:

- a discrete pregeometric structure can generate an effective physical metric;
- physical geometry is a non-invertible compression of the emergent structure;

- emergent correlation determines local geometric coherence;
- emergent distance determines global metric structure;
- the projection Φ_2 realizes the ontological transition from emergence to physicality.

This minimal example constitutes the first explicit realization of the geometric physical domain in the PSE and prepares the construction of the dynamical, statistical, and field domains.

8.3 Dynamical Domain

The dynamical domain describes how the regularized emergent structure

$$E_{\text{reg}} = \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}$$

is projected into dynamical laws, equations of motion, and physical temporal structures through projection class Φ_3 , activated by threshold Θ_3 .

$$\Phi_3: E_{\text{reg}} \rightarrow E_{\text{din}}.$$

Definition Φ_3 — Equations of Motion $\mathcal{L}$

The equations of motion emerge as a compressive image of the regularized emergent domain:

$$\mathcal{L}(p) = \Phi_3(E_{\text{reg}}), \quad p \in M.$$

Properties of the Equations of Motion

- **Dynamical monotonicity:** variations in $D_S^{(e)}$ produce monotonic variations in $\mathcal{L}$.
- **Dynamical rigidity:**

$$| \delta \mathcal{L} | \leq K | \delta D_S^{(e)} |.$$

- **Regularity:** the equations of motion are regular and free of uncontrolled divergences.
- **Non-invertibility:**

$$\exists \Psi: \mathcal{L} \rightarrow E_{\text{reg}}.$$

- **Dynamical saturation:** compressivity implies

$$| \mathcal{L}(p) | \leq \mathcal{L}_{\text{max}}.$$

Definition Φ_4 — Physical Temporal Structure T

The physical temporal component emerges from the regularized emergent correlation:

$$T(p) = \Phi_3(C_e).$$

Properties of the Temporal Structure

- **Temporal monotonicity:** higher correlation $\rightarrow$ higher temporal coherence.
- **Regularity:** T is continuous and differentiable.
- **Temporal saturation:**

$$| T(p) | \leq T_{\text{max}}.$$

Emergent Dynamical Physical Domain

The dynamical domain describes the effective physical representation of the emergent constraints generated by the regularized pregeometric structure E_{reg} . Physical dynamics are not fundamental: they are a compressive projection of emergent structural variation, consistent with Axioms Φ_1 – Φ_{10} and with the ontological chain of the PSE.

1. Emergent Dynamics as Structural Transformation

Physical dynamics are defined by the map:

$$\Phi_3: (X, D_S^{(e)}, C_e)_{\text{reg}} \rightarrow (\hat{H}, \hat{U}, (T)\rho),$$

where:

- $\hat{H}$ is an effective operator (emergent Hamiltonian),
- $\hat{U}(T)$ is a physical evolution in physical time T ,
- ρ is an effective physical state.

Physical time T is not ontological; it emerges as a compressive representation of structural constraints. Dynamics do not exist in the fundamental domain.

2. Emergent Structural Variation

Structural variation is encoded by the pair $(D_S^{(e)}, C_e)$. In the minimal example:

- within each emergent region, the structure is homogeneous;
- between the two regions, the structure exhibits a coherent discontinuity.

Structural variation is therefore described by:

$$\begin{aligned}\Delta D_S^{(e)}(x, y) &= D_S^{(e)}(x, y) - D_S^{(e)}(x, y) \mid_{\text{region}}, \\ \Delta C_e(x, y) &= C_e(x, y) - C_e(x, y) \mid_{\text{region}}.\end{aligned}$$

These variations are pregeometric, non-physical, and constitute the input of the dynamical projection Φ_3 .

3. Dynamical Projection Φ_3

We define the dynamical projection as:

$$\Phi_3(E_{\text{reg}}) = \hat{H},$$

where the emergent Hamiltonian is a compressive function of the emergent distance and correlation:

$$\hat{H}(x, y) = \begin{cases} h_\alpha & D_S^{(e)}(x, y) = a, \\ h_\beta & D_S^{(e)}(x, y) = b, \end{cases} \quad h_\alpha < h_\beta.$$

Interpretation

- pairs within emergent regions have dynamical contribution h_α ;
- pairs between regions have contribution h_β ;

- physical dynamics are a non-invertible compression of the emergent structure.

4. Effective Physical Evolution

Physical evolution is defined by:

$$\hat{U}(T) = e^{-i\hat{H}T},$$

where T is physical time.

In the minimal example:

- within each emergent region, evolution is homogeneous;
- between regions, evolution is discontinuous but coherent;
- dynamics are fully determined by the emergent structure.

5. Emergent Physical State

The effective physical state is defined as:

$$\rho = \Phi_3(C_e),$$

with:

$$\rho(x, y) = \begin{cases} \rho_\alpha & C_e(x, y) = 1, \\ \rho_\beta & C_e(x, y) = 0, \rho_\alpha > \rho_\beta. \end{cases}$$

The physical state is therefore a compression of the emergent correlation.

6. Properties of the Dynamical Domain

The emergent dynamical domain satisfies:

- **non-invertibility:** from $(\hat{H}, \hat{U}, (T)\rho)$ it is not possible to recover E_{reg} ;
- **compressivity:** physical dynamics contain less information than the emergent structure;
- **monotonicity:** monotonic structural variations generate monotonic dynamical variations;
- **non-physicality:** dynamics do not exist in the fundamental domain;
- **joint dependence:** dynamics depend simultaneously on $D_S^{(e)}$ and C_e .

7. Synthesis of the Dynamical Domain

In the minimal example:

- physical dynamics are an effective representation of emergent constraints;
- the emergent Hamiltonian is a compression of the emergent distance;
- the physical state is a compression of the emergent correlation;
- physical evolution is determined by the emergent structure;
- the projection Φ_3 realizes the ontological transition from structure to dynamics.

The dynamical domain completes the second physical phase of the PSE ontological chain and prepares the construction of the statistical, field, and discrete domains.

8.4 Statistical Domain

Projection class Φ_4 , activated by threshold Θ_4 , projects the regularized emergent domain

$$E_{\text{reg}} = (X, D_S^{(e)}, C_e)_{\text{reg}}$$

into effective physical statistical quantities:

$$\Phi_4: E_{\text{reg}} \rightarrow \rho(p), P(p, q), \sigma(p).$$

Definition Φ_5 — Statistical Density ρ

$$\rho(p) = \Phi_4(D_S^{(e)}, C_e).$$

Properties of the Density

- **Statistical monotonicity:** variations in $D_S^{(e)}$ and C_e produce monotonic variations in ρ .
- **Regularity:** ρ is continuous and bounded.
- **Statistical saturation:**

$$\rho(p) \leq \rho_{\text{max}}.$$

Definition Φ_6 — Statistical Correlations P

$$P(p, q) = \Phi_4(D_S^{(e)}, (x, y)C_e(x, y)).$$

Properties of Statistical Correlations

- **Correlative monotonicity:** larger emergent distance $\rightarrow$ smaller statistical correlation.
- **Regularity:** P is continuous and bounded.
- **Statistical saturation:**

$$P(p, q) \leq P_{\text{max}}.$$

Emergent Statistical Physical Domain

The statistical domain describes the effective physical representation of emergent fluctuations of the regularized domain E_{reg} . Physical statistics are not fundamental: they are a compressive projection of structural and dynamical variations, consistent with Axioms Φ_1 – Φ_{10} .

1. Emergent Structural Fluctuations

In the regularized emergent domain, the structure is not perfectly rigid: the quantities $D_S^{(e)}$ and C_e may exhibit local fluctuations due to fundamental reorganization.

We define structural fluctuations as:

$$\delta D_S^{(e)}(x, y) = D_S^{(e)}(x, y) - \langle D_S^{(e)}(x, y) \rangle,$$

$$\delta C_e(x, y) = C_e(x, y) - \langle C_e(x, y) \rangle,$$

where $\langle \cdot \rangle$ denotes an emergent mean value.

In the minimal example:

- within emergent regions, fluctuations are zero;
- in the transition zone, fluctuations are maximal.

2. Statistical Projection Φ_4

We define the statistical projection as:

$$\Phi_4: E_{\text{reg}} \rightarrow (\sigma, P),$$

where:

- σ is an emergent statistical deviation,
- P is an effective probability density.

The projection is defined as:

$$\sigma(x, y) = \Phi_4 \left(D_S^{(e)} \right) (x, y),$$

$$P(x, y) = \Phi_4(C_e)(x, y).$$

3. Emergent Statistical Deviation

We set:

$$\sigma(x, y) = \begin{cases} \sigma_\alpha & D_S^{(e)}(x, y) = a, \\ \sigma_\beta & D_S^{(e)}(x, y) = b, \end{cases} \sigma_\alpha < \sigma_\beta.$$

Interpretation

- within emergent regions, the structure is stable $\rightarrow$ small deviation σ_α ;
- between regions, the structure is unstable $\rightarrow$ large deviation σ_β .

Physical statistics are therefore a compression of emergent structural stability.

4. Emergent Probability Density

We define the effective probability density as:

$$P(x, y) = \begin{cases} p_\alpha & C_e(x, y) = 1, \\ p_\beta & C_e(x, y) = 0, \end{cases} p_\alpha > p_\beta.$$

Interpretation

- coherent pairs $\rightarrow$ high probability;
- incoherent pairs $\rightarrow$ low probability.

Physical statistics are a compression of emergent correlation.

5. Emergent Statistical Distribution

The emergent statistical distribution is defined as:

$$P_{\text{phys}}(x, y) = \frac{1}{Z} \exp \left(-\frac{g(x, y)}{\sigma(x, y)} \right),$$

where:

- $g(x, y)$ is the physical metric emerging from the geometric domain Φ_2 ,
- $\sigma(x, y)$ is the emergent statistical deviation,
- Z is a normalization constant.

In the minimal example:

- within emergent regions: $g = \alpha, \sigma = \sigma_\alpha$;
- between regions: $g = \beta, \sigma = \sigma_\beta$.

Thus:

$$P_{\text{phys}}(x, y) = \begin{cases} \frac{1}{Z} \exp\left(-\frac{\alpha}{\sigma_\alpha}\right) & x, y \in [0, 1/2] \text{ or } x, y \in [1, 2/3], \\ \frac{1}{Z} \exp\left(-\frac{\beta}{\sigma_\beta}\right) & x \in [0, 1/2], y \in [1, 2/3] \text{ or vice versa.} \end{cases}$$

6. Properties of the Statistical Domain

The emergent statistical domain satisfies:

- **non-invertibility:** from (σ, P) it is not possible to recover E_{reg} ;
- **compressivity:** physical statistics contain less information than the emergent structure;
- **monotonicity:** monotonic structural variations generate monotonic statistical variations;
- **joint dependence:** statistics depend simultaneously on $D_S^{(e)}, C_e, g$;
- **non-fundamentality:** statistics do not exist in the fundamental domain.

7. Synthesis of the Statistical Domain

In the minimal example:

- physical statistics are an effective representation of emergent stability;
- statistical deviation is a compression of emergent distance;
- probability density is a compression of emergent correlation;
- the physical distribution is determined by the emergent metric;
- projection Φ_4 realizes the ontological transition from structure to statistics.

The statistical domain completes the third phase of the PSE ontological chain and prepares the construction of the field and discrete domains.

8.5 Field Domain

Projection class Φ_5 , activated by threshold Θ_5 , projects the regularized emergent domain

$$E_{\text{reg}} = (X, D_S^{(e)}, C_e)_{\text{reg}}$$

into emergent physical field potentials and field intensities:

$$\Phi_5: E_{\text{reg}} \rightarrow A_\mu(p), F_{\mu\nu}(p).$$

Definition $\Phi 7$ — Field Potential A_μ

$$A_\mu(p) = \Phi_5 \left(D_S^{(e)}, C_e \right).$$

Properties of the Field Potential

- **Field monotonicity:** structural variations produce monotonic variations of the potential.
- **Regularity:** A_μ is continuous and differentiable.
- **Potential saturation:**

$$| A_\mu(p) | \leq A_{\max}.$$

Definition $\Phi 8$ — Field Intensity $F_{\mu\nu}$

$$F_{\mu\nu}(p) = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

Properties of Field Intensity

- **Regularity:** $F_{\mu\nu}$ is continuous and bounded.
- **Field saturation:**

$$| F_{\mu\nu}(p) | \leq F_{\max}.$$

Emergent Physical Field Domain

The field domain describes the effective physical representation of local variations of the regularized emergent domain E_{reg} . Physical fields are not fundamental: they are compressed images of the pregeometric structure, obtained through projection Φ_5 and consistent with Axioms $\Phi 1$ – $\Phi 10$.

1. Local Variations of the Emergent Structure

The emergent structure exhibits local variations of distance and correlation:

$$\partial_x D_S^{(e)}(x, y), \partial_x C_e(x, y).$$

In the minimal example:

- within emergent regions, local variations are zero;
- in the transition zone, variations are maximal.

These variations are pregeometric, non-physical, and constitute the input of the field projection Φ_5 .

2. Field Projection Φ_5

We define the field projection as:

$$\Phi_5: E_{\text{reg}} \rightarrow F_\mu,$$

where F_μ is an emergent physical field.

The projection is defined as a local transformation of the emergent distance:

$$F_\mu(x) = \Phi_5 \left(D_S^{(e)} \right) (x) = \partial_x D_S^{(e)}(x, x).$$

Interpretation

- the physical field is a compression of local variations of the emergent structure;

- the field does not exist in the fundamental domain;
- the field is non-invertible: it does not allow reconstruction of $D_S^{(e)}$.

3. Emergent Physical Field in the Minimal Case

In the minimal example:

$$D_S^{(e)}(x, y) = \begin{cases} a & x, y \in [0, 1/2] \text{ or } x, y \in [1, 21], \\ b & x \in [0, 1/2], y \in [1, 21] \text{ or vice versa,} \end{cases}$$

The local derivative is:

$$\partial_x D_S^{(e)}(x, x) = \begin{cases} 0 & x \in [0, 1/2] \text{ or } x \in [1, 21], \\ \kappa & x = 1/2, \end{cases}$$

where κ represents the structural variation concentrated at the transition between the two regions.

Thus the emergent physical field is:

$$F_\mu(x) = \begin{cases} 0 & \text{internal region,} \\ \kappa & \text{transition zone.} \end{cases}$$

Interpretation

- the field is zero in homogeneous regions;
- the field is maximal in the transition zone;
- the field is a non-invertible compression of structural variation.

4. Field as Image of Emergent Correlation

Analogously, we define:

$$G_\mu(x) = \Phi_5(C_e)(x) = \partial_x C_e(x, x).$$

In the minimal case:

$$G_\mu(x) = \begin{cases} 0 & \text{internal region,} \\ \gamma & \text{transition zone,} \end{cases}$$

with γ an emergent constant.

Properties of G_μ

- it is a compression of correlation variation;
- it is non-invertible;
- it is non-fundamental.

5. Physical Interpretation of the Field Domain

The emergent field domain shows that:

- physical fields are compressed images of local variations of the emergent structure;
- the fundamental domain contains no fields;
- fields emerge only after projection Φ_5 ;

- fields are local, non-invertible, and non-fundamental;
- the transition zone between emergent regions generates effective physical fields.

6. Synthesis of the Field Domain

In the minimal example:

- the physical field is a compression of local variations of $D_S^{(e)}$ and C_e ;
- the field is zero in homogeneous regions;
- the field is maximal in the transition zone;
- the field is an emergent, non-fundamental physical quantity;
- projection Φ_5 realizes the ontological transition from structure to field.

The field domain completes the fourth phase of the PSE ontological chain and prepares the discrete domain.

8.6 Quantum Domain

Projection class Φ_6 , activated by threshold Θ_6 , projects the regularized emergent domain

$$E_{\text{reg}} = \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}$$

into the emergent physical quantum domain:

$$\Phi_6: E_{\text{reg}} \rightarrow Q = (\mathcal{H}, A, \hbar).$$

The quantum domain describes the effective physical representation of emergent correlational constraints. Quantum structure is **not fundamental**: it is a compressed image of the emergent structure, consistent with Axioms $\Phi 1$ – $\Phi 10$.

Definition $\Phi 11$ — Emergent State Space

$$\mathcal{H} = \Phi_6(X).$$

The state space is the continuous representation of coherent configurations generated by the emergent correlation C_e .

Properties of the State Space

- **continuity**: inherited from the continuity of $D_S^{(e)}$;
- **saturation**: the state space possesses a minimal granularity;
- **structural nonlocality**: reflects the nonlocality of C_e ;
- **non-invertibility**: from $\mathcal{H}$ it is not possible to recover $(D_S^{(e)}, C_e)$.

Definition $\Phi 12$ — Emergent Action

$$A = \Phi_6(D_S^{(e)}, C_e).$$

The action is the continuous representation of emergent correlational constraints.

Properties of the Emergent Action

- **monotonicity:** monotonic structural variations generate monotonic variations of the action;
- **regularity:** the action is finite and free of divergences;
- **saturation:** a minimal action threshold exists.

Definition Φ_{13} — Emergent Quantum Threshold

$$\hbar = \Phi_6(C_f).$$

The quantum threshold is the compression of the fundamental binary granularity.

Properties of the Quantum Threshold

- **minimal granularity:** $\hbar$ represents the minimal correlation scale;
- **compressed nonlocality:** reflects the nonlocality of C_e ;
- **non-fundamentality:** $\hbar$ does not exist in the ontological domain.

Quantum Rigidity and Continuous Projection

The emergent correlation C_e contains highly coherent regions. The quantum rigidity associated with such regions defines a continuous parameter via:

$$q_{\text{cont}} = \Phi_{6,\text{rig-quant}}(C_e).$$

This parameter is consistent with Axioms Φ_1 – Φ_{10} : it is monotonic, compressive, and non-invertible.

Emergent Physical Quantum Domain

The quantum domain describes:

- the state space $\mathcal{H}$;
- the continuous action A ;
- the minimal threshold $\hbar$;
- physical nonlocality as a compressed image of emergent nonlocality;
- quantization as an effect of the binary granularity of C_f .

1. Emergent Quantum Configurations

An emergent quantum configuration is a coherent region of the emergent structure:

$$Q = \{X_\alpha, X_\beta, \dots\}.$$

Emergent correlation satisfies:

$$C_e(x, y) \approx 1 \text{ if } x, y \in X_\alpha.$$

These coherent regions are the structural analogue of quantum states.

2. Quantum Projection Φ_6

$$\Phi_6: \left(X, D_S^{(e)}, C_e \right)_{\text{reg}} \rightarrow (\mathcal{H}, A, \hbar).$$

The quantum projection assigns:

- a state $\psi \in \mathcal{H}$ to each coherent region;

- an action $A(\psi)$ to the configuration;
- a minimal threshold $\hbar$ to correlational granularity.

3. Emergent Quantum Quantities

$$Q = (\mathcal{H}, A, \hbar)$$

These represent:

- quantum states,
- amplitudes,
- minimal thresholds,
- nonlocal correlations,
- operational structures of quantum mechanics.

4. Physical Quantum Configurations

A physical quantum configuration is:

$$Q_{\text{phys}} = \Phi_6(Q).$$

It is a compressed image of the emergent structure.

5. Properties of the Quantum Domain

- **non-invertibility**
- **compressivity**
- **monotonicity**
- **joint dependence on $D_S^{(e)}$ and C_e**
- **non-fundamentality**

6. Synthesis of the Quantum Domain

- quantum structure is a compressed image of the emergent structure;
- the state space, action, and minimal threshold emerge from coherent regions;
- projection Φ_6 is monotonic, compressive, and non-invertible;
- the quantum domain completes the ontological chain of the PSE.

8.7 Common Properties of All Physical Domains

Every emergent physical quantity generated by a projection class Φ_k , activated by threshold Θ_k , satisfies:

(1) Saturation (derived from compressivity)

$$Q(x, y) \leq Q_{\text{max}}.$$

Saturation arises from the compressivity of the operators F_λ, G_λ , of the regularity operator R , and of the projection Φ_k .

(2) Rigidity

$$|\delta Q| \leq K |\delta D_S^{(e)}|.$$

Rigidity is a structural property of the regularized emergent domain E_{reg} .

(3) Non-invertibility

$$\nexists \Phi_{k,Q}^{-1}.$$

No physical quantity allows reconstruction of the emergent domain.

(4) Non-retroactivity

$$\frac{\partial (D_S^{(e)}, C_e)}{\partial Q} = 0.$$

The physical domain cannot influence the emergent structure.

(5) Total dependence on the emergent structure

$$Q = \Phi_{k,Q}(E_{\text{reg}}) = \Phi_{k,Q}(X, D_S^{(e)}, C_e)_{\text{reg}}.$$

Every physical quantity depends exclusively on the regularized emergent domain and on the threshold Θ_k .

8.8 General Projection Φ

The general projection Φ is the family of ontological maps

$$\Phi = \{\Phi_1, \Phi_2, \Phi_3, \Phi_4, \Phi_5, \Phi_6\},$$

which transforms the regularized emergent structure

$$E_{\text{reg}} = R(X, D_S^{(e)}, C_e)$$

into observable physical domains. Φ is the **only point** of the ontological chain where physical quantities appear.

1. Domain of Definition of the Projection

The projection acts on the regularized emergent domain:

$$E_{\text{reg}} = (X, D_S^{(e)}, C_e)_{\text{reg}},$$

where:

- X is the pregeometric emergent domain;
- $D_S^{(e)}: X \times X \rightarrow \mathbb{R}^+$ is the emergent distance;
- $C_e: X \times X \rightarrow \mathbb{R}^+$ is the emergent correlation;
- R is the structural regularity operator.

The emergent domain is non-physical, non-metric, non-dynamical, and non-invertible.

2. Codomain of the Projection

The projection generates the physical domains:

$$D_{\text{physic}}(k) = \Phi_k(E_{\text{reg}}),$$

where the set of emergent physical quantities is:

$$\mathcal{O}_{\text{physic}} = \{g_{\mu\nu}, d, \hat{H}, \rho, \sigma, P, F_\mu, \mathcal{N}, \dots\}.$$

Each class Φ_k generates a distinct physical domain:

- discrete,
- geometric,
- dynamical,
- statistical,
- field,
- quantum.

3. General Definition of the Projection

The general projection is defined as:

$$\Phi_k: E_{\text{reg}} \rightarrow D_{\text{physic}}(k),$$

$$\Phi_k(D_S^{(e)}, C_e) = \mathcal{O}_{\text{physic}}(k).$$

Every emergent physical quantity is a compressive function of the pair $(D_S^{(e)}, C_e)$:

$$\mathcal{O}_{\text{physic}} = \Phi_{k,\mathcal{O}}(D_S^{(e)}, C_e).$$

4. General Properties of the Projection Φ

($\Phi 1$) Non-locality

$$\Phi_k(D_S^{(e)}, C_e)(x) = F(D_S^{(e)}, (U_x)C_e(U_x)),$$

where $U_x \subset X$ is an emergent region.

($\Phi 2$) Structural monotonicity

$$D_S^{(e)}(x, y) < D_S^{(e)}(x, z) \Rightarrow \Phi_k(x, y) < \Phi_k(x, z).$$

($\Phi 3$) Non-invertibility

$$\exists \Phi_k^{-1}: D_{\text{physic}}(k) \rightarrow E_{\text{reg}}.$$

($\Phi 4$) Non-physicality

Φ_k does not represent dynamics or interactions: it is an ontological transformation, not a physical one.

($\Phi 5$) Joint dependence

$$\mathcal{O}_{\text{physic}} = \Phi_k(D_S^{(e)}, C_e).$$

(Φ6) Compressivity

$$I(D_S^{(e)}, C_e) > I(\mathcal{O}_{\text{physic}}).$$

(Φ7) Saturation

$$\Phi_k(D_S^{(e)}) = \text{constant in coherent regions.}$$

(Φ8) Non-arbitrariness

The form of Φ_k is completely determined by the regularized emergent structure.

(Φ9) Non-modifiability

Modifying Φ_k would mean modifying the ontological chain itself.

(Φ10) Structural compatibility

$$C_e(x, y) \approx C_e(x, z) \Rightarrow \Phi_k(x, y) \approx \Phi_k(x, z).$$

5. General Form of Emergent Physical Quantities**5.1 Physical Metric**

$$g_{\mu\nu}(x, y) = \Phi_{2,g}(D_S^{(e)}, C_e)(x, y).$$

5.2 Physical Distance

$$d(x, y) = \Phi_{2,d}(D_S^{(e)})(x, y).$$

5.3 Physical Dynamics

$$\hat{H} = \Phi_3(D_S^{(e)}, C_e).$$

5.4 Physical State

$$\rho = \Phi_{3,\rho}(C_e).$$

5.5 Physical Statistics

$$P(x, y) = \Phi_{4,\text{stat}}(D_S^{(e)}, C_e)(x, y).$$

5.6 Physical Field

$$F_\mu(x) = \Phi_{5,\text{field}}(\partial_x, D_S^{(e)} \partial_x C_e).$$

5.7 Discrete Quantities

$$\mathcal{N}(x) = \Phi_{1,\text{disc}}(D_S^{(e)}, C_e)(x).$$

6. Synthesis of the General Projection

The projection Φ :

- is the only point of the ontological chain that produces physical quantities;
- is non-local, non-invertible, compressive, and non-arbitrary;

- depends simultaneously on emergent distance and emergent correlation;
- generates all physical domains: discrete, geometric, dynamical, statistical, field, quantum;
- realizes the ontological transition from emergence to physicality.

The complete chain is:

$$S \rightarrow (D_S^{(f)}, C_f) \rightarrow (D_S^{(e)}, C_e) \rightarrow R \rightarrow \Theta_k \rightarrow \Phi_k \rightarrow D_{\text{physic}}(k).$$

9. Comparison with Bell and Zanghì

This section places the PSE within the conceptual framework of quantum nonlocality, comparing it with two fundamental references:

1. **Bell's theorem**, which establishes precise limits for local hidden-variable theories;
2. **Nino Zanghì's ontological approach**, which interprets quantum mechanics as a theory of nonlocal correlations.

The comparison is **structural**, not interpretative: the PSE does not modify or extend the results of Bell or Zanghì, but aligns with them, providing a pregeometric formalization consistent with their conclusions.

9.1 Structural Compatibility with Bell's Theorem

Bell's theorem shows that:

- no local hidden-variable theory can reproduce quantum correlations;
- quantum correlations require a form of structural nonlocality;
- nonlocality cannot be eliminated through local geometric or dynamical mechanisms.

Contribution of the PSE

The PSE is compatible with Bell because it introduces a **primary nonlocal ontological structure**:

$$S \rightarrow D_S^{(f)} \rightarrow C_f,$$

where:

- the fundamental correlation C_f is binary and not metrically constrained;
- the threshold Δ selects coherent correlations without any notion of physical distance;
- nonlocality is **structural**, not dynamical.

The emergent correlation C_e , defined by

$$C_e(x, y) = F(D_S^{(e)}(x, y)), F' < 0,$$

does not live in spacetime and is not limited by any physical metric. It constitutes a form of **structural nonlocality** compatible with quantum nonlocal correlations.

9.2 Comparison with Nino Zanghì's Ontological Approach

Zanghì's ontological approach interprets quantum mechanics as a theory of nonlocal correlations, where observable physical structure is an effective representation of more primitive relations.

Contribution of the PSE

The PSE is compatible with this view because:

- the emergent correlation C_e is a pregeometric nonlocal relation;
- the emergent distance $D_S^{(e)}$ is not a physical distance;
- the projection map Φ produces physical correlations as compressed images of the emergent structural correlations, without introducing new physical content.

Formally:

$$\Phi(D_S^{(e)}, C_e) = \mathcal{O}_{phys},$$

where $\mathcal{O}_{phys}$ includes nonlocal physical correlations as a particular case.

9.3 Role of the PSE in the Landscape of Nonlocality

The PSE provides a **pregeometric ontological basis** for quantum nonlocality:

1. Fundamental nonlocality

$$C_f(i, j) = 1 \Leftrightarrow D_S^{(f)}(i, j) \leq \Delta,$$

with no notion of physical distance.

2. Emergent nonlocality

$$C_e(x, y) = F(D_S^{(e)}(x, y)),$$

where F is monotonic decreasing.

3. Physical nonlocality

$$\Phi_{stat}(D_S^{(e)}, C_e)$$

generates nonlocal physical correlations as compressed images of the emergent structure.

Structural consequence

Physical nonlocality is the **compressed image** of structural nonlocality encoded in C_e . No superluminal dynamics is required: nonlocality is **ontological**, not physical.

2. Fundamental Theorems of the PSE

This section collects the mathematical properties that follow from the axioms of the fundamental, emergent, and physical domains. The theorems are formulated in a neutral way, without assuming any specific physical structure (metric, causality, dynamics). They describe the inevitable consequences of the ontological chain:

$$S \xrightarrow{\mathfrak{D}} D_S^{(f)} \xrightarrow{\Delta} C_f \xrightarrow{F_\lambda} D_S^{(e)} \xrightarrow{G_\lambda} C_e \xrightarrow{R} (X, D_S^{(e)}, C_e)_{\text{reg}} \xrightarrow{\Theta_k} (X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)} \xrightarrow{\Phi} \mathcal{D}_{\text{physical}}$$

Theorem 1 — Existence of the Emergent Structural Distance

Let S be a non-empty set (fundamental domain) and let

$$D_S^{(f)}: S \times S \rightarrow \mathbb{R}^+$$

be a discrete and bounded structural distance.

Let X be a non-empty set (emergent domain) and let

$$\pi: S \rightarrow X$$

be an aggregation map.

Denote by $D(S)$ the space of distances on S and by $D(X)$ the space of distances on X . For every parameter $\lambda \geq 0$, let a regularization operator be defined:

$$F_\lambda: D(S) \rightarrow D(X),$$

and set

$$D_\lambda(x, y) := F_\lambda(D_S^{(f)})(x, y), \quad x, y \in X.$$

Assume:

1. Monotonicity in λ

For all $\lambda_1, \lambda_2 \geq 0$ with $\lambda_1 < \lambda_2$,

$$D_{\lambda_1}(x, y) \leq D_{\lambda_2}(x, y), \quad \forall (x, y) \in X \times X.$$

2. Uniform boundedness

There exists $M > 0$ such that

$$D_\lambda(x, y) \leq M, \quad \forall \lambda \geq 0, \quad \forall (x, y) \in X \times X.$$

3. Structural stability (Cauchy in λ in the sup norm)

Consider the space

$$\mathcal{C}(X \times X) := \{f: X \times X \rightarrow \mathbb{R} \text{ continuous}\},$$

equipped with the norm

$$\|f\|_\infty := \sup_{(x, y) \in X \times X} |f(x, y)|.$$

Assume the family

$$\{D_\lambda\}_{\lambda \geq 0} \subset C(X \times X)$$

is Cauchy in $\|\cdot\|_\infty$, i.e.:

$$\forall \varepsilon > 0 \exists \Lambda \geq 0: \lambda, \mu \geq \Lambda \Rightarrow \|D_\lambda - D_\mu\|_\infty < \varepsilon.$$

Equivalently:

$$\forall \varepsilon > 0 \exists \Lambda \geq 0: \lambda, \mu \geq \Lambda \Rightarrow |D_\lambda(x, y) - D_\mu(x, y)| < \varepsilon, \forall (x, y) \in X \times X.$$

4. Equicontinuity

The family

$$\{D_\lambda\}_{\lambda \geq 0} \subset C(X \times X)$$

is equicontinuous, i.e.:

$$\forall \varepsilon > 0 \exists \delta > 0: d_{X \times X}((x, y), (x', y')) < \delta \Rightarrow |D_\lambda(x, y) - D_\lambda(x', y')| < \varepsilon, \forall \lambda \geq 0.$$

5. Compactness of the emergent domain

The metric space $X \times X$, with the product metric $d_{X \times X}$, is compact.

Statement

Define, for every $(x, y) \in X \times X$,

$$D_S^{(e)}(x, y) := \lim_{\lambda \rightarrow \infty} D_\lambda(x, y),$$

where the limit is taken in the real sense.

Then:

1. The limit exists for every $(x, y) \in X \times X$, and

$$D_S^{(e)}: X \times X \rightarrow \mathbb{R}^+$$

is well-defined and bounded.

2. $D_S^{(e)}$ is continuous on $X \times X$.
3. $D_S^{(e)}$ inherits the structural properties of a distance (positivity, symmetry, separation, and possibly the triangle inequality if assumed for each D_λ).
4. For every $(x, y) \in X \times X$,

$$D_S^{(e)}(x, y) = \sup_{\lambda \geq 0} D_\lambda(x, y),$$

i.e., the emergent distance is **saturating**.

Proof

1. Existence of the pointwise limit and boundedness

Fix $(x, y) \in X \times X$. By structural stability in the sup norm, the family $\{D_\lambda\}_{\lambda \geq 0}$ is Cauchy in $C(X \times X)$. Thus, for every $\varepsilon > 0$ there exists Λ such that

$$\lambda, \mu \geq \Lambda \Rightarrow |D_\lambda(x, y) - D_\mu(x, y)| < \varepsilon.$$

Hence the real sequence $\{D_\lambda(x, y)\}_{\lambda \geq 0} \subset \mathbb{R}$ is Cauchy. Since $\mathbb{R}$ is complete, the limit exists:

$$\lim_{\lambda \rightarrow \infty} D_\lambda(x, y) =: D_S^{(e)}(x, y) \in \mathbb{R}.$$

Uniform boundedness implies:

$$|D_\lambda(x, y)| \leq M \Rightarrow |D_S^{(e)}(x, y)| \leq M.$$

Since each $D_\lambda(x, y) \geq 0$, the limit is non-negative:

$$D_S^{(e)}(x, y) \geq 0.$$

Thus,

$$D_S^{(e)}: X \times X \rightarrow \mathbb{R}^+$$

is well-defined and bounded.

2. Uniform convergence and continuity of the limit

Since $X \times X$ is compact, $C(X \times X)$ is complete under $\|\cdot\|_\infty$. Because $\{D_\lambda\}$ is Cauchy in $\|\cdot\|_\infty$, there exists $D_S^{(e)} \in C(X \times X)$ such that

$$\|D_\lambda - D_S^{(e)}\|_\infty \rightarrow 0 \text{ as } \lambda \rightarrow \infty.$$

Thus convergence is uniform:

$$\forall \varepsilon > 0 \exists \Lambda: \lambda \geq \Lambda \Rightarrow |D_\lambda(x, y) - D_S^{(e)}(x, y)| < \varepsilon, \forall (x, y).$$

Uniform convergence plus continuity of each D_λ implies continuity of $D_S^{(e)}$. Hence:

$$D_S^{(e)} \in C(X \times X).$$

3. Inheritance of structural distance properties

Assume each D_λ satisfies:

- **positivity:** $D_\lambda(x, y) \geq 0$;
- **symmetry:** $D_\lambda(x, y) = D_\lambda(y, x)$;
- **separation:** $D_\lambda(x, x) = 0$;
- **triangle inequality** (optional).

Passing to the limit:

- positivity:

$$D_S^{(e)}(x, y) = \lim_{\lambda \rightarrow \infty} D_\lambda(x, y) \geq 0;$$

- symmetry:

$$D_S^{(e)}(x, y) = D_S^{(e)}(y, x);$$

- separation:

$$D_S^{(e)}(x, x) = 0;$$

- triangle inequality (if assumed):

$$D_S^{(e)}(x, z) \leq D_S^{(e)}(x, y) + D_S^{(e)}(y, z).$$

Thus $D_S^{(e)}$ is a regular emergent distance on X .

4. Saturation: representation as a supremum

Fix $(x, y) \in X \times X$. By monotonicity in λ , the sequence $\{D_\lambda(x, y)\}$ is monotone increasing:

$$\lambda_1 < \lambda_2 \Rightarrow D_{\lambda_1}(x, y) \leq D_{\lambda_2}(x, y).$$

By uniform boundedness, it is bounded above by M . Thus it is monotone and bounded, so:

$$\lim_{\lambda \rightarrow \infty} D_\lambda(x, y) = \sup_{\lambda \geq 0} D_\lambda(x, y).$$

Hence:

$$D_S^{(e)}(x, y) = \sup_{\lambda \geq 0} D_\lambda(x, y).$$

The emergent distance is therefore **saturating**.

Conclusion

The function

$$D_S^{(e)}: X \times X \rightarrow \mathbb{R}^+, \quad D_S^{(e)}(x, y) = \lim_{\lambda \rightarrow \infty} D_\lambda(x, y),$$

is:

- well-defined and bounded;
- continuous on $X \times X$;
- structurally a distance (positive, symmetric, separating, and possibly satisfying the triangle inequality);
- **saturating**, i.e.

$$D_S^{(e)}(x, y) = \sup_{\lambda \geq 0} D_\lambda(x, y).$$

The theorem is proven.

Theorem 2 — Existence of the Emergent Correlation

Let S be a non-empty set (fundamental domain) and let

$$D_S^{(f)}: S \times S \rightarrow \mathbb{R}^+$$

be a discrete and bounded structural distance.

Define the binary fundamental correlation

$$C_f: S \times S \rightarrow \{0, 1\}$$

through a threshold $\Delta > 0$ applied to the fundamental distance:

$$C_f(i, j) = 1 \Leftrightarrow D_S^{(f)}(i, j) \leq \Delta, C_f(i, j) = 0 \Leftrightarrow D_S^{(f)}(i, j) > \Delta.$$

Let X be the emergent domain and let $\pi: S \rightarrow X$ be an aggregation map. Denote by $C(S)$ the space of correlations on S and by $C(X)$ the space of correlations on X .

For every parameter $\lambda \geq 0$, let a regularization operator be defined:

$$G_\lambda: C(S) \rightarrow C(X),$$

and set

$$C_\lambda(x, y) := G_\lambda(C_f)(x, y), \quad x, y \in X.$$

Assume:

1. Monotonicity in λ

For all $\lambda_1, \lambda_2 \geq 0$ with $\lambda_1 < \lambda_2$,

$$C_{\lambda_1}(x, y) \leq C_{\lambda_2}(x, y), \quad \forall (x, y) \in X \times X.$$

2. Uniform boundedness

For all $\lambda \geq 0$ and all $(x, y) \in X \times X$,

$$0 \leq C_\lambda(x, y) \leq 1.$$

3. Structural stability (Cauchy in λ in the sup norm)

Consider the space

$$C(X \times X) := \{f: X \times X \rightarrow \mathbb{R} \text{ continuous}\},$$

with norm

$$\|f\|_\infty := \sup_{(x, y) \in X \times X} |f(x, y)|.$$

Assume the family

$$\{C_\lambda\}_{\lambda \geq 0} \subset C(X \times X)$$

is Cauchy in $\|\cdot\|_\infty$, i.e.:

$$\forall \varepsilon > 0 \exists \Lambda \geq 0: \lambda, \mu \geq \Lambda \Rightarrow \|C_\lambda - C_\mu\|_\infty < \varepsilon.$$

Equivalently:

$$\forall \varepsilon > 0 \exists \Lambda \geq 0: \lambda, \mu \geq \Lambda \Rightarrow |C_\lambda(x, y) - C_\mu(x, y)| < \varepsilon, \forall (x, y).$$

4. Equicontinuity

The family

$$\{C_\lambda\}_{\lambda \geq 0}$$

is equicontinuous, i.e.:

$$\forall \varepsilon > 0 \exists \delta > 0: d_{X \times X}((x, y), (x', y')) < \delta \Rightarrow |C_\lambda(x, y) - C_\lambda(x', y')| < \varepsilon, \forall \lambda \geq 0.$$

5. Compactness of the emergent domain

The metric space $X \times X$, with the product metric $d_{X \times X}$, is compact.

Statement

Define, for every $(x, y) \in X \times X$,

$$C_e(x, y) := \lim_{\lambda \rightarrow \infty} C_\lambda(x, y),$$

where the limit is taken in the real sense.

Then:

1. The limit exists for every $(x, y) \in X \times X$, and

$$C_e: X \times X \rightarrow [0, 1]$$

is well-defined and bounded.

2. C_e is continuous on $X \times X$.
3. C_e is regular as an emergent correlation (range in $[0, 1]$, possible symmetry, structural coherence with the emergent metric).
4. For every $(x, y) \in X \times X$,

$$C_e(x, y) = \sup_{\lambda \geq 0} C_\lambda(x, y),$$

i.e., the emergent correlation is **saturating**.

Proof

1. Existence of the pointwise limit and range $[0, 1]$

Fix $(x, y) \in X \times X$. By structural stability in the sup norm, the family $\{C_\lambda\}$ is Cauchy in $\mathcal{C}(X \times X)$. Thus, for every $\varepsilon > 0$ there exists Λ such that

$$\lambda, \mu \geq \Lambda \Rightarrow |C_\lambda(x, y) - C_\mu(x, y)| < \varepsilon.$$

Hence the real sequence $\{C_\lambda(x, y)\}$ is Cauchy. Since $\mathbb{R}$ is complete, the limit exists:

$$\lim_{\lambda \rightarrow \infty} C_\lambda(x, y) =: C_e(x, y) \in \mathbb{R}.$$

Uniform boundedness implies:

$$0 \leq C_\lambda(x, y) \leq 1 \Rightarrow 0 \leq C_e(x, y) \leq 1.$$

Thus:

$$C_e: X \times X \rightarrow [0, 1]$$

is well-defined.

2. Uniform convergence and continuity of the limit

As in Theorem 1, since $X \times X$ is compact, $\mathcal{C}(X \times X)$ is complete under $\|\cdot\|_\infty$. Because $\{C_\lambda\}$ is Cauchy, there exists $C_e \in \mathcal{C}(X \times X)$ such that

$$\|C_\lambda - C_e\|_\infty \rightarrow 0 \text{ as } \lambda \rightarrow \infty.$$

Thus convergence is uniform:

$$\forall \varepsilon > 0 \exists \Lambda: \lambda \geq \Lambda \Rightarrow |C_\lambda(x, y) - C_e(x, y)| < \varepsilon, \forall (x, y).$$

Uniform convergence of continuous functions implies continuity of the limit. Hence:

$$C_e \in \mathcal{C}(X \times X).$$

3. Regularity and coherence with the emergent metric

Assume each C_λ satisfies:

- **range constraint:** $C_\lambda(x, y) \in [0, 1]$;
- **possible symmetry:** $C_\lambda(x, y) = C_\lambda(y, x)$;
- **structural dependence on C_f via G_λ .**

Then:

- the range $[0, 1]$ passes to the limit;
- symmetry passes to the limit:

$$C_e(x, y) = C_e(y, x);$$

- coherence with $D_S^{(e)}$ follows from the structural relation between F_λ and G_λ : C_e is the continuous, regularized image of the binary correlation induced by Δ .

4. Saturation: representation as a supremum

Fix $(x, y) \in X \times X$. By monotonicity in λ , the sequence $\{C_\lambda(x, y)\}$ is monotone increasing:

$$\lambda_1 < \lambda_2 \Rightarrow C_{\lambda_1}(x, y) \leq C_{\lambda_2}(x, y).$$

By boundedness:

$$0 \leq C_\lambda(x, y) \leq 1.$$

Thus the sequence is monotone and bounded, so:

$$\lim_{\lambda \rightarrow \infty} C_\lambda(x, y) = \sup_{\lambda \geq 0} C_\lambda(x, y).$$

Hence:

$$C_e(x, y) = \sup_{\lambda \geq 0} C_\lambda(x, y).$$

The emergent correlation is therefore **saturating**.

Conclusion

The function

$$C_e: X \times X \rightarrow [0, 1], \quad C_e(x, y) = \lim_{\lambda \rightarrow \infty} C_\lambda(x, y),$$

is:

- well-defined and bounded;
- continuous on $X \times X$;
- regular as an emergent correlation (range $[0, 1]$, possible symmetry);
- **saturating**, i.e.

$$C_e(x, y) = \sup_{\lambda \geq 0} C_\lambda(x, y).$$

The theorem is proven.

Theorem 3 — Correlative Monotonicity

Let $(X, D_S^{(e)})$ be an emergent metric space, with

$$D_S^{(e)}: X \times X \rightarrow [0, \infty)$$

continuous. Let

$$C_e: X \times X \rightarrow \mathbb{R}$$

be a continuous emergent correlation.

Assume:

(A1) Structural homogeneity and isotropy

For every pair $(x, y), (x', y') \in X \times X$,

$$D_S^{(e)}(x, y) = D_S^{(e)}(x', y') \implies C_e(x, y) = C_e(x', y').$$

That is, C_e is constant on metric spheres.

(A2) Functional dependence on distance

There exists a function

$$F: [0, \infty) \rightarrow \mathbb{R}$$

such that

$$C_e(x, y) = F(D_S^{(e)}(x, y)), \quad \forall (x, y) \in X \times X.$$

(A3) Decreasing monotonicity

For all $r_1, r_2 \in [0, \infty)$,

$$r_1 < r_2 \implies F(r_1) \geq F(r_2).$$

(A4) Continuity and regularity

The function F is continuous on $[0, \infty)$ and differentiable on $(0, \infty)$.

Statement

Under assumptions (A1)–(A4), the emergent correlation is of the form

$$C_e(x, y) = F(D_S^{(e)}(x, y)),$$

where F is strictly decreasing, and its derivative satisfies

$$F'(r) < 0, \quad \forall r > 0.$$

Proof

1. Existence and well-definition of F

For each $r \geq 0$, consider the set

$$S_r := \{(x, y) \in X \times X : D_S^{(e)}(x, y) = r\}.$$

By (A1), the function C_e is constant on each S_r . Define

$$F(r) := C_e(x, y) \text{ for any } (x, y) \in S_r.$$

This definition is well-posed: homogeneity/isotropy ensures the value does not depend on the choice of (x, y) .

By construction,

$$C_e(x, y) = F(D_S^{(e)}(x, y)),$$

so (A2) is automatically satisfied.

2. Decreasing monotonicity

By (A3), if $r_1 < r_2$ then

$$F(r_1) \geq F(r_2).$$

Thus F is monotone decreasing on $[0, \infty)$.

3. Continuity of F

The function C_e is continuous on $X \times X$. The emergent distance $D_S^{(e)}$ is continuous on $X \times X$.

The composition

$$(x, y) \mapsto D_S^{(e)}(x, y) \mapsto F(D_S^{(e)}(x, y))$$

is continuous. If F were not continuous, the continuity of C_e would be violated.

Thus F is continuous on $[0, \infty)$.

4. Differentiability and sign of the derivative

By (A4), F is differentiable on $(0, \infty)$.

Since F is monotone decreasing, for every $r > 0$,

$$F'(r) \leq 0.$$

In the PSE, emergent correlation **strictly** decreases with distance (not merely non-increasing). This implies strict monotonicity:

$$r_1 < r_2 \implies F(r_1) > F(r_2).$$

For strictly decreasing differentiable functions, the derivative is strictly negative:

$$F'(r) < 0, \quad \forall r > 0.$$

Conclusion

There exists a continuous, differentiable, strictly decreasing function

$$F: [0, \infty) \rightarrow \mathbb{R}$$

such that

$$C_e(x, y) = F(D_S^{(e)}(x, y)),$$

and

$$F'(r) < 0 \text{ for all } r > 0.$$

The theorem is proven.

Theorem 4 — Existence of the Projection Map Φ

Let X be a non-empty set (emergent domain) and let

$$D_S^{(e)}: X \times X \rightarrow [0, \infty)$$

be a continuous, regular, saturating, and monotonic emergent distance (in the PSE sense). Let

$$C_e: X \times X \rightarrow \mathbb{R}$$

be a continuous, regular, coherent, and saturating emergent correlation.

Define the structural emergent domain:

$$E := (X, D_S^{(e)}, C_e).$$

Let $\mathcal{D}_{phys}$ be a space of physical structures (fields, metrics, energy distributions, etc.) endowed with a natural topology (e.g., one induced by a norm or a weak topology).

Space of Emergent Profiles

Let

$$\mathcal{F} := \mathcal{C}(X) \times \mathcal{C}(X)$$

be the product space of continuous functions on X , equipped with the product topology (e.g., that induced by the sup norm on each factor).

For each $x \in X$, define the emergent profile of x as

$$P(x) := (D_S^{(e)}(x, \cdot), C_e(x, \cdot)) \in \mathcal{F}.$$

Structural Assumptions

Assume:

1. **Physical functional dependence** The physical structure associated with an emergent point $x \in X$ depends only on its emergent profile $P(x)$.
2. **Existence of a continuous and compressive physical functional** There exists an operator

$$H: \mathcal{F} \rightarrow \mathcal{D}_{phys}$$

such that:

- H is continuous;
- H is regular (it respects the distance/correlation structure encoded in $P(x)$);
- H is monotonic with respect to a natural order on $\mathcal{F}$ (more “intense” profiles $\Rightarrow$ more “intense” physical structures);

- H is compressive, i.e., non-injective: there exist $P_1 \neq P_2$ in $\mathcal{F}$ such that

$$H(P_1) = H(P_2).$$

Statement

Under these assumptions, there exists a map

$$\Phi: (X, D_S^{(e)}, C_e) \rightarrow \mathcal{D}_{phys}$$

such that:

- Φ is well-defined and continuous;
- Φ is regular with respect to the emergent structures;
- Φ is monotonic in the physical sense;
- Φ is non-invertible (compressive).

Proof

1. Explicit construction of Φ

Consider the map

$$P: X \rightarrow \mathcal{F}, P(x) := (D_S^{(e)}(x, \cdot), C_e(x, \cdot)).$$

Since $D_S^{(e)}$ and C_e are continuous in (x, y) , for each fixed x the functions $y \mapsto D_S^{(e)}(x, y)$ and $y \mapsto C_e(x, y)$ are continuous on X . Thus the map

$$x \mapsto P(x)$$

is continuous from X to $\mathcal{F}$.

Define now

$$\Phi: X \rightarrow \mathcal{D}_{phys}, \Phi(x) := H(P(x)).$$

This is a constructive definition: Φ is the composition of two well-defined maps,

$$\Phi = H \circ P.$$

2. Continuity and regularity of Φ

The map $P: X \rightarrow \mathcal{F}$ is continuous. By assumption, $H: \mathcal{F} \rightarrow \mathcal{D}_{phys}$ is continuous.

Thus the composition

$$\Phi: X \rightarrow \mathcal{D}_{phys}$$

is continuous.

Regularity follows because:

- $P(x)$ encodes the emergent profile of x in a regular way (thanks to the regularity of $D_S^{(e)}$ and C_e);
- H is constructed to transform regular emergent profiles into regular physical structures.

Hence regular variations in $x \mapsto P(x)$ induce regular variations in $x \mapsto \Phi(x)$.

3. Physical monotonicity of Φ

In the PSE, monotonicity of $D_S^{(e)}$ and C_e with respect to distance (Theorem 3) induces a natural order on $\mathcal{F}$. For example, define

$$P_1 \geq P_2$$

if, pointwise or on average, $D_S^{(e)}$ is smaller and C_e is larger for P_1 than for P_2 (a “more intense” profile).

If an emergent point x_1 has a more intense profile than x_2 , then

$$P(x_1) \geq P(x_2).$$

By assumption, H is monotonic:

$$P_1 \geq P_2 \implies H(P_1) \geq H(P_2),$$

where the order on the right is a physical order (e.g., stronger field intensity, higher energy density, etc.).

Applying this to $P(x_1)$ and $P(x_2)$:

$$P(x_1) \geq P(x_2) \implies \Phi(x_1) = H(P(x_1)) \geq H(P(x_2)) = \Phi(x_2).$$

Thus Φ is physically monotonic: emergent points with more intense profiles produce more intense physical structures.

4. Compressivity and non-invertibility of Φ

By assumption, H is compressive (non-injective): there exist $P_1, P_2 \in \mathcal{F}$ with

$$P_1 \neq P_2, \quad H(P_1) = H(P_2).$$

Moreover, the map $P: X \rightarrow \mathcal{F}$ is generally non-injective: different emergent points may have identical or nearly identical profiles.

Thus there exist $x_1, x_2 \in X$ such that

$$P(x_1) \neq P(x_2), \quad H(P(x_1)) = H(P(x_2)).$$

By definition of Φ :

$$\Phi(x_1) = H(P(x_1)) = H(P(x_2)) = \Phi(x_2).$$

Hence Φ is not injective. In particular, no map

$$\Phi^{-1}: \mathcal{D}_{phys} \rightarrow X$$

can reconstruct the emergent point from the physical structure.

This formalizes the ontological compression of the PSE: many emergent configurations (profiles $P(x)$) correspond to the same physical structure $\Phi(x)$.

Conclusion

The map

$$\Phi: X \rightarrow \mathcal{D}_{phys}, \Phi(x) = H(D_S^{(e)}(x, \cdot), C_e(x, \cdot)),$$

is:

- well-defined;
- continuous;
- regular with respect to emergent structures;
- physically monotonic;
- non-injective (compressive).

Thus there exists a projection map

$$\Phi: (X, D_S^{(e)}, C_e) \rightarrow \mathcal{D}_{phys}$$

that constructs physical structures which are regular, continuous, monotonic, and compressive.

Theorem 5 — Non-invertibility of the Projection Φ

Let

$$\Phi: (X, D_S^{(e)}, C_e) \rightarrow \mathcal{D}_{phys}$$

be the projection map constructed in Theorem 4, defined by

$$\Phi(x) := H(P(x)), \quad x \in X,$$

where:

- $P: X \rightarrow \mathcal{F}$ is the emergent profile

$$P(x) := (D_S^{(e)}(x, \cdot), C_e(x, \cdot)),$$

- $H: \mathcal{F} \rightarrow \mathcal{D}_{phys}$ is a continuous, regular, monotonic, and compressive physical functional.

Assume:

1. **Compressivity of H** There exist $P_1, P_2 \in \mathcal{F}$ such that

$$P_1 \neq P_2, \quad H(P_1) = H(P_2).$$

2. **Emergent saturation** The functions $D_S^{(e)}$ and C_e are saturated limits of the families $\{D_\lambda\}$ and $\{C_\lambda\}$. This implies that many distinct emergent configurations produce the same limiting profile.

3. **Non-injectivity of the emergent profile** There exist $x_1, x_2 \in X$ such that

$$x_1 \neq x_2, \quad P(x_1) = P(x_2).$$

Statement

The map

$$\Phi: (X, D_S^{(e)}, C_e) \rightarrow \mathcal{D}_{phys}$$

is **not invertible**: there exists no map

$$\Phi^{-1}: \mathcal{D}_{phys} \rightarrow (X, D_S^{(e)}, C_e)$$

that reconstructs the emergent point uniquely from the physical structure.

Proof

1. Non-injectivity of Φ due to compressivity of H

By assumption, H is compressive: there exist $P_1, P_2 \in \mathcal{F}$ with

$$P_1 \neq P_2, \quad H(P_1) = H(P_2).$$

If there exist $x_1, x_2 \in X$ such that

$$P(x_1) = P_1, \quad P(x_2) = P_2,$$

then

$$\Phi(x_1) = H(P(x_1)) = H(P_1) = H(P_2) = H(P(x_2)) = \Phi(x_2).$$

Thus Φ is not injective: there exist distinct points $x_1 \neq x_2$ with identical physical images.

2. Stronger non-injectivity from emergent saturation

From Theorems 1 and 2:

- $D_S^{(e)}$ is the saturated limit of $\{D_\lambda\}$;
- C_e is the saturated limit of $\{C_\lambda\}$.

Saturation implies that many distinct emergent configurations (different microstructures, different fundamental sets) produce the same limiting distance and correlation.

Thus there exist $x_1, x_2 \in X$ such that

$$x_1 \neq x_2, \quad D_S^{(e)}(x_1, \cdot) = D_S^{(e)}(x_2, \cdot), \quad C_e(x_1, \cdot) = C_e(x_2, \cdot),$$

i.e.

$$P(x_1) = P(x_2).$$

Then

$$\Phi(x_1) = H(P(x_1)) = H(P(x_2)) = \Phi(x_2).$$

Thus Φ identifies even distinct emergent points with identical profiles.

3. Impossibility of an inverse map Φ^{-1}

A function $\Phi^{-1}: \mathcal{D}_{phys} \rightarrow X$ can exist only if Φ is bijective, in particular injective.

But we have shown that there exist $x_1 \neq x_2$ such that

$$\Phi(x_1) = \Phi(x_2).$$

Any hypothetical inverse would have to assign two distinct values of X to the same physical structure, which is impossible for a function.

Thus:

$$\nexists \Phi^{-1}: \mathcal{D}_{phys} \rightarrow X.$$

Conclusion

The non-invertibility of Φ follows directly from:

- **compressivity of H** $\Rightarrow$ information loss at the profile level;
- **emergent saturation** $\Rightarrow$ many emergent configurations yield the same limiting profile;
- **non-injectivity of P** $\Rightarrow$ distinct emergent points share identical profiles.

Thus the chain

$$S \rightarrow D_S^{(f)} \rightarrow C_f \rightarrow D_S^{(e)} \rightarrow C_e \rightarrow P \rightarrow H \rightarrow \Phi$$

ends in a projection map Φ that is **intrinsically non-invertible**.

The theorem is proven.

Theorem 6 — Existence of Emergent Physical Quantities

Let the emergent domain be

$$E = (X, D_S^{(e)}, C_e),$$

where:

- $D_S^{(e)}: X \times X \rightarrow \mathbb{R}^+$ is continuous, bounded, regular, and saturating;
- $C_e: X \times X \rightarrow \mathbb{R}$ is continuous, bounded, regular, and coherent with $D_S^{(e)}$.

Let $\mathcal{F}$ be a functional space containing the emergent pair

$$\mathcal{F} \ni (D_S^{(e)}, C_e).$$

For each class of physical quantity Q , assume the existence of a projection operator

$$\Phi_Q: \mathcal{F} \rightarrow V_Q,$$

where V_Q is the space of the physical quantity (scalars, fields, tensors, densities, etc.), and Φ_Q is:

- well-defined on the domain of continuous and bounded functions;
- continuous;
- regular (introduces no divergences);
- compressive (reduces degrees of freedom relative to the emergent structure, hence non-injective).

Statement

For every projection class Φ_Q , there exists at least one emergent physical quantity

$$Q := \Phi_Q(D_S^{(e)}, C_e) \in V_Q.$$

Proof

1. Well-definedness

By assumption, $D_S^{(e)}$ and C_e :

- are continuous on $X \times X$;
- are bounded;

- belong to the functional space $\mathcal{F}$.

Since Φ_Q is defined on $\mathcal{F}$, the expression

$$Q := \Phi_Q(D_S^{(e)}, C_e)$$

is well-defined: the argument $(D_S^{(e)}, C_e)$ lies in the domain of Φ_Q .

2. Existence (direct construction)

The emergent physical quantity Q is constructed explicitly as the image of the emergent structure under the operator Φ_Q :

$$Q = \Phi_Q(D_S^{(e)}, C_e).$$

This is not an existence by contradiction, but a **direct construction**: given $(D_S^{(e)}, C_e) \in \mathcal{F}$ and $\Phi_Q: \mathcal{F} \rightarrow V_Q$, the element $Q \in V_Q$ exists **by definition**.

3. Continuity and regularity of Q

Since:

- $D_S^{(e)}$ and C_e are continuous and bounded;
- Φ_Q is continuous and regular;

the composition

$$(D_S^{(e)}, C_e) \mapsto Q = \Phi_Q(D_S^{(e)}, C_e)$$

is continuous with respect to the topology of $\mathcal{F}$ and of V_Q .

Regularity of Q follows from the regularity of Φ_Q : a regular operator applied to continuous and bounded functions **cannot generate divergences or uncontrolled discontinuities**.

Thus Q is a regular physical quantity in the sense of the PSE.

4. Effect of compressivity

Compressivity of Φ_Q implies:

- Φ_Q is non-injective: there exist $(f_1, g_1), (f_2, g_2) \in \mathcal{F}$ with

$$(f_1, g_1) \neq (f_2, g_2), \quad \Phi_Q(f_1, g_1) = \Phi_Q(f_2, g_2);$$
- the physical quantity Q depends on **fewer degrees of freedom** than the full emergent structure $(D_S^{(e)}, C_e)$;
- the image of Φ_Q is more constrained, simpler, and more physically interpretable.

Mathematically, the image of Φ_Q is a regular subset of V_Q , free of divergences and compatible with the continuity and boundedness of its argument.

This is precisely the “finite and regular image” characteristic of the PSE.

Conclusion

For every physical projection class

$$\Phi_Q: \mathcal{F} \rightarrow V_Q,$$

we have:

- the argument $(D_S^{(e)}, C_e)$ is valid;
- the operator is well-defined;
- the emergent physical quantity

$$Q = \Phi_Q(D_S^{(e)}, C_e)$$

exists;

- Q is continuous, regular, and compressed (fewer degrees of freedom than the emergent structure);
- Q is mathematically constructed, not merely postulated.

The theorem is proven.

Theorem 7 — Saturation of Emergent Physical Quantities

Let the emergent domain be

$$E = (X, D_S^{(e)}, C_e),$$

where:

- $D_S^{(e)}: X \times X \rightarrow \mathbb{R}^+$ is a bounded emergent distance:

$$\exists D_{\max} > 0: D_S^{(e)}(x, y) \leq D_{\max}, \quad \forall (x, y) \in X \times X;$$

- $C_e: X \times X \rightarrow \mathbb{R}$ is a bounded emergent correlation:

$$\exists C_{\max} > 0: |C_e(x, y)| \leq C_{\max}, \quad \forall (x, y) \in X \times X.$$

Let $\mathcal{F}$ be a functional space containing the emergent pair $(D_S^{(e)}, C_e)$.

For each class of emergent physical quantity Q , assume the existence of a projection operator

$$\Phi_Q: \mathcal{F} \rightarrow V_Q,$$

where V_Q is the space of the physical quantity, and Φ_Q is:

- regular;
- compressive; in particular, it maps bounded subsets of $\mathcal{F}$ into bounded subsets of V_Q .

The emergent physical quantity is defined by

$$Q := \Phi_Q(D_S^{(e)}, C_e) \in V_Q.$$

Statement

Every emergent physical quantity Q is **saturating**, i.e., there exists a finite constant $Q_{\max} > 0$ such that

$$Q(x, y) \leq Q_{\max}, \quad \forall (x, y) \in X \times X.$$

Proof

1. Bounded input in $\mathcal{F}$

By assumption, $D_S^{(e)}$ and C_e are bounded:

$$D_S^{(e)}(x, y) \leq D_{\max}, \quad |C_e(x, y)| \leq C_{\max}, \quad \forall (x, y) \in X \times X.$$

Thus the pair $(D_S^{(e)}, C_e)$ belongs to a bounded subset of $\mathcal{F}$ (e.g., with respect to the sup norm on $X \times X$).

2. Bounded image of Φ_Q

By assumption, Φ_Q is regular and compressive, and in particular:

- on bounded subsets of $\mathcal{F}$, it produces bounded subsets of V_Q .

Applying Φ_Q to the bounded subset containing $(D_S^{(e)}, C_e)$, we obtain that the image

$$Q = \Phi_Q(D_S^{(e)}, C_e)$$

is a bounded function on $X \times X$. Hence there exists a finite constant $Q_{\max} > 0$ such that

$$Q(x, y) \leq Q_{\max}, \quad \forall (x, y) \in X \times X.$$

3. Saturation

Define

$$Q_{\max} := \sup_{(x, y) \in X \times X} Q(x, y).$$

From the previous step, $Q_{\max} < \infty$. This constant represents the maximum physically attainable value of the emergent quantity Q on the domain $X \times X$.

Consequently, Q is **saturating**: it cannot exceed the bound $Q_{\max}$, and is entirely contained within a finite interval.

Conclusion

For every regular and compressive physical projection operator

$$\Phi_Q: \mathcal{F} \rightarrow V_Q,$$

applied to the bounded emergent structure $(D_S^{(e)}, C_e)$, the emergent physical quantity

$$Q = \Phi_Q(D_S^{(e)}, C_e)$$

is:

- well-defined;
- bounded;
- **saturating**:

$$Q(x, y) \leq Q_{\max} < \infty, \quad \forall (x, y) \in X \times X.$$

The theorem is proven.

Theorem 8 — Rigidity of Emergent Physical Quantities

Let

$$D_S^{(e)}: X \times X \rightarrow \mathbb{R}^+$$

be a continuous emergent distance, monotonic and stable with respect to structural variations: small variations in the emergent structure produce small variations in $D_S^{(e)}$.

Let Q be an emergent physical quantity obtained through a projection

$$Q = \Phi_Q(D_S^{(e)}, C_e),$$

where Φ_Q is:

- continuous;
- monotonic (in the physical sense);
- compressive (does not amplify degrees of freedom).

Assume furthermore that Φ_Q is Lipschitz with respect to $D_S^{(e)}$: there exists a constant $K > 0$ such that, for two emergent distances D_1, D_2 ,

$$|\Phi_Q(D_1) - \Phi_Q(D_2)| \leq K |D_1 - D_2|.$$

Statement

For small variations of the emergent distance, the variations of the emergent physical quantity are bounded by

$$|\delta Q| \leq K |\delta D_S^{(e)}|.$$

Proof

1. Variation of the emergent distance

Consider two nearby emergent configurations, with distances

$$D_S^{(e)} \text{ and } D_S^{(e)} + \delta D_S^{(e)}.$$

By emergent stability, the variation $\delta D_S^{(e)}$ is small and controlled (e.g., in the sup norm on $X \times X$).

2. Variation of the physical quantity

The physical quantity corresponding to the first configuration is

$$Q = \Phi_Q(D_S^{(e)}),$$

while for the second configuration it is

$$Q + \delta Q = \Phi_Q(D_S^{(e)} + \delta D_S^{(e)}).$$

Thus,

$$\delta Q = \Phi_Q(D_S^{(e)} + \delta D_S^{(e)}) - \Phi_Q(D_S^{(e)}).$$

3. Lipschitz estimate and rigidity

By the Lipschitz assumption on Φ_Q ,

$$|\delta Q| = |\Phi_Q(D_S^{(e)} + \delta D_S^{(e)}) - \Phi_Q(D_S^{(e)})| \leq K |\delta D_S^{(e)}|.$$

Monotonicity of Φ_Q prevents uncontrolled amplification (no “infinite derivative”), emergent stability of $D_S^{(e)}$ excludes divergences, and compressivity ensures that the physical response is more rigid than the emergent structure.

Conclusion

Variations of an emergent physical quantity Q are rigidly controlled by variations of the emergent distance $D_S^{(e)}$:

$$|\delta Q| \leq K |\delta D_S^{(e)}|.$$

Theorem 9 — Non-Retroactivity

Let the emergent domain be

$$E = (X, D_S^{(e)}, C_e),$$

where $D_S^{(e)}$ and C_e are emergent quantities obtained as saturated limits of the regularized families $\{D_\lambda\}$ and $\{C_\lambda\}$.

Let Q be an emergent physical quantity obtained through a projection

$$Q = \Phi_Q(D_S^{(e)}, C_e),$$

where:

- $\Phi_Q: \mathcal{F} \rightarrow V_Q$ is a one-directional operator (from emergent domain to physical domain);
- Φ_Q is non-invertible (Theorem 5);
- Φ_Q is compressive (reduces degrees of freedom);
- the emergent structure $(D_S^{(e)}, C_e)$ is defined as a saturated limit, hence stable and independent of any physical quantity.

Statement

No physical quantity can influence the emergent structure:

$$\frac{\partial(D_S^{(e)}, C_e)}{\partial Q} = 0.$$

Proof

1. One-directionality of the projection

By definition,

$$Q = \Phi_Q(D_S^{(e)}, C_e)$$

is a function of the emergent structure. There exists no inverse map

$$\Phi_Q^{-1}: V_Q \rightarrow \mathcal{F},$$

as proven in Theorem 5.

Thus no functional dependence of the form

$$(D_S^{(e)}, C_e) = \Psi(Q)$$

can exist. The projection acts only as

$$\mathcal{F} \rightarrow V_Q,$$

never the reverse.

2. Irreversible information loss (compressivity)

Compressivity implies:

- many distinct emergent structures produce the same physical quantity;
- Φ_Q eliminates degrees of freedom;
- the physical image does not contain enough information to reconstruct the emergent structure.

Formally, there exist $P_1 \neq P_2 \in \mathcal{F}$ such that

$$\Phi_Q(P_1) = \Phi_Q(P_2).$$

Thus no variation of Q can select or modify a specific variation of $(D_S^{(e)}, C_e)$. Inverse dependence is mathematically impossible.

3. Emergent stability of saturated limits

The emergent structure is defined as a limit:

$$D_S^{(e)} = \lim_{\lambda \rightarrow \infty} D_\lambda, \quad C_e = \lim_{\lambda \rightarrow \infty} C_\lambda.$$

These limits:

- depend only on emergent dynamics;
- are guaranteed by structural stability (Theorems 1–3);
- do not depend on any physical quantity.

Physical quantities Q are constructed **after** saturation, via compressive projection. Thus:

$$\delta Q \text{ cannot modify } \delta D_S^{(e)} \text{ nor } \delta C_e.$$

Conclusion

Since:

- the projection Φ_Q is one-directional;
- compressivity eliminates information;
- the emergent structure is defined prior to and independently of Q ;

it follows that:

$$\frac{\partial(D_S^{(e)}, C_e)}{\partial Q} = 0.$$

Theorem 10 — Structural Determination of Physical Quantities

Let the emergent domain be

$$E = (X, D_S^{(e)}, C_e),$$

where $D_S^{(e)}$ and C_e are continuous, regular, saturating, and stabilized (Theorems 1–3).

Let $\mathcal{F}$ be a functional space containing the emergent structure $(D_S^{(e)}, C_e)$. For each emergent physical quantity Q , there exists a projection operator

$$\Phi_Q: \mathcal{F} \rightarrow V_Q,$$

with:

- **one-directionality:** Φ_Q acts only from emergent $\rightarrow$ physical;
- **non-invertibility:** no $\Phi_Q^{-1}: V_Q \rightarrow \mathcal{F}$ exists (Theorem 5);
- **compressivity:** Φ_Q reduces degrees of freedom;
- **monotonicity:** the order induced by $D_S^{(e)}$ and C_e is transferred to Q .

The physical quantity is defined pointwise by

$$Q(x, y) = \Phi_Q(D_S^{(e)}(x, y), C_e(x, y)), \quad (x, y) \in X \times X.$$

Statement

Every emergent physical quantity is completely determined by the emergent structure:

$$Q(x, y) = \Phi_Q(D_S^{(e)}(x, y), C_e(x, y)), \quad \forall (x, y) \in X \times X.$$

Proof

1. Monotonicity $\Rightarrow$ induced order

From Theorem 3, the emergent correlation C_e depends monotonically on the emergent distance $D_S^{(e)}$. This induces a structural order on $X \times X$: for $(x_1, y_1), (x_2, y_2) \in X \times X$,

$$D_S^{(e)}(x_1, y_1) < D_S^{(e)}(x_2, y_2) \Rightarrow C_e(x_1, y_1) \geq C_e(x_2, y_2).$$

Since Φ_Q is monotonic with respect to the order induced by $(D_S^{(e)}, C_e)$, the order is transferred to Q . Thus Q cannot depend on any information external to the emergent structure.

2. Compressivity $\Rightarrow$ no additional information

Compressivity of Φ_Q implies:

- Φ_Q eliminates degrees of freedom;
- it cannot introduce new variables;

- it cannot depend on external physical quantities.

Formally, there exist emergent profiles $P_1 \neq P_2 \in \mathcal{F}$ such that

$$\Phi_Q(P_1) = \Phi_Q(P_2).$$

Thus Q is entirely determined by the emergent profile

$$P(x) := (D_S^{(e)}(x, \cdot), C_e(x, \cdot)),$$

and cannot contain information not already encoded in $(D_S^{(e)}, C_e)$.

3. Non-invertibility $\Rightarrow$ impossibility of inverse physical dependence

From Theorem 5, Φ_Q is non-invertible:

$$\exists \Phi_Q^{-1}: V_Q \rightarrow \mathcal{F}.$$

Thus:

- no physical quantity can reconstruct the emergent structure;
- no physical quantity can influence it (Theorem 9);
- no physical quantity can serve as an independent variable for defining $D_S^{(e)}$ or C_e .

Therefore, Q cannot depend on external physical magnitudes: it depends only on $(D_S^{(e)}, C_e)$.

Conclusion

The form

$$Q(x, y) = \Phi_Q(D_S^{(e)}(x, y), C_e(x, y))$$

is not a choice, but the **only possible form** compatible with:

- structural monotonicity;
- compressivity;
- non-invertibility;
- one-directionality (emergent $\rightarrow$ physical).

Every emergent physical quantity is completely determined by the emergent structure $(D_S^{(e)}, C_e)$.

3. Structural Theorems of Emergent Coherence

Theorem 11 — Emergent Separation

Hypotheses. Let $(X, D_S^{(e)})$ be an emergent metric space, with

$$D_S^{(e)}: X \times X \rightarrow [0, \infty)$$

continuous, regular, and saturating (Theorem 1). Assume:

- for every $x \in X$, $D_S^{(e)}(x, x) = 0$;
- for every $x \neq y$, there exists λ such that $D_\lambda(x, y) > 0$.

Statement. The emergent distance is separating:

$$D_S^{(e)}(x, y) = 0 \Rightarrow x = y.$$

Proof. Let $x \neq y$. By hypothesis, there exists λ_0 such that $D_{\lambda_0}(x, y) > 0$. By monotonicity and saturation,

$$D_S^{(e)}(x, y) = \lim_{\lambda \rightarrow \infty} D_\lambda(x, y) \geq D_{\lambda_0}(x, y) > 0.$$

Thus it is impossible that $D_S^{(e)}(x, y) = 0$ with $x \neq y$. Conversely, $D_S^{(e)}(x, x) = 0$ for all x by stability of the limit. Separation follows.

Theorem 12 — Emergent Completeness

Hypotheses. Let $(X, D_S^{(e)})$ be an emergent metric space with $X \times X$ compact (Theorem 1) and $D_S^{(e)}$ continuous.

Statement. The metric space $(X, D_S^{(e)})$ is complete.

Proof. Compactness of $X \times X$ implies that X is compact with respect to the topology induced by $D_S^{(e)}$. Every compact metric space is complete: every Cauchy sequence has a convergent subsequence and, by compactness, converges in X . Thus $(X, D_S^{(e)})$ is complete.

Theorem 13 — Metric–Correlational Compatibility

Hypotheses. Let $(X, D_S^{(e)}, C_e)$ be the emergent domain, with:

- $D_S^{(e)}$ continuous and separating (Theorems 1, 11);
- C_e continuous and monotonic with respect to $D_S^{(e)}$ (Theorem 3).

Assume there exists $F: [0, \infty) \rightarrow \mathbb{R}$ such that

$$C_e(x, y) = F(D_S^{(e)}(x, y)).$$

Statement. The emergent correlation is metrically compatible with $D_S^{(e)}$: for every (x, y) ,

$$D_S^{(e)}(x, y) = 0 \Rightarrow C_e(x, y) = F(0),$$

and the structure $(D_S^{(e)}, C_e)$ is fully described by $D_S^{(e)}$ and F .

Proof. By definition,

$$C_e(x, y) = F(D_S^{(e)}(x, y)).$$

If $D_S^{(e)}(x, y) = 0$, then $C_e(x, y) = F(0)$. Since F is monotonic and continuous, the correlation is a radial function of the distance, and all correlational information is metrically encoded. Compatibility follows.

Theorem 14 — Radial Continuity of Physical Quantities

Hypotheses. Let $(X, D_S^{(e)}, C_e)$ be as above, and let Q be an emergent physical quantity

$$Q = \Phi_Q(D_S^{(e)}, C_e),$$

with Φ_Q continuous and monotonic. Assume $C_e(x, y) = F(D_S^{(e)}(x, y))$.

Statement. The emergent physical quantity Q is continuous with respect to the emergent distance: there exists $G: [0, \infty) \rightarrow \mathbb{R}$ such that

$$Q(x, y) = G(D_S^{(e)}(x, y)),$$

with G continuous.

Proof. Since C_e depends only on $D_S^{(e)}$, the pair $(D_S^{(e)}, C_e)$ can be viewed as a function of $r = D_S^{(e)}(x, y)$. The operator Φ_Q acts on $(D_S^{(e)}, C_e)$ continuously and monotonically, so there exists a function G such that

$$Q(x, y) = G(D_S^{(e)}(x, y)).$$

Continuity of Φ_Q and of $D_S^{(e)}$ implies continuity of G .

Theorem 15 — Uniform Regularity of Projections

Hypotheses. Let $\mathcal{F}$ be the functional space of emergent pairs $(D_S^{(e)}, C_e)$, with sup norm. Let

$$\Phi_Q: \mathcal{F} \rightarrow V_Q$$

be a continuous, regular, and compressive projection operator, defined on bounded subsets of $\mathcal{F}$.

Statement. On every bounded subset of $\mathcal{F}$, Φ_Q is uniformly regular: there exists a constant $L > 0$ such that, for any $(f_1, g_1), (f_2, g_2)$ in the subset,

$$\| \Phi_Q(f_1, g_1) - \Phi_Q(f_2, g_2) \| \leq L \| (f_1, g_1) - (f_2, g_2) \|.$$

Proof. Continuity of Φ_Q on compact/bounded subsets of $\mathcal{F}$ implies, by the Heine–Cantor theorem, that Φ_Q is uniformly continuous on such subsets. Regularity and compressivity ensure that the response does not diverge. Thus there exists $L > 0$ providing a uniform bound on variations, as stated.

Theorem 16 — Stability of Emergent Physical Quantities

Hypotheses. Let $D_S^{(e)}$ be stable with respect to structural variations (Theorem 8), and let

$$Q = \Phi_Q(D_S^{(e)}, C_e)$$

with Φ_Q Lipschitz with respect to $D_S^{(e)}$, with constant $K > 0$.

Statement. Every emergent physical quantity is stable: small variations of $D_S^{(e)}$ produce small variations of Q , with

$$\| \delta Q \| \leq K \| \delta D_S^{(e)} \|.$$

Proof. Consider two nearby emergent structures, with distances $D_S^{(e)}$ and $D_S^{(e)} + \delta D_S^{(e)}$. The corresponding physical quantities are

$$Q = \Phi_Q(D_S^{(e)}), Q + \delta Q = \Phi_Q(D_S^{(e)} + \delta D_S^{(e)}).$$

Thus,

$$\delta Q = \Phi_Q(D_S^{(e)} + \delta D_S^{(e)}) - \Phi_Q(D_S^{(e)}).$$

By the Lipschitz assumption,

$$\| \delta Q \| \leq K \| \delta D_S^{(e)} \|.$$

Stability of $D_S^{(e)}$ ensures that $\| \delta D_S^{(e)} \|^$ is small and controlled, hence δQ is also small: Q is stable.

4. Structural Theorems of Emergence

Theorem 17 — Existence of the Emergent Limit

Let X be a metric space, and let $\{F_\lambda\}_{\lambda \geq 0}$ be a family of functions

$$F_\lambda: X \times X \rightarrow \mathbb{R},$$

satisfying:

1. **Monotonicity in λ .** For every $(x, y) \in X \times X$, the function

$$\lambda \mapsto F_\lambda(x, y)$$

is monotonic (either non-increasing or non-decreasing), i.e.

$$\lambda_1 \leq \lambda_2 \Rightarrow F_{\lambda_1}(x, y) \leq F_{\lambda_2}(x, y)$$

(or the reversed inequality).

2. **Uniform boundedness.** There exists $M < \infty$ such that

$$|F_\lambda(x, y)| \leq M, \forall \lambda, \quad \forall (x, y) \in X \times X.$$

3. **Equicontinuity.** For every $\varepsilon > 0$ there exists $\delta > 0$ such that, for every λ ,

$$\|(x, y) - (x', y')\| < \delta \Rightarrow |F_\lambda(x, y) - F_\lambda(x', y')| < \varepsilon.$$

4. **Structural stability (uniform Cauchy in λ).** For every $\varepsilon > 0$ there exists Λ such that, for every $\lambda_1, \lambda_2 > \Lambda$,

$$|F_{\lambda_1}(x, y) - F_{\lambda_2}(x, y)| < \varepsilon, \quad \forall (x, y) \in X \times X.$$

5. **Continuity of each F_λ .** Each F_λ is continuous on $X \times X$.

Statement

The emergent limit

$$F_e(x, y) := \lim_{\lambda \rightarrow \infty} F_\lambda(x, y)$$

exists for every $(x, y) \in X \times X$, the function F_e is continuous on $X \times X$, and the limit is unique.

Proof

1. Existence of the pointwise limit

Fix $(x, y) \in X \times X$. The real sequence $\{F_\lambda(x, y)\}_\lambda$ is:

- monotonic (Hypothesis 1),
- bounded (Hypothesis 2).

By the classical monotone convergence theorem, every monotone and bounded real sequence converges. Thus the limit

$$F_e(x, y) := \lim_{\lambda \rightarrow \infty} F_\lambda(x, y)$$

exists. This defines a function $F_e: X \times X \rightarrow \mathbb{R}$.

2. Uniform convergence

We now show that the convergence is uniform on $X \times X$.

Let $\varepsilon > 0$. By structural stability (Hypothesis 4), there exists Λ such that, for every $\lambda_1, \lambda_2 > \Lambda$,

$$|F_{\lambda_1}(x, y) - F_{\lambda_2}(x, y)| < \varepsilon, \quad \forall (x, y) \in X \times X.$$

Fix $\lambda > \Lambda$. By definition of the pointwise limit,

$$F_e(x, y) = \lim_{\mu \rightarrow \infty} F_\mu(x, y).$$

The uniform Cauchy condition implies that, for every $\mu > \Lambda$,

$$|F_\mu(x, y) - F_\lambda(x, y)| < \varepsilon, \quad \forall (x, y).$$

Passing to the limit as $\mu \rightarrow \infty$,

$$|F_e(x, y) - F_\lambda(x, y)| \leq \varepsilon, \quad \forall (x, y) \in X \times X.$$

Thus the convergence $F_\lambda \rightarrow F_e$ is uniform.

3. Continuity of the limit

Each F_λ is continuous (Hypothesis 5). Since the convergence is uniform, the uniform limit of continuous functions is continuous. Therefore:

$$F_e \text{ is continuous on } X \times X.$$

4. Uniqueness of the limit

Suppose there exist two functions F_e and F'_e such that

$$\lim_{\lambda \rightarrow \infty} F_\lambda(x, y) = F_e(x, y), \quad \lim_{\lambda \rightarrow \infty} F_\lambda(x, y) = F'_e(x, y).$$

By uniform Cauchy stability, for every $\varepsilon > 0$ there exists Λ such that, for every $\lambda > \Lambda$,

$$|F_\lambda(x, y) - F_e(x, y)| < \varepsilon/2, \quad |F_\lambda(x, y) - F'_e(x, y)| < \varepsilon/2.$$

Thus,

$$|F_e(x, y) - F'_e(x, y)| \leq |F_e(x, y) - F_\lambda(x, y)| + |F_\lambda(x, y) - F'_e(x, y)| < \varepsilon.$$

Since ε is arbitrary,

$$F_e(x, y) = F'_e(x, y), \quad \forall (x, y) \in X \times X.$$

Conclusion

We have shown that:

- the pointwise limit exists,
- the convergence is uniform,
- the limit is continuous,
- the limit is unique.

Therefore, the emergent limit F_e exists, is continuous, and is unique, as stated.

Theorem 18 — Uniqueness of the Emergent Domain

Hypotheses. Let F be the fundamental structure. Let $\{F_\lambda\}_{\lambda \geq 0}$ and $\{F'_\lambda\}_{\lambda \geq 0}$ be two regularization families

$$F_\lambda: X \times X \rightarrow \mathbb{R}, \quad F'_\lambda: X \times X \rightarrow \mathbb{R},$$

each satisfying the hypotheses of Theorem 17:

- monotonicity in λ ;
- uniform boundedness;
- equicontinuity;
- structural stability (uniform Cauchy condition in λ);
- continuity of every F_λ and F'_λ .

Define the two emergent limits:

$$F_e(x, y) := \lim_{\lambda \rightarrow \infty} F_\lambda(x, y), \quad F'_e(x, y) := \lim_{\lambda \rightarrow \infty} F'_\lambda(x, y).$$

Assume further:

- **structural coherence:** each family regularizes the same fundamental structure F ;
- **parameterization independence:** if λ is renumbered through a monotone transformation $\sigma(\lambda)$, the family remains equivalent;
- **isotropy of the regularization:** no direction or label of the fundamental domain is privileged.

Statement

The emergent limit is unique:

$$F_e(x, y) = F'_e(x, y), \quad \forall (x, y) \in X \times X.$$

Proof

1. Existence and continuity of the limits

By Theorem 17, both families admit an emergent limit:

$$F_e(x, y) = \lim_{\lambda \rightarrow \infty} F_\lambda(x, y), \quad F'_e(x, y) = \lim_{\lambda \rightarrow \infty} F'_\lambda(x, y),$$

and these limits are continuous and unique.

2. Uniform convergence

By structural stability, there exist $\Lambda, \Lambda' > 0$ such that, for all $\lambda_1, \lambda_2 > \Lambda$,

$$|F_{\lambda_1}(x, y) - F_{\lambda_2}(x, y)| < \varepsilon,$$

and similarly for the family F'_λ . Thus both families converge uniformly to their respective limits.

3. Compatibility with the same fundamental structure

Since both families regularize the same fundamental structure F , and the parameterization is irrelevant, there exists a monotone transformation $\sigma(\lambda)$ such that, for every $\varepsilon > 0$, there is Λ_ε with:

$$|F_\lambda(x, y) - F'_{\sigma(\lambda)}(x, y)| < \varepsilon, \quad \forall \lambda > \Lambda_\varepsilon.$$

4. Passage to the limit

Fix $(x, y) \in X \times X$. For any $\varepsilon > 0$, choose λ sufficiently large:

$$|F_e(x, y) - F'_e(x, y)| \leq |F_e(x, y) - F_\lambda(x, y)| + |F_\lambda(x, y) - F'_{\sigma(\lambda)}(x, y)| + |F'_{\sigma(\lambda)}(x, y) - F'_e(x, y)|.$$

Each term is $< \varepsilon/3$ for large λ . Thus:

$$|F_e(x, y) - F'_e(x, y)| < \varepsilon.$$

Since ε is arbitrary:

$$F_e(x, y) = F'_e(x, y).$$

Conclusion

The emergent limit is unique, independently of:

- the chosen regularization family,
- the parameterization λ ,
- the renumbering of fundamental elements,
- the direction or order in which the limit is taken.

This rigorously formalizes that the emergent structure does not depend on technical details of the regularization, but only on the fundamental structure F .

Theorem 19 — Emergent Stability

Let F be the fundamental structure, and let

$$E = (D_S^{(e)}, C_e)$$

be the emergent domain obtained as the limit of the regularized families $\{F_\lambda\}$ (Theorem 17).

Assume:

1. **Stability of the regularization.** For every $\varepsilon > 0$ there exists $\delta > 0$ such that, if two fundamental structures F and F' satisfy

$$\|F - F'\| < \delta,$$

then their regularizations satisfy

$$\|F_\lambda - F'_\lambda\| < \varepsilon, \forall \lambda.$$

2. **Uniform convergence of the limits.** The families $\{F_\lambda\}$ and $\{F'_\lambda\}$ converge uniformly to their respective emergent limits:

$$E = \lim_{\lambda \rightarrow \infty} F_\lambda, E' = \lim_{\lambda \rightarrow \infty} F'_\lambda.$$

3. **Monotonicity and compressivity.** The maps $F \mapsto F_\lambda$ and $F_\lambda \mapsto E$ are monotone and compressive, and therefore do not amplify variations.

Statement

Small perturbations of the fundamental structure produce bounded perturbations in the emergent domain:

$$\| E - E' \| \leq K \| F - F' \|$$

for some constant $K > 0$ independent of F and F' .

Proof

1. Stability of the regularization

By hypothesis, if $\| F - F' \| < \delta$, then

$$\| F_\lambda - F'_\lambda \| < \varepsilon, \forall \lambda.$$

Thus the microscopic perturbation $\| F - F' \|$ is not amplified during regularization: the difference between F_λ and F'_λ remains controlled by ε .

2. Uniform convergence

Since convergence is uniform,

$$E = \lim_{\lambda \rightarrow \infty} F_\lambda, E' = \lim_{\lambda \rightarrow \infty} F'_\lambda,$$

there exists $\Lambda > 0$ such that, for every $\lambda > \Lambda$,

$$\| F_\lambda - E \| < \varepsilon, \| F'_\lambda - E' \| < \varepsilon.$$

3. Estimate of the emergent difference

For every $\lambda > \Lambda$,

$$\| E - E' \| \leq \| E - F_\lambda \| + \| F_\lambda - F'_\lambda \| + \| F'_\lambda - E' \|.$$

Using the hypotheses:

- $\| E - F_\lambda \| < \varepsilon,$
- $\| F_\lambda - F'_\lambda \| < \varepsilon,$
- $\| F'_\lambda - E' \| < \varepsilon,$

we obtain:

$$\| E - E' \| < 3\varepsilon.$$

Now, by stability of the regularization, we may choose ε proportional to $\| F - F' \|$. In particular, there exists a constant $K' > 0$ such that, for $\| F - F' \|$ sufficiently small,

$$\varepsilon \leq K' \| F - F' \|.$$

Thus:

$$\| E - E' \| < 3K' \| F - F' \|.$$

Setting $K := 3K'$, we obtain:

$$\| E - E' \| \leq K \| F - F' \|.$$

Conclusion

The map sending the fundamental structure F to the emergent domain E is stable:

- small variations in F produce proportionally small variations in E ;
- no uncontrolled amplification of microscopic perturbations occurs;
- the emergent structure is rigidly controlled by the fundamental structure.

Theorem 20 — Composition of the Maps

Formal context

Let:

- F be the fundamental structure, an element of a normed space $(\mathcal{F}, \|\cdot\|_{\mathcal{F}})$;
- E the emergent domain, an element of a normed space $(E, \|\cdot\|_E)$;
- P the physical domain, an element of a normed space $(\mathcal{P}, \|\cdot\|_{\mathcal{P}})$.

The PSE maps are:

$$\Psi: \mathcal{F} \rightarrow E, \Phi: E \rightarrow \mathcal{P},$$

and the composition is:

$$\Phi \circ \Psi: \mathcal{F} \rightarrow \mathcal{P}.$$

Assumptions

1. **Monotonicity** There exists a partial order $\leq$ on $\mathcal{F}, E, \mathcal{P}$ such that:

$$F_1 \leq F_2 \Rightarrow \Psi(F_1) \leq \Psi(F_2),$$

$$E_1 \leq E_2 \Rightarrow \Phi(E_1) \leq \Phi(E_2).$$

2. **Continuity** The maps are continuous:

$$\|F - F'\|_{\mathcal{F}} \rightarrow 0 \Rightarrow \|\Psi(F) - \Psi(F')\|_E \rightarrow 0,$$

$$\|E - E'\|_E \rightarrow 0 \Rightarrow \|\Phi(E) - \Phi(E')\|_{\mathcal{P}} \rightarrow 0.$$

3. **Compressivity** There exist constants $K_{\Psi}, K_{\Phi} < 1$ such that:

$$\|\Psi(F) - \Psi(F')\|_E \leq K_{\Psi} \|F - F'\|_{\mathcal{F}},$$

$$\|\Phi(E) - \Phi(E')\|_{\mathcal{P}} \leq K_{\Phi} \|E - E'\|_E.$$

Statement

The composition

$$\Phi \circ \Psi: \mathcal{F} \rightarrow \mathcal{P}$$

is:

- monotone,
- continuous,
- compressive, with compressivity constant

$$K = K_{\Phi} K_{\Psi} < 1.$$

Proof

1. Monotonicity of the composition

Let $F_1 \leq F_2$. By monotonicity of Ψ :

$$\Psi(F_1) \leq \Psi(F_2).$$

By monotonicity of Φ :

$$\Phi(\Psi(F_1)) \leq \Phi(\Psi(F_2)).$$

Thus:

$$F_1 \leq F_2 \Rightarrow (\Phi \circ \Psi)(F_1) \leq (\Phi \circ \Psi)(F_2).$$

The composition is monotone.

2. Continuity of the composition

Let $\|F - F'\|_{\mathcal{F}} \rightarrow 0$. By continuity of Ψ :

$$\|\Psi(F) - \Psi(F')\|_E \rightarrow 0.$$

By continuity of Φ :

$$\|\Phi(\Psi(F)) - \Phi(\Psi(F'))\|_{\mathcal{P}} \rightarrow 0.$$

Thus:

$$\|(\Phi \circ \Psi)(F) - (\Phi \circ \Psi)(F')\|_{\mathcal{P}} \rightarrow 0.$$

The composition is continuous.

3. Compressivity of the composition

For any $F, F' \in \mathcal{F}$:

$$\|\Psi(F) - \Psi(F')\|_E \leq K_{\Psi} \|F - F'\|_{\mathcal{F}}.$$

Applying Φ :

$$\|\Phi(\Psi(F)) - \Phi(\Psi(F'))\|_{\mathcal{P}} \leq K_{\Phi} \|\Psi(F) - \Psi(F')\|_E.$$

Combining:

$$\|\Phi(\Psi(F)) - \Phi(\Psi(F'))\|_{\mathcal{P}} \leq K_{\Phi} K_{\Psi} \|F - F'\|_{\mathcal{F}}.$$

Since $K_{\Phi} < 1$ and $K_{\Psi} < 1$, the composite constant

$$K := K_{\Phi} K_{\Psi} < 1$$

is again a compressivity constant.

Thus:

$$\|(\Phi \circ \Psi)(F) - (\Phi \circ \Psi)(F')\|_{\mathcal{P}} \leq K \|F - F'\|_{\mathcal{F}}.$$

The composition is compressive.

Conclusion

The full PSE map

$$F \rightarrow P$$

automatically inherits the fundamental properties:

- **monotonicity** (the structural order is preserved),
- **continuity** (no discontinuities are introduced),
- **compressivity** (the physical domain is always “more rigid” than the emergent domain).

Theorem 21 — Universality of the Physical Domain

Formal Context

In the PSE, the complete ontological chain is:

$$F \xrightarrow{\Psi} E \xrightarrow{R} E_{\text{reg}} \xrightarrow{\Theta_k} E_{\text{reg}} \xrightarrow{\Phi_k} D_{\text{phys}}(k),$$

where:

- $F \in \mathcal{F}$ is the fundamental structure, an element of the normed space $(\mathcal{F}, \|\cdot\|_{\mathcal{F}})$;
- $E = \Psi(F) \in E$ is the emergent domain, an element of the normed space $(E, \|\cdot\|_E)$;
- $E_{\text{reg}} = R(E) \in E_{\text{reg}}$ is the regularized emergent domain;
- $\Theta_k: E_{\text{reg}} \rightarrow E_{\text{reg}}$ is the structural complexity threshold of level k ;
- $\Phi_k: E_{\text{reg}} \rightarrow D_{\text{phys}}(k)$ is the physical projection of level k ;
- $D_{\text{phys}}(k)$ is the physical domain corresponding to projection class k .

The overall physical map is:

$$\Phi_k \circ \Theta_k \circ R \circ \Psi: F \rightarrow D_{\text{phys}}(k).$$

Assumptions

1. Uniqueness of the emergent domain

Two regularization families $\{F_\lambda\}$ and $\{F'_\lambda\}$ satisfy:

- monotonicity in λ ,
- uniform boundedness,
- equicontinuity,
- structural stability (uniform Cauchy),
- continuity of each F_λ ,

and therefore, by Theorem 17, the emergent limits:

$$E = \lim_{\lambda \rightarrow \infty} F_\lambda, \quad E' = \lim_{\lambda \rightarrow \infty} F'_\lambda$$

exist, are continuous, and are unique.

By Theorem 18:

$$E = E'.$$

2. Properties of the physical projection

The map $\Phi_k: E_{\text{reg}} \rightarrow D_{\text{phys}}(k)$ is:

- monotone,
- continuous,
- compressive,
- non-invertible,
- isotropic (invariant under renumberings, permutations, and relabelings of the emergent domain).

Formally, for every isotropic transformation σ :

$$\Phi_k(E_{\text{reg}}) = \Phi_k(E_{\text{reg}}^\sigma).$$

Statement

The physical domain produced by the PSE is unique:

$$D_{\text{phys}}(k) = D'_{\text{phys}}(k).$$

Every physical domain obtained from the PSE arises from the same fundamental structure F , independently of the regularization procedure, the complexity threshold, or the projection class.

Proof

1. Uniqueness of the emergent domain

By Theorem 18:

$$E = E'.$$

Thus any regularization procedure compatible with F yields the same emergent domain.

2. Uniqueness of the regularized emergent domain

Since R is continuous, monotone, isotropic, and contractive:

$$E_{\text{reg}} = R(E), E'_{\text{reg}} = R(E').$$

Substituting $E = E'$:

$$E_{\text{reg}} = E'_{\text{reg}}.$$

3. Application of the complexity threshold

Since Θ_k is continuous, monotone, and isotropic:

$$\Theta_k(E_{\text{reg}}) = \Theta_k(E'_{\text{reg}}).$$

4. Application of the physical projection

By definition:

$$D_{\text{phys}}(k) = \Phi_k(\Theta_k(E_{\text{reg}})), D'_{\text{phys}}(k) = \Phi_k(\Theta_k(E'_{\text{reg}})).$$

Substituting:

$$E_{\text{reg}} = E'_{\text{reg}}, \quad \Theta_k(E_{\text{reg}}) = \Theta_k(E'_{\text{reg}}),$$

we obtain:

$$D_{\text{phys}}(k) = \Phi_k(\Theta_k(E_{\text{reg}})) = \Phi_k(\Theta_k(E'_{\text{reg}})) = D'_{\text{phys}}(k).$$

5. Role of isotropy and non-invertibility

Isotropy

The projection Φ_k is invariant under:

- renumberings,
- permutations,
- relabelings,
- monotone transformations of regularization parameters.

Formally:

$$\Phi_k(E_{\text{reg}}) = \Phi_k(E_{\text{reg}}^\sigma).$$

Non-invertibility

Compressivity implies that Φ_k is not invertible:

$$\nexists \Phi_k^{-1}: D_{\text{phys}}(k) \rightarrow E_{\text{reg}}.$$

Thus:

- no two distinct physical domains correspond to the same regularized emergent domain;
- the projection cannot split or branch the physical domain.

Formally:

$$\Phi_k(E_{\text{reg},1}) = \Phi_k(E_{\text{reg},2}) \Rightarrow E_{\text{reg},1} = E_{\text{reg},2}.$$

Applying this to $E_{\text{reg}} = E'_{\text{reg}}$:

$$D_{\text{phys}}(k) = D'_{\text{phys}}(k).$$

Conclusion

The physical domain of the PSE is universal:

- it does not depend on the regularization procedure,
- it does not depend on parametrization,
- it does not depend on renumberings or permutations,
- it does not depend on technical or conventional choices,
- it does not depend on any freedom in the emergent construction.

It depends solely on the fundamental structure F .

$$D_{\text{phys}}(k) = (\Phi_k \circ \Theta_k \circ R \circ \Psi)(F) \text{ is unique.}$$

The emergent physics is deterministic, universal, and unambiguous.

Theorem 22 — Invariance Under Fundamental Isomorphisms

Formal Context

Let:

- S be a non-empty set of fundamental elements;
- $F: S \times S \rightarrow \mathbb{R}$ the fundamental structure (distances, correlations, structural functions);
- $\sigma: S \rightarrow S$ an isomorphism of the fundamental domain (permutation, renumbering, relabeling).

Define the renumbered structure:

$$F^\sigma(x, y) := F(\sigma(x), \sigma(y)).$$

Let $\{F_\lambda\}_{\lambda \geq 0}$ be a regularization family of F , and let:

$$E := \lim_{\lambda \rightarrow \infty} F_\lambda$$

be the emergent domain (Theorems 17–18).

Similarly:

$$E^\sigma := \lim_{\lambda \rightarrow \infty} (F^\sigma)_\lambda.$$

The regularized emergent domain is:

$$E_{\text{reg}} = R(E), \quad E_{\text{reg}}^\sigma = R(E^\sigma),$$

where $R: E \rightarrow E_{\text{reg}}$ is the regularization operator.

The complexity threshold is:

$$\Theta_k: E_{\text{reg}} \rightarrow E_{\text{reg}}, \quad \Theta_k(E_{\text{reg}}^\sigma) = \Theta_k(E_{\text{reg}})^\sigma.$$

The physical projection is:

$$\Phi_k: E_{\text{reg}} \rightarrow D_{\text{phys}}(k).$$

Assumptions

1. Isotropy of the regularization

The regularization does not depend on the labels of the fundamental elements:

$$(F^\sigma)_\lambda = (F_\lambda)^\sigma, \quad \forall \lambda.$$

2. Independence from parametrization

If λ is renumbered through a monotone transformation $\tau(\lambda)$, the family remains equivalent:

$$F_{\tau(\lambda)} \sim F_\lambda.$$

3. Uniqueness of the emergent domain (Theorem 18)

The emergent limit is unique:

$$E = \lim_{\lambda \rightarrow \infty} F_\lambda.$$

4. Isotropy of the threshold and of the physical projection

The threshold and the projection are invariant under isomorphisms of the emergent domain:

$$\Theta_k(E_{\text{reg}}^\sigma) = \Theta_k(E_{\text{reg}}), \quad \Phi_k(E_{\text{reg}}^\sigma) = \Phi_k(E_{\text{reg}}).$$

Statement

The PSE is completely invariant under isomorphisms of the fundamental domain:

$$\Phi_k(E_{\text{reg}}^\sigma) = \Phi_k(E_{\text{reg}}).$$

In particular, the physical domain is invariant:

$$D_{\text{phys}}^\sigma(k) = D_{\text{phys}}(k).$$

Proof

1. Renumbering of the fundamental structure

By definition:

$$F^\sigma(x, y) = F(\sigma, (x)\sigma(y)).$$

This is a mere relabeling of the fundamental elements: it does not modify the mathematical structure, only its representation.

2. Isotropic regularization

By assumption:

$$(F^\sigma)_\lambda = (F_\lambda)^\sigma.$$

Thus the regularization procedure is equivariantly compatible with the isomorphism σ : it introduces no dependence on labels.

3. Passage to the emergent limit

From Theorem 17 (existence of the limit) and Theorem 18 (uniqueness of the emergent limit):

$$E = \lim_{\lambda \rightarrow \infty} F_\lambda, \quad E^\sigma = \lim_{\lambda \rightarrow \infty} (F^\sigma)_\lambda.$$

Using isotropy of the regularization:

$$E^\sigma = \lim_{\lambda \rightarrow \infty} (F^\sigma)_\lambda = \lim_{\lambda \rightarrow \infty} (F_\lambda)^\sigma = \left(\lim_{\lambda \rightarrow \infty} F_\lambda \right)^\sigma = E^\sigma.$$

Thus E^σ is simply a relabeling of the emergent domain E .

4. Regularization of the emergent domain

Since R is isotropic:

$$E_{\text{reg}}^\sigma = R(E^\sigma) = R(E)^\sigma = E_{\text{reg}}^\sigma.$$

Hence:

$$E_{\text{reg}}^\sigma = E_{\text{reg}}.$$

5. Isotropic threshold

Since Θ_k is isotropic:

$$\Theta_k(E_{\text{reg}}^\sigma) = \Theta_k(E_{\text{reg}}).$$

6. Isotropic projection

By assumption:

$$\Phi_k(E_{\text{reg}}^\sigma) = \Phi_k(E_{\text{reg}}).$$

Therefore the physical domain is invariant:

$$D_{\text{phys}}^\sigma(k) = \Phi_k(E_{\text{reg}}^\sigma) = \Phi_k(E_{\text{reg}}) = D_{\text{phys}}(k).$$

Conclusion

The PSE is completely invariant under fundamental isomorphisms:

- renumberings,
- permutations,
- relabelings,
- arbitrary reorderings.

Such transformations do **not** alter:

- the emergent domain E ,
- the regularized emergent domain E_{reg} ,
- the physical domain $D_{\text{phys}}(k)$.

The entire ontological chain of the PSE:

$$F \rightarrow E \rightarrow E_{\text{reg}} \xrightarrow{\Theta_k} E_{\text{reg}} \xrightarrow{\Phi_k} D_{\text{phys}}(k)$$

is invariant under isomorphisms.

The emergent physics depends solely on the fundamental structure F , not on its representation.

Theorem 23 — Stability of the Physical Projection

Formal Context

In the PSE, the physical domain of level k is obtained through the composite map:

$$F \xrightarrow{\Psi} E \xrightarrow{R} E_{\text{reg}} \xrightarrow{\Theta_k} E_{\text{reg}} \xrightarrow{\Phi_k} D_{\text{phys}}(k),$$

where:

- $E = \Psi(F)$ is the emergent domain,
- $E_{\text{reg}} = R(E)$ is the regularized emergent domain,

- $\Theta_k(E_{\text{reg}})$ is the structural complexity threshold,
- $D_{\text{phys}}(k) = \Phi_k(\Theta_k(E_{\text{reg}}))$ is the physical domain.

Let:

$$E = \Psi(F), E' = \Psi(F'),$$

be the emergent domains obtained from two fundamental structures $F, F' \in \mathcal{F}$.

Let:

$$E_{\text{reg}} = R(E), E'_{\text{reg}} = R(E'),$$

be the corresponding regularized domains.

Finally, let:

$$D_{\text{phys}}(k) = \Phi_k(\Theta_k(E_{\text{reg}})), D'_{\text{phys}}(k) = \Phi_k(\Theta_k(E'_{\text{reg}})).$$

Assumptions

1. Stability of the emergent domain (Theorem 19)

There exists a constant $K_\Psi > 0$ such that:

$$\|E - E'\|_E \leq K_\Psi \|F - F'\|_{\mathcal{F}}.$$

2. Stability of the regularization

There exists a constant $K_R > 0$ such that:

$$\|E_{\text{reg}} - E'_{\text{reg}}\|_{E_{\text{reg}}} \leq K_R \|E - E'\|_E.$$

3. Stability of the complexity threshold

There exists a constant $K_\Theta > 0$ such that:

$$\|\Theta_k(E_{\text{reg}}) - \Theta_k(E'_{\text{reg}})\|_{E_{\text{reg}}} \leq K_\Theta \|E_{\text{reg}} - E'_{\text{reg}}\|_{E_{\text{reg}}}.$$

4. Continuity and contractivity of the physical projection

The map Φ_k is continuous and contractive: there exists $K_\Phi < 1$ such that:

$$\|\Phi_k(X) - \Phi_k(Y)\|_{\mathcal{P}} \leq K_\Phi \|X - Y\|_{E_{\text{reg}}}.$$

5. Monotonicity

If $X_1 \leq X_2$, then:

$$\Phi_k(X_1) \leq \Phi_k(X_2).$$

6. Isotropy

For every isomorphism σ :

$$\Phi_k(E_{\text{reg}}^\sigma) = \Phi_k(E_{\text{reg}}).$$

Statement

The physical projection of the PSE is stable: small variations in the fundamental structure produce proportionally small variations in the physical domain.

Formally, there exists a constant $K > 0$ such that:

$$\| D_{\text{phys}}(k) - D'_{\text{phys}}(k) \|_{\mathcal{P}} \leq K \| F - F' \|_{\mathcal{F}}.$$

In particular, the composite map:

$$F \mapsto D_{\text{phys}}(k)$$

is **Lipschitz-stable**.

Proof

1. Stability of the emergent domain

From Theorem 19:

$$\| E - E' \|_E \leq K_{\Psi} \| F - F' \|_{\mathcal{F}}.$$

Thus the map $F \mapsto E$ does not amplify microscopic perturbations.

2. Stability of the regularization

By assumption:

$$\| E_{\text{reg}} - E'_{\text{reg}} \|_{E_{\text{reg}}} \leq K_R \| E - E' \|_E.$$

3. Stability of the complexity threshold

By assumption:

$$\| \Theta_k(E_{\text{reg}}) - \Theta_k(E'_{\text{reg}}) \|_{E_{\text{reg}}} \leq K_{\Theta} \| E_{\text{reg}} - E'_{\text{reg}} \|_{E_{\text{reg}}}.$$

4. Contractivity of the physical projection

By assumption:

$$\| D_{\text{phys}}(k) - D'_{\text{phys}}(k) \|_{\mathcal{P}} \leq K_{\Phi} \| \Theta_k(E_{\text{reg}}) - \Theta_k(E'_{\text{reg}}) \|_{E_{\text{reg}}}.$$

5. Composition of the estimates

Combining the four inequalities:

$$\| D_{\text{phys}}(k) - D'_{\text{phys}}(k) \|_{\mathcal{P}} \leq K_{\Phi} K_{\Theta} K_R K_{\Psi} \| F - F' \|_{\mathcal{F}}.$$

Define:

$$K := K_{\Phi} K_{\Theta} K_R K_{\Psi}.$$

Since:

- $K_{\Psi} > 0$,
- $K_R > 0$,
- $K_{\Theta} > 0$,
- $K_{\Phi} < 1$,

it follows that:

$$0 < K < K_\Psi.$$

6. Role of isotropy

For every isomorphism σ :

$$E^\sigma = E \Rightarrow E_{\text{reg}}^\sigma = E_{\text{reg}} \Rightarrow D_{\text{phys}}^\sigma(k) = D_{\text{phys}}(k).$$

Thus stability is invariant under renumberings, permutations, and relabelings of the emergent domain.

Conclusion

The physical projection of the PSE is:

- stable,
- continuous,
- monotone,
- contractive,
- isotropic.

In particular:

$$\| D_{\text{phys}}(k) - D'_{\text{phys}}(k) \|_{\mathcal{P}} \leq K \| F - F' \|_{\mathcal{F}}$$

The emergent physics of the PSE:

- does not amplify microscopic perturbations,
- does not depend on technical choices,
- is robust under variations of the fundamental structure,
- is invariant under isomorphisms.

Theorem 24 — Functional Determination of the Physical Domain

Formal Context

In the PSE, the complete ontological chain is:

$$F \xrightarrow{\Psi} E \xrightarrow{R} E_{\text{reg}} \xrightarrow{\Theta_k} E_{\text{reg}} \xrightarrow{\Phi_k} D_{\text{phys}}(k),$$

where:

- $F \in \mathcal{F}$ is the fundamental structure;
- $E = \Psi(F) \in E$ is the emergent domain;
- $E_{\text{reg}} = R(E) \in E_{\text{reg}}$ is the regularized emergent domain;
- $\Theta_k: E_{\text{reg}} \rightarrow E_{\text{reg}}$ is the structural complexity threshold of level k ;
- $\Phi_k: E_{\text{reg}} \rightarrow D_{\text{phys}}(k)$ is the physical projection of level k ;
- $D_{\text{phys}}(k)$ is the physical domain corresponding to the projection class k .

The overall physical map is therefore:

$$F \mapsto D_{\text{phys}}(k) \text{ via } \Phi_k \circ \Theta_k \circ R \circ \Psi.$$

The previous theorems have established:

- existence of the emergent limit (Theorem 17),
- uniqueness of the emergent domain (Theorem 18),
- stability of the emergent domain (Theorem 19),
- universality of the physical domain (Theorem 21),
- invariance under fundamental isomorphisms (Theorem 22),
- stability of the physical projection (Theorem 23).

In this context, the physical domain of level k is defined as:

$$D_{\text{phys}}(k) = \Phi_k \left(\Theta_k \left(R \left(\Psi(F) \right) \right) \right).$$

We now require the theorem that closes the ontological chain:

The overall map $F \mapsto D_{\text{phys}}(k)$ is completely determined by F . No residual freedom exists.

Assumptions

Let:

- $\Psi: F \rightarrow E$ be the emergent map,
- $R: E \rightarrow E_{\text{reg}}$ the regularization operator,
- $\Theta_k: E_{\text{reg}} \rightarrow E_{\text{reg}}$ the complexity threshold of level k ,
- $\Phi_k: E_{\text{reg}} \rightarrow D_{\text{phys}}(k)$ the physical projection.

We assume:

1. Uniqueness of the emergent domain

For every F , the emergent limit is unique:

$$E = \Psi(F).$$

2. Universality of the physical domain

For every E_{reg} , the projection is unique:

$$D_{\text{phys}}(k) = \Phi_k(E_{\text{reg}}).$$

3. Continuity and contractivity

All maps are continuous and contractive:

$$\| \Psi(F) - \Psi(F') \|_E \leq K_\Psi \| F - F' \|_{\mathcal{F}},$$

$$\| R(E) - R(E') \|_{E_{\text{reg}}} \leq K_R \| E - E' \|_E,$$

$$\begin{aligned}\|\Theta_k(E_{\text{reg}}) - \Theta_k(E'_{\text{reg}})\|_{E_{\text{reg}}} &\leq K_\Theta \|E_{\text{reg}} - E'_{\text{reg}}\|_{E_{\text{reg}}}, \\ \|\Phi_k(E_{\text{reg}}) - \Phi_k(E'_{\text{reg}})\|_{\mathcal{P}} &\leq K_\Phi \|E_{\text{reg}} - E'_{\text{reg}}\|_{E_{\text{reg}}},\end{aligned}$$

with:

$$K_\Psi, K_R, K_\Theta, K_\Phi < 1.$$

4. Isotropy

For every fundamental isomorphism σ :

$$\Psi(F^\sigma) = \Psi(F)^\sigma, R(E^\sigma) = R(E)^\sigma, \Theta_k(E_{\text{reg}}^\sigma) = \Theta_k(E_{\text{reg}})^\sigma, \Phi_k(E_{\text{reg}}^\sigma) = \Phi_k(E_{\text{reg}}).$$

5. Non-invertibility

Φ_k is contractive and non-invertible: no two distinct physical domains correspond to the same regularized emergent domain.

Statement

The overall PSE map:

$$\Phi_k \circ \Theta_k \circ R \circ \Psi: F \rightarrow D_{\text{phys}}(k)$$

is completely determined by the fundamental structure F .

In particular:

$$F = F' \Rightarrow D_{\text{phys}}(k) = D'_{\text{phys}}(k),$$

and more generally:

$$D_{\text{phys}}(k) = (\Phi_k \circ \Theta_k \circ R \circ \Psi)(F)$$

is the **unique** physical domain compatible with F .

No alternative choices, conventions, hidden parameters, or bifurcations exist.

Proof

1. Determination of the emergent domain

From Theorem 18:

$$\Psi(F) = E \text{ is unique.}$$

Thus no two distinct emergent domains can arise from F .

2. Determination of the regularized emergent domain

Since R is continuous, contractive, and isotropic:

$$E_{\text{reg}} = R(E) \text{ is unique.}$$

3. Determination of the complexity threshold

Since Θ_k is continuous, contractive, and isotropic:

$$\Theta_k(E_{\text{reg}}) \text{ is unique.}$$

4. Determination of the physical domain

From Theorem 21 (universality of the physical domain):

$$\Phi_k(E_{\text{reg}}) = D_{\text{phys}}(k) \text{ is unique.}$$

Thus no two distinct physical domains can arise from E_{reg} .

5. Composition of uniqueness results

Let F be a fundamental structure.

Then:

$$E = \Psi(F) \text{ unique,}$$

$$E_{\text{reg}} = R(E) \text{ unique,}$$

$$\Theta_k(E_{\text{reg}}) \text{ unique,}$$

$$D_{\text{phys}}(k) = \Phi_k(\Theta_k(E_{\text{reg}})) \text{ unique.}$$

Therefore:

$$D_{\text{phys}}(k) = (\Phi_k \circ \Theta_k \circ R \circ \Psi)(F) \text{ is unique.}$$

6. Elimination of residual freedom

Assume, for contradiction, that two distinct physical domains $D_{\text{phys}}(k)$ and $D'_{\text{phys}}(k)$ can be obtained from F :

$$D_{\text{phys}}(k) = (\Phi_k \circ \Theta_k \circ R \circ \Psi)(F),$$

$$D'_{\text{phys}}(k) = (\Phi_k \circ \Theta_k \circ R \circ \Psi')(F),$$

with $\Psi' \neq \Psi$.

But from Theorem 18:

$$\Psi(F) = \Psi'(F).$$

Hence:

$$D_{\text{phys}}(k) = \Phi_k(\Theta_k(R(\Psi(F)))) = \Phi_k(\Theta_k(R(\Psi'(F)))) = D'_{\text{phys}}(k).$$

Contradiction.

7. Invariance under isomorphisms

For every isomorphism σ :

$$F^\sigma = F \Rightarrow E^\sigma = E \Rightarrow E_{\text{reg}}^\sigma = E_{\text{reg}} \Rightarrow \Theta_k(E_{\text{reg}}^\sigma) = \Theta_k(E_{\text{reg}}) \Rightarrow D_{\text{phys}}^\sigma(k) = D_{\text{phys}}(k).$$

Thus the determination is isomorphism-invariant.

Conclusion

The overall PSE map:

$$F \longrightarrow D_{\text{phys}}(k)$$

is:

- deterministic,
- unique,
- isotropic,
- stable,
- contractive,
- unambiguous,
- non-bifurcating.

In particular:

$D_{\text{phys}}(k) = (\Phi_k \circ \Theta_k \circ R \circ \Psi)(F)$ is completely determined by F .
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5. Physical Theorems of Emergence

Theorem 25 — Existence of the Emergent Physical Metric

Formal Context

In the PSE, the emergent domain is:

$$E = (D_S^{(e)}, C_e),$$

where:

- $D_S^{(e)}: X \times X \rightarrow \mathbb{R}_{\geq 0}$ is the emergent distance (Theorems 11–12),
- $C_e: X \times X \rightarrow \mathbb{R}$ is the emergent correlation (Theorem 13),

both of which are:

- continuous,
- stable,
- unique,
- isotropic,
- deterministic (Theorems 17–24).

The regularized emergent domain is:

$$E_{\text{reg}} = R(E),$$

where $R: E \rightarrow E_{\text{reg}}$ is the regularization operator.

The structural complexity threshold is:

$$\Theta_k: E_{\text{reg}} \rightarrow E_{\text{reg}}.$$

The physical domain of level k is obtained through the projection:

$$D_{\text{phys}}(k) = \Phi_k(\Theta_k(E_{\text{reg}})).$$

The emergent physical metric $g_{\mu\nu}$ is one of the components of the physical domain:

$$D_{\text{phys}}(k) = (g_{\mu\nu}, \text{ other physical fields}).$$

Assumptions

1. Continuity and regularity of the emergent domain

The functions $D_S^{(e)}$ and C_e are continuous on $X \times X$.

2. Metric–correlation compatibility (Theorem 13)

There exists a continuous function $F: [0, \infty) \rightarrow \mathbb{R}$ such that:

$$C_e(x, y) = F(D_S^{(e)}(x, y)).$$

3. Regularity of the physical projection

The projection Φ_k is:

- monotone,
- continuous,
- compressive,
- isotropic,
- deterministic.

4. Geometric structure of the physical domain

The physical metric is obtained through an operator:

$$\Gamma: E_{\text{reg}} \rightarrow \mathcal{G},$$

where $\mathcal{G}$ is the space of physical metrics (symmetric rank-2 tensors).

Assume that Γ is:

- continuous,
- monotone,
- regular,
- isotropic,
- non-amplifying.

Statement

There exists an emergent physical metric:

$$g_{\mu\nu} = \Gamma(E_{\text{reg}}),$$

and it is:

- well-defined,
- continuous,
- regular,
- isotropic,
- stable,
- unique.

In particular:

$$g_{\mu\nu}(x) = G\left(D_S^{(e)}, (x, \cdot) C_e(x, \cdot)\right),$$

for a continuous function G determined by the PSE.

Proof

1. Existence of the emergent domain

From Theorems 17–18:

$$E = (D_S^{(e)}, C_e)$$

exists, is unique, and is continuous.

2. Radial compatibility

From Theorem 13:

$$C_e(x, y) = F(D_S^{(e)}(x, y)),$$

with F continuous.

Thus the entire emergent domain is determined by $D_S^{(e)}$.

3. Regularization

Since R is continuous and isotropic:

$$E_{\text{reg}} = R(E)$$

is well-defined, continuous, and unique.

4. Continuity of the metric operator

The metric operator Γ is continuous:

$$E_{\text{reg},n} \rightarrow E_{\text{reg}} \Rightarrow \Gamma(E_{\text{reg},n}) \rightarrow \Gamma(E_{\text{reg}}).$$

Since E_{reg} is continuous, $\Gamma(E_{\text{reg}})$ is continuous.

5. Regularity and isotropy

Since Γ is isotropic:

$$\Gamma(E_{\text{reg}}^\sigma) = \Gamma(E_{\text{reg}}),$$

for every isomorphism σ of the emergent domain.

Thus the metric does not depend on renumberings or permutations.

6. Stability

From Theorem 19:

$$\|E - E'\| \leq K_\Psi \|F - F'\|.$$

Since R , Θ_k , and Γ are compressive:

$$\|\Gamma(E_{\text{reg}}) - \Gamma(E'_{\text{reg}})\| \leq K_\Gamma K_\Theta K_R K_\Psi \|F - F'\|.$$

Thus the metric is stable.

7. Uniqueness

From Theorem 24, the composite map:

$$F \mapsto D_{\text{phys}}(k)$$

is deterministic.

Therefore:

$$g_{\mu\nu} = \Gamma(E_{\text{reg}})$$

is unique.

Conclusion

There exists an emergent physical metric:

$$\boxed{g_{\mu\nu} = \Gamma(E_{\text{reg}})}$$

which is:

- continuous,
- regular,
- stable,
- isotropic,
- unique,

and fully determined by the fundamental structure F

Theorem 26 — Regularity and Continuity of the Emergent Physical Metric

Formal Context

From Theorem 25, an emergent physical metric exists:

$$g_{\mu\nu} = \Gamma(E_{\text{reg}}),$$

where:

- $E = (D_S^{(e)}, C_e)$ is the emergent domain,
- $E_{\text{reg}} = R(E)$ is the regularized emergent domain,
- $\Gamma: E_{\text{reg}} \rightarrow \mathcal{G}$ is the metric operator,
- $\mathcal{G}$ is the space of physical metrics (symmetric rank-2 tensors).

We must now prove that the physical metric is **regular**, meaning:

- continuous,
- uniformly continuous,
- Lipschitz-stable,
- non-amplifying,
- isotropic,

- deterministic.

This theorem is essential because the metric is the foundation of all emergent physics: geometry, fields, dynamics, and observables.

Assumptions

1. Continuity of the emergent domain

$$D_S^{(e)}, C_e \in C^0(X \times X).$$

2. Metric–correlation compatibility (Theorem 13)

There exists a continuous function $F: [0, \infty) \rightarrow \mathbb{R}$ such that:

$$C_e(x, y) = F\left(D_S^{(e)}(x, y)\right).$$

3. Continuity of the metric operator

$$E_{\text{reg},n} \rightarrow E_{\text{reg}} \Rightarrow \Gamma(E_{\text{reg},n}) \rightarrow \Gamma(E_{\text{reg}}).$$

4. Contractivity of the metric operator

There exists $K_\Gamma < 1$ such that:

$$\|\Gamma(E_{\text{reg}}) - \Gamma(E'_{\text{reg}})\|_{\mathcal{G}} \leq K_\Gamma \|E_{\text{reg}} - E'_{\text{reg}}\|_{E_{\text{reg}}}.$$

5. Isotropy

For every isomorphism σ :

$$\Gamma(E_{\text{reg}}^\sigma) = \Gamma(E_{\text{reg}}).$$

6. Emergent stability (Theorem 19)

$$\|E - E'\|_E \leq K_\Psi \|F - F'\|_{\mathcal{F}}.$$

Statement

The emergent physical metric:

$$g_{\mu\nu} = \Gamma(E_{\text{reg}})$$

is:

- continuous on the entire physical domain,
- uniformly continuous,
- Lipschitz-stable with respect to variations in E_{reg} ,
- Lipschitz-stable with respect to variations in F ,
- isotropic,
- deterministic,
- non-amplifying.

In particular:

$$\|g_{\mu\nu} - g'_{\mu\nu}\|_{\mathcal{G}} \leq K_{\Gamma} K_{\Psi} \|F - F'\|_{\mathcal{F}}.$$

Proof

1. Continuity of the metric

Since E is continuous (Theorems 17–18) and Γ is continuous:

$$E_{\text{reg},n} \rightarrow E_{\text{reg}} \Rightarrow \Gamma(E_{\text{reg},n}) \rightarrow \Gamma(E_{\text{reg}}).$$

Thus:

$$g_{\mu\nu} = \Gamma(E_{\text{reg}}) \in C^0.$$

2. Uniform continuity

The emergent domain is defined on a compact set (Theorem 15). A continuous function on a compact set is uniformly continuous.

Therefore:

$$g_{\mu\nu} \text{ is uniformly continuous.}$$

3. Lipschitz stability with respect to E_{reg}

From contractivity:

$$\|\Gamma(E_{\text{reg}}) - \Gamma(E'_{\text{reg}})\|_{\mathcal{G}} \leq K_{\Gamma} \|E_{\text{reg}} - E'_{\text{reg}}\|_{E_{\text{reg}}}.$$

Thus the metric does not amplify variations of the emergent domain.

4. Lipschitz stability with respect to F

From Theorem 19:

$$\|E - E'\|_E \leq K_{\Psi} \|F - F'\|_{\mathcal{F}}.$$

Since R , Θ_k and Γ are contractive:

$$\|g_{\mu\nu} - g'_{\mu\nu}\|_{\mathcal{G}} \leq K_{\Gamma} K_{\Psi} \|F - F'\|_{\mathcal{F}}.$$

Thus the metric is stable with respect to the fundamental structure.

5. Isotropy

For every isomorphism σ :

$$E_{\text{reg}}^{\sigma} = E_{\text{reg}} \Rightarrow \Gamma(E_{\text{reg}}^{\sigma}) = \Gamma(E_{\text{reg}}).$$

Hence:

$$g_{\mu\nu}^{\sigma} = g_{\mu\nu}.$$

The metric is invariant under renumberings, permutations, and relabelings.

6. Functional determination

From Theorem 24:

$$D_{\text{phys}}(k) = (\Phi_k \circ \Theta_k \circ R \circ \Psi)(F)$$

is unique.

Since $g_{\mu\nu}$ is a component of $D_{\text{phys}}(k)$:

$g_{\mu\nu}$ is unique.

Conclusion

The emergent physical metric:

$$g_{\mu\nu} = \Gamma(E_{\text{reg}})$$

is:

- continuous,
- uniformly continuous,
- stable,
- isotropic,
- deterministic,
- non-amplifying,

and completely determined by the fundamental structure F .

Theorem 27 — Existence of Emergent Physical Fields

Formal Context

From Theorems 25 and 26 we know that:

- an emergent physical metric exists

$$g_{\mu\nu} = \Gamma(E_{\text{reg}}),$$

- the metric is continuous, stable, isotropic, and deterministic.

We now introduce the **emergent physical fields**, i.e., all physical quantities other than the metric:

- scalar fields,
- vector fields,
- tensor fields,
- densities,
- potentials,
- currents,
- derived geometric structures.

Formally, let:

$$\mathcal{F}_{\text{phys}} = \{\phi, A_\mu, T_{\mu\nu}, J_\mu, \rho, \dots\}$$

be the space of physical fields.

In the PSE, physical fields arise from a map:

$$\Xi: E_{\text{reg}} \rightarrow \mathcal{F}_{\text{phys}},$$

which takes the regularized emergent domain and produces all physical fields.

The physical domain of level k is:

$$D_{\text{phys}}(k) = (g_{\mu\nu}, \mathcal{F}_{\text{phys}}) = \Phi_k(\Theta_k(E_{\text{reg}})).$$

Assumptions

1. Continuity of the emergent domain

$$E = (D_S^{(e)}, C_e) \in C^0(X \times X).$$

2. Metric–correlation compatibility

$$C_e(x, y) = F(D_S^{(e)}(x, y)),$$

with F continuous.

3. Continuity of the metric

$$g_{\mu\nu} = \Gamma(E_{\text{reg}}) \in C^0.$$

4. Geometric structure of the physical domain

$$D_{\text{phys}}(k) = (g_{\mu\nu}, \mathcal{F}_{\text{phys}}).$$

5. Continuity of the field operator

The field operator Ξ is:

- continuous,
- monotone,
- isotropic,
- compressive,
- deterministic.

Formally:

$$\|\Xi(E_{\text{reg}}) - \Xi(E'_{\text{reg}})\|_{\mathcal{F}_{\text{phys}}} \leq K_{\Xi} \|E_{\text{reg}} - E'_{\text{reg}}\|_{E_{\text{reg}}}, \quad K_{\Xi} < 1.$$

6. Emergent stability (Theorem 19)

$$\|E - E'\|_E \leq K_{\Psi} \|F - F'\|_{\mathcal{F}}.$$

7. Functional determination (Theorem 24)

The composite map:

$$F \mapsto D_{\text{phys}}(k)$$

is unique and deterministic.

Statement

Emergent physical fields exist:

$$\mathcal{F}_{\text{phys}} = \Xi(E_{\text{reg}}),$$

and these fields are:

- well-defined,
- continuous,
- regular,
- stable,
- isotropic,
- deterministic,
- non-amplifying,
- completely determined by the fundamental structure F .

In particular:

$$\|\Xi(E_{\text{reg}}) - \Xi(E'_{\text{reg}})\|_{\mathcal{F}_{\text{phys}}} \leq K_{\Xi} K_{\Psi} \|F - F'\|_{\mathcal{F}}.$$

Proof

1. Continuity of the emergent domain

From Theorems 17–18:

$$E \in C^0.$$

2. Continuity of the metric

From Theorem 26:

$$g_{\mu\nu} = \Gamma(E_{\text{reg}}) \in C^0.$$

The metric is continuous and regular.

3. Definition of physical fields

Physical fields are defined through:

$$\Xi: E_{\text{reg}} \rightarrow \mathcal{F}_{\text{phys}}.$$

Since Ξ is continuous:

$$E_{\text{reg},n} \rightarrow E_{\text{reg}} \Rightarrow \Xi(E_{\text{reg},n}) \rightarrow \Xi(E_{\text{reg}}).$$

Thus:

$$\Xi(E_{\text{reg}}) \in C^0.$$

4. Stability with respect to the regularized emergent domain

By contractivity:

$$\| \Xi(E_{\text{reg}}) - \Xi(E'_{\text{reg}}) \|_{\mathcal{F}_{\text{phys}}} \leq K_{\Xi} \| E_{\text{reg}} - E'_{\text{reg}} \|_{E_{\text{reg}}}.$$

Thus the fields do not amplify variations of the emergent domain.

5. Stability with respect to the fundamental structure

From Theorem 19:

$$\| E - E' \|_E \leq K_{\Psi} \| F - F' \|_{\mathcal{F}}.$$

Since R , Θ_k and Ξ are contractive:

$$\| \Xi(E_{\text{reg}}) - \Xi(E'_{\text{reg}}) \|_{\mathcal{F}_{\text{phys}}} \leq K_{\Xi} K_{\Psi} \| F - F' \|_{\mathcal{F}}.$$

Thus the fields are stable with respect to microscopic variations of F .

6. Isotropy

For every isomorphism σ :

$$E_{\text{reg}}^{\sigma} = E_{\text{reg}} \Rightarrow \Xi(E_{\text{reg}}^{\sigma}) = \Xi(E_{\text{reg}}).$$

Thus the fields are invariant under renumberings, permutations, and relabelings.

7. Functional determination

From Theorem 24:

$$D_{\text{phys}}(k) = (\Phi_k \circ \Theta_k \circ R \circ \Psi)(F)$$

is unique.

Since the fields are a component of $D_{\text{phys}}(k)$:

$$\Xi(E_{\text{reg}}) \text{ is unique.}$$

Conclusion

Emergent physical fields exist:

$$\mathcal{F}_{\text{phys}} = \Xi(E_{\text{reg}})$$

and they are:

- continuous,
- regular,
- stable,
- isotropic,
- deterministic,
- non-amplifying,

and fully determined by the fundamental structure F

Theorem 28 — Stability of Emergent Physical Fields

Formal Context

From Theorem 27 we know that emergent physical fields exist:

$$\mathcal{F}_{\text{phys}} = \Xi(E_{\text{reg}}),$$

where:

- $E = (D_S^{(e)}, C_e)$ is the emergent domain,
- $\Xi: E_{\text{reg}} \rightarrow \mathcal{F}_{\text{phys}}$ is the physical-field operator.

We must now prove that emergent physical fields are **stable**, meaning:

- they do not amplify microscopic perturbations,
- they are Lipschitz-stable with respect to variations in E_{reg} ,
- they are Lipschitz-stable with respect to variations in F ,
- they are isotropic,
- they are deterministic,
- they are regular.

This theorem is essential because it guarantees that emergent physics is robust: small variations in the fundamental structure do not produce instabilities in physical fields.

Assumptions

1. Emergent stability (Theorem 19)

There exists $K_\Psi > 0$ such that:

$$\|E - E'\|_E \leq K_\Psi \|F - F'\|_{\mathcal{F}}.$$

2. Continuity and contractivity of the field operator

There exists $K_\Xi < 1$ such that:

$$\|\Xi(E_{\text{reg}}) - \Xi(E'_{\text{reg}})\|_{\mathcal{F}_{\text{phys}}} \leq K_\Xi \|E_{\text{reg}} - E'_{\text{reg}}\|_{E_{\text{reg}}}.$$

3. Continuity and regularity of the emergent domain

$$E \in C^0(X \times X).$$

4. Isotropy

For every isomorphism σ :

$$\Xi(E_{\text{reg}}^\sigma) = \Xi(E_{\text{reg}}).$$

5. Functional determination (Theorem 24)

The composite map:

$$F \mapsto D_{\text{phys}}(k)$$

is unique and deterministic.

Statement

The emergent physical fields:

$$\mathcal{F}_{\text{phys}} = \Xi(E_{\text{reg}})$$

are:

- stable,
- Lipschitz-continuous with respect to E_{reg} ,
- Lipschitz-continuous with respect to F ,
- isotropic,
- deterministic,
- non-amplifying,
- regular.

In particular:

$$\| \Xi(E_{\text{reg}}) - \Xi(E'_{\text{reg}}) \|_{\mathcal{F}_{\text{phys}}} \leq K_{\Xi} K_{\Psi} \| F - F' \|_{\mathcal{F}}.$$

Proof

1. Stability with respect to the emergent domain

By contractivity of the field operator:

$$\| \Xi(E_{\text{reg}}) - \Xi(E'_{\text{reg}}) \|_{\mathcal{F}_{\text{phys}}} \leq K_{\Xi} \| E_{\text{reg}} - E'_{\text{reg}} \|_{E_{\text{reg}}}.$$

Thus the fields do not amplify variations of the emergent domain.

2. Stability with respect to the fundamental structure

From Theorem 19:

$$\| E - E' \|_E \leq K_{\Psi} \| F - F' \|_{\mathcal{F}}.$$

Composing with contractivity of R , Θ_k and Ξ :

$$\| \Xi(E_{\text{reg}}) - \Xi(E'_{\text{reg}}) \|_{\mathcal{F}_{\text{phys}}} \leq K_{\Xi} K_{\Psi} \| F - F' \|_{\mathcal{F}}.$$

Thus the fields are stable with respect to microscopic variations of F .

3. Continuity and regularity

Since:

- E is continuous,
- Ξ is continuous,

it follows that:

$$\Xi(E_{\text{reg}}) \in \mathcal{C}^0.$$

The physical fields are regular and present no discontinuities.

4. Isotropy

For every isomorphism σ :

$$E_{\text{reg}}^\sigma = E_{\text{reg}} \Rightarrow \Xi(E_{\text{reg}}^\sigma) = \Xi(E_{\text{reg}}).$$

Thus the fields are invariant under renumberings, permutations, and relabelings.

5. Functional determination

From Theorem 24:

$$D_{\text{phys}}(k) = (\Phi_k \circ \Theta_k \circ R \circ \Psi)(F)$$

is unique.

Since the fields are a component of $D_{\text{phys}}(k)$:

$$\Xi(E_{\text{reg}}) \text{ is unique.}$$

Conclusion

The emergent physical fields:

$$\mathcal{F}_{\text{phys}} = \Xi(E_{\text{reg}})$$

are:

- stable,
- regular,
- isotropic,
- deterministic,
- non-amplifying,

and completely determined by the fundamental structure F .

Theorem 29 — Functional Determination of Physical Quantities

Formal Context

From Theorems 25–28 we have established that:

- an emergent physical metric exists

$$g_{\mu\nu} = \Gamma(E_{\text{reg}}),$$

- emergent physical fields exist

$$\mathcal{F}_{\text{phys}} = \Xi(E_{\text{reg}}),$$

- both the metric and the fields are continuous, stable, isotropic, and deterministic.

We must now prove that **all observable physical quantities**:

- physical scalars,
- densities,
- currents,

- geometric invariants,
- curvature tensors,
- energetic quantities,
- dynamical quantities,

are deterministic functions of the physical domain, and therefore of the fundamental structure F .

Formally, let:

$$\mathcal{Q}_{\text{phys}} = \{Q_1, Q_2, \dots\}$$

be the space of observable physical quantities.

Their origin in the PSE is a map:

$$\Omega: D_{\text{phys}}(k) \rightarrow \mathcal{Q}_{\text{phys}},$$

where:

$$D_{\text{phys}}(k) = (g_{\mu\nu}, \mathcal{F}_{\text{phys}}).$$

Assumptions

1. Functional determination of the physical domain (Theorem 24)

The composite map:

$$F \mapsto D_{\text{phys}}(k)$$

is unique and deterministic.

2. Continuity and regularity of the metric (Theorem 26)

$$g_{\mu\nu} \in \mathcal{C}^0.$$

3. Continuity and stability of physical fields (Theorem 28)

$$\mathcal{F}_{\text{phys}} = \Xi(E_{\text{reg}}) \in \mathcal{C}^0.$$

4. Continuity of the physical-quantity operator

There exists an operator:

$$\Omega: D_{\text{phys}}(k) \rightarrow \mathcal{Q}_{\text{phys}}$$

which is:

- continuous,
- monotone,
- isotropic,
- compressive,
- deterministic.

Formally:

$$\| \Omega(P) - \Omega(P') \|_{\mathcal{Q}_{\text{phys}}} \leq K_{\Omega} \| P - P' \|_{\mathcal{P}}, \quad K_{\Omega} < 1.$$

5. Emergent stability (Theorem 19)

$$\| E - E' \|_E \leq K_{\Psi} \| F - F' \|_{\mathcal{F}}.$$

6. Stability of the physical projection (Theorem 23)

$$\| P - P' \|_{\mathcal{P}} \leq K_{\Phi} K_{\Psi} \| F - F' \|_{\mathcal{F}}.$$

Statement

Every emergent physical quantity:

$$Q = \Omega(P)$$

is:

- well-defined,
- continuous,
- regular,
- stable,
- isotropic,
- deterministic,
- non-amplifying,
- completely determined by the fundamental structure F .

In particular:

$$\| \Omega(P) - \Omega(P') \|_{\mathcal{Q}_{\text{phys}}} \leq K_{\Omega} K_{\Phi} K_{\Psi} \| F - F' \|_{\mathcal{F}}.$$

Proof

1. Determination of the physical domain

From Theorem 24:

$$D_{\text{phys}}(k) = (\Phi_k \circ \Theta_k \circ R \circ \Psi)(F)$$

is unique.

Thus:

P is completely determined by F .

2. Continuity and regularity of the metric

From Theorem 26:

$$g_{\mu\nu} = \Gamma(E_{\text{reg}}) \in C^0.$$

3. Continuity and regularity of physical fields

From Theorem 28:

$$\mathcal{F}_{\text{phys}} = \Xi(E_{\text{reg}}) \in \mathcal{C}^0.$$

4. Continuity of the physical-quantity operator

Since Ω is continuous:

$$P_n \rightarrow P \Rightarrow \Omega(P_n) \rightarrow \Omega(P).$$

Thus:

$$Q = \Omega(P) \in \mathcal{C}^0.$$

5. Stability with respect to the physical domain

By contractivity:

$$\| \Omega(P) - \Omega(P') \|_{\mathcal{Q}_{\text{phys}}} \leq K_{\Omega} \| P - P' \|_{\mathcal{P}}.$$

6. Stability with respect to the fundamental structure

From Theorem 23:

$$\| P - P' \|_{\mathcal{P}} \leq K_{\Phi} K_{\Psi} \| F - F' \|_{\mathcal{F}}.$$

Composing:

$$\| \Omega(P) - \Omega(P') \|_{\mathcal{Q}_{\text{phys}}} \leq K_{\Omega} K_{\Phi} K_{\Psi} \| F - F' \|_{\mathcal{F}}.$$

Thus physical quantities are stable with respect to microscopic variations of the fundamental structure.

7. Isotropy

For every isomorphism σ :

$$P^{\sigma} = P \Rightarrow \Omega(P^{\sigma}) = \Omega(P).$$

Thus physical quantities are invariant under renumberings, permutations, and relabelings.

8. Functional determination

Since:

$$P = (\Phi_k \circ \Theta_k \circ R \circ \Psi)(F)$$

is unique, and Ω is deterministic:

$$Q = \Omega(P)$$

is unique.

Conclusion

Every emergent physical quantity:

$$Q = \Omega(P)$$

is:

- continuous,
- regular,

- stable,
- isotropic,
- deterministic,
- non-amplifying,

and completely determined by the fundamental structure F .

Theorem 30 — Emergent Dynamical Coherence

Formal Context

From Theorems 25–29 we have established that:

- an emergent physical metric exists

$$g_{\mu\nu} = \Gamma(E_{\text{reg}}),$$

- emergent physical fields exist

$$\mathcal{F}_{\text{phys}} = \Xi(E_{\text{reg}}),$$

- all observable physical quantities

$$Q = \Omega(P)$$

are determined by the fundamental structure F .

We must now prove that the **emergent physical dynamics**, i.e., the temporal evolution of physical quantities, is:

- well-defined,
- continuous,
- stable,
- isotropic,
- deterministic,
- non-amplifying,
- coherent with the emergent geometric structure.

Formally, the dynamics is a map:

$$\Delta: P \rightarrow \mathcal{D},$$

where:

- $P = D_{\text{phys}}(k)$ is the physical domain,
- $\mathcal{D}$ is the space of physical dynamics (equations of motion, evolution operators, physical laws).

Assumptions

1. Functional determination of the physical domain (Theorem 24)

$$P = (\Phi_k \circ \Theta_k \circ R \circ \Psi)(F)$$

is unique.

2. Continuity and regularity of the metric (Theorem 26)

$$g_{\mu\nu} \in \mathcal{C}^0.$$

3. Continuity and stability of physical fields (Theorem 28)

$$\mathcal{F}_{\text{phys}} = \Xi(E_{\text{reg}}) \in \mathcal{C}^0.$$

4. Functional determination of physical quantities (Theorem 29)

$$Q = \Omega(P)$$

is unique and stable.

5. Continuity and contractivity of the dynamical operator

There exists an operator:

$$\Delta: P \rightarrow \mathcal{D}$$

which is:

- continuous,
- monotone,
- isotropic,
- compressive,
- deterministic.

Formally:

$$\| \Delta(P) - \Delta(P') \|_{\mathcal{D}} \leq K_{\Delta} \| P - P' \|_{\mathcal{P}}, \quad K_{\Delta} < 1.$$

6. Stability of the physical projection (Theorem 23)

$$\| P - P' \|_{\mathcal{P}} \leq K_{\Phi} K_{\Psi} \| F - F' \|_{\mathcal{F}}.$$

Statement

The emergent physical dynamics:

$$\mathcal{D}_{\text{phys}} = \Delta(P)$$

is:

- well-defined,
- continuous,
- regular,
- stable,
- isotropic,
- deterministic,

- non-amplifying,
- completely determined by the fundamental structure F .

In particular:

$$\| \Delta(P) - \Delta(P') \|_{\mathcal{D}} \leq K_{\Delta} K_{\Phi} K_{\Psi} \| F - F' \|_{\mathcal{F}}.$$

Proof

1. Determination of the physical domain

From Theorem 24:

$$P = (\Phi_k \circ \Theta_k \circ R \circ \Psi)(F)$$

is unique.

Thus:

P is completely determined by F .

2. Continuity and regularity of physical structures

From Theorem 26:

$$g_{\mu\nu} \in \mathcal{C}^0.$$

From Theorem 28:

$$\mathcal{F}_{\text{phys}} \in \mathcal{C}^0.$$

From Theorem 29:

$$Q = \Omega(P) \in \mathcal{C}^0.$$

Thus all physical structures are regular.

3. Continuity of the dynamical operator

Since Δ is continuous:

$$P_n \rightarrow P \Rightarrow \Delta(P_n) \rightarrow \Delta(P).$$

Thus:

$$\Delta(P) \in \mathcal{C}^0.$$

4. Stability with respect to the physical domain

By contractivity:

$$\| \Delta(P) - \Delta(P') \|_{\mathcal{D}} \leq K_{\Delta} \| P - P' \|_{\mathcal{P}}.$$

5. Stability with respect to the fundamental structure

From Theorem 23:

$$\| P - P' \|_{\mathcal{P}} \leq K_{\Phi} K_{\Psi} \| F - F' \|_{\mathcal{F}}.$$

Composing:

$$\| \Delta(P) - \Delta(P') \|_{\mathcal{D}} \leq K_{\Delta} K_{\Phi} K_{\Psi} \| F - F' \|_{\mathcal{F}}.$$

Thus the dynamics is stable with respect to microscopic variations of the fundamental structure.

6. Isotropy

For every isomorphism σ :

$$P^\sigma = P \Rightarrow \Delta(P^\sigma) = \Delta(P).$$

Thus the dynamics is invariant under renumberings, permutations, and relabelings.

7. Functional determination

Since:

$$P = (\Phi_k \circ \Theta_k \circ R \circ \Psi)(F)$$

is unique, and Δ is deterministic:

$$\Delta(P)$$

is unique.

Conclusion

The emergent physical dynamics:

$$\mathcal{D}_{\text{phys}} = \Delta(P)$$

is:

- continuous,
- regular,
- stable,
- isotropic,
- deterministic,
- non-amplifying,

and completely determined by the fundamental structure F .

6. Discrete Emergence Theorems

Formal Context

In the PSE, the emergent domain is:

$$E = (D_S^{(e)}, C_e),$$

and the regularized emergent domain is:

$$E_{\text{reg}} = R(E).$$

The structural complexity threshold is:

$$\Theta_k: E_{\text{reg}} \rightarrow E_{\text{reg}}.$$

The discrete emergent domain at level k is obtained through the discrete projection:

$$D_{\text{disc}}(k) = \Phi_{k,\text{disc}}(\Theta_k(E_{\text{reg}})).$$

A discrete emergent quantity is a component of $D_{\text{disc}}(k)$, interpreted as a **coherent partition** induced by the emergent correlation.

Disc1 Theorem — Emergence of Discrete Quantities

Statement

Given the discrete projection:

$$\Phi_{k,\text{disc}}: E_{\text{reg}} \rightarrow \mathcal{N},$$

there exists an emergent discrete quantity:

$$\mathcal{N} = \Phi_{k,\text{disc}}(E_{\text{reg}}),$$

which is:

- monotone,
- compressive,
- well-defined.

Conclusion

Discrete quantities emerge as **coherent partitions** induced by the regularized emergent correlation.

Disc2 Theorem — Emergent Discrete Stability

Statement

If the emergent structure is stable (Theorems 17–19), then:

$$\delta E_{\text{reg}} \rightarrow 0 \implies \delta \mathcal{N} \rightarrow 0.$$

Conclusion

Discrete stability is guaranteed by emergent stability and by the compressivity of the discrete projection.

Disc3 Theorem — Emergent Discrete Continuity

Statement

If $D_S^{(e)}$ is continuous and C_e is regular, then the discrete projection is continuous:

$$\lim_{x \rightarrow y} C_e(x, y) = 1 \Rightarrow \Phi_{k, \text{disc}}(x) = \Phi_{k, \text{disc}}(y).$$

Conclusion

Discrete continuity is interpreted as **stability of emergent partitions**.

Disc4 Theorem — Functional Determination of Discrete Quantities

Statement

Discrete emergent quantities are fully determined by the regularized emergent structure:

$$\mathcal{N} = F_{\text{disc}}(E_{\text{reg}}) = F_{\text{disc}}(D_S^{(e)}, C_e).$$

Conclusion

Discrete quantities are not fundamental: they are **effective representations** of the emergent structure.

7. Geometric Emergence Theorems

Formal Context

In the PSE, physical geometry is not fundamental: it emerges from the regularized correlational structure. The ontological chain is:

$$F \xrightarrow{\Psi} E \xrightarrow{R} E_{\text{reg}} \xrightarrow{\Theta_k} E_{\text{reg}} \xrightarrow{\Gamma} g_{\mu\nu}.$$

The emergent physical metric is defined as:

$$g_{\mu\nu} = \Gamma(E_{\text{reg}}),$$

and constitutes the geometric physical domain:

$$D_{\text{geom}}(k) = g_{\mu\nu}.$$

G1 — Emergence of the Physical Metric

Statement

Given the geometric projection:

$$\Gamma: E_{\text{reg}} \rightarrow g_{\mu\nu},$$

there exists an emergent physical metric:

$$g_{\mu\nu} = \Gamma(E_{\text{reg}}),$$

which is:

- monotone,
- compressive,
- non-invertible,
- functionally determined by the emergent structure.

Formally:

$$g_{\mu\nu} = F_{\text{geom}}(E_{\text{reg}}).$$

Conclusion

The emergent metric is **regular, continuous, and free of divergences**, as it is a compression of the regularized correlational structure.

G2 — Emergent Geometric Stability

Statement

If the emergent structure is stable (Theorems 17–19), then:

$$\delta E_{\text{reg}} \rightarrow 0 \Rightarrow \delta g_{\mu\nu} \rightarrow 0.$$

Conclusion

The physical metric inherits the stability of the emergent structure and does not amplify microscopic perturbations.

G3 — Emergent Geometric Continuity

Statement

If $D_S^{(e)}$ is continuous and C_e is regular, then the geometric projection is continuous:

$$\lim_{x \rightarrow y} C_e(x, y) = 1 \Rightarrow g_{\mu\nu}(x) \rightarrow g_{\mu\nu}(y).$$

Conclusion

Geometric continuity is a **compressed image** of the emergent continuity of the correlational domain.

G4 — Functional Determination of the Metric

Statement

The emergent metric is fully determined by the regularized emergent structure:

$$g_{\mu\nu} = F_{\text{geom}}(E_{\text{reg}}) = F_{\text{geom}}(D_S^{(e)}, C_e).$$

Conclusion

Geometry is not fundamental: it is an **effective representation** of the emergent structure.

8. Dynamical Emergence Theorems

Formal Context

In the PSE, physical dynamics is not fundamental: it emerges from the regularized correlational structure. The ontological chain is:

$$F \xrightarrow{\Psi} E \xrightarrow{R} E_{\text{reg}} \xrightarrow{\Theta_k} E_{\text{reg}} \xrightarrow{\Delta} \mathcal{D}_{\text{phys}}.$$

The emergent dynamics is defined as:

$$\mathcal{D}_{\text{phys}} = \Delta(E_{\text{reg}}),$$

and constitutes a component of the physical domain at level k .

D1 — Emergence of Physical Dynamics

Statement

Given the dynamical projection:

$$\Delta: E_{\text{reg}} \rightarrow \mathcal{D}_{\text{phys}},$$

there exists an emergent physical dynamics:

$$\mathcal{D}_{\text{phys}} = \Delta(E_{\text{reg}}),$$

which is:

- monotone,
- compressive,
- non-invertible,
- well-defined.

Conclusion

Physical dynamics is a **functional image** of the regularized emergent structure.

D2 — Emergent Dynamical Stability

Statement

If the emergent structure is stable (Theorems 17–19), then:

$$\delta E_{\text{reg}} \rightarrow 0 \Rightarrow \delta \mathcal{D}_{\text{phys}} \rightarrow 0.$$

Conclusion

Emergent dynamics is stable with respect to monotone variations of the emergent structure.

D3 — Emergent Dynamical Continuity

Statement

If $D_S^{(e)}$ is continuous and C_e is regular, then the dynamical projection is continuous:

$$\lim_{x \rightarrow y} D_S^{(e)}(x, y) = 0 \Rightarrow \mathcal{D}_{\text{phys}}(x) \rightarrow \mathcal{D}_{\text{phys}}(y).$$

Conclusion

Dynamical continuity follows from the continuity of the regularized emergent domain.

D4 — Functional Determination of Dynamics

Statement

Emergent dynamics is fully determined by the regularized emergent structure:

$$\mathcal{D}_{\text{phys}} = F_{\text{dyn}}(E_{\text{reg}}) = F_{\text{dyn}}(D_S^{(e)}, C_e).$$

Conclusion

Dynamics is not autonomous: it is an **effective representation** of the emergent structure.

9. Statistical Emergence Theorems

Formal Context

In the PSE, physical statistics is not fundamental: it emerges from the regularized correlational structure. The ontological chain is:

$$F \xrightarrow{\Psi} E \xrightarrow{R} E_{\text{reg}} \xrightarrow{\Theta_k} E_{\text{reg}} \xrightarrow{\Phi_{k,\text{stat}}} \rho.$$

The emergent statistical distribution is defined as:

$$\rho = \Phi_{k,\text{stat}}(E_{\text{reg}}),$$

and constitutes the statistical physical domain at level k .

S1 — Emergence of Statistical Quantities

Statement

Given the statistical projection:

$$\Phi_{k,\text{stat}}: E_{\text{reg}} \rightarrow \rho,$$

there exists an emergent statistical distribution:

$$\rho = \Phi_{k,\text{stat}}(E_{\text{reg}}),$$

which is:

- monotone,
- compressive,
- well-defined.

Conclusion

Emergent physical statistics is an **effective representation** of the regularized emergent correlation.

S2 — Emergent Statistical Stability

Statement

If the emergent structure is stable (Theorems 17–19), then:

$$\delta E_{\text{reg}} \rightarrow 0 \Rightarrow \delta \rho \rightarrow 0.$$

Conclusion

The emergent distribution is stable with respect to the emergent structure and does not amplify microscopic perturbations.

S3 — Emergent Statistical Continuity

Statement

If $D_S^{(e)}$ is continuous and C_e is regular, then the statistical projection is continuous:

$$\lim_{x \rightarrow y} C_e(x, y) = 1 \Rightarrow \rho(x) \rightarrow \rho(y).$$

Conclusion

Statistical continuity is **induced by the emergent continuity** of the regularized correlational domain.

S4 — Functional Determination of Statistical Quantities

Statement

The emergent statistical distribution is fully determined by the regularized emergent structure:

$$\rho = F_{\text{stat}}(E_{\text{reg}}) = F_{\text{stat}}\left(D_S^{(e)}, C_e\right).$$

Conclusion

Emergent statistics is not autonomous: it is an **effective representation** of the emergent structure.

10. Field Emergence Theorems

Formal Context

In the PSE, physical fields are not fundamental: they emerge from the regularized correlational structure. The ontological chain is:

$$F \xrightarrow{\Psi} E \xrightarrow{R} E_{\text{reg}} \xrightarrow{\Theta_k} E_{\text{reg}} \xrightarrow{\Xi} \mathcal{F}_{\text{phys}}.$$

The emergent physical field is defined as:

$$\mathcal{F}_{\text{phys}} = \Xi(E_{\text{reg}}),$$

and constitutes a component of the physical domain at level k .

C1 — Emergence of Physical Fields

Statement

Given the field projection:

$$\Xi: E_{\text{reg}} \rightarrow \mathcal{F}_{\text{phys}},$$

there exists an emergent physical field:

$$\mathcal{F}_{\text{phys}} = \Xi(E_{\text{reg}}),$$

which is:

- monotone,
- compressive,
- non-invertible,
- well-defined.

Conclusion

Physical fields emerge as **functional images** of the regularized emergent structure.

C2 — Stability of Emergent Fields

Statement

If the emergent structure is stable (Theorems 17–19), then:

$$\delta E_{\text{reg}} \rightarrow 0 \Rightarrow \delta \mathcal{F}_{\text{phys}} \rightarrow 0.$$

Conclusion

Field stability follows directly from emergent stability and from the compressivity of the field projection.

C3 — Continuity of Emergent Fields

Statement

If $D_S^{(e)}$ is continuous and C_e is regular, then the field projection is continuous:

$$\lim_{x \rightarrow y} C_e(x, y) = 1 \Rightarrow \mathcal{F}_{\text{phys}}(x) \rightarrow \mathcal{F}_{\text{phys}}(y).$$

Conclusion

Field continuity is **induced by the emergent continuity** of the regularized correlational domain.

C4 — Functional Determination of Fields

Statement

Emergent fields are fully determined by the regularized emergent structure:

$$\mathcal{F}_{\text{phys}} = F_{\text{campo}}(E_{\text{reg}}) = F_{\text{campo}}(D_S^{(e)}, C_e).$$

Conclusion

Fields are not fundamental: they are **effective representations** of the emergent structure.

11. Quantum Emergence Theorems

Formal Context

In the PSE, the quantum structure is not fundamental: it emerges from the regularized correlational structure. The ontological chain is:

$$F \xrightarrow{\Psi} E \xrightarrow{R} E_{\text{reg}} \xrightarrow{\Theta_k} E_{\text{reg}} \xrightarrow{\Phi_{k,\text{quant}}} (\mathcal{H}, A, \hbar).$$

The emergent quantum domain is:

$$D_{\text{quant}}(k) = (\mathcal{H}, A, \hbar).$$

Q1 — Emergence of the Quantum Structure

Statement

Given the regularized emergent structure E_{reg} , there exists a quantum projection:

$$\Phi_{k,\text{quant}}: E_{\text{reg}} \rightarrow (\mathcal{H}, A, \hbar),$$

which is monotone, compressive, and non-invertible, such that:

1. Emergent state space

$$\mathcal{H} = \Phi_{k,\text{quant}}(E_{\text{reg}})$$

represents the **coherent regions** induced by the emergent correlation.

2. Emergent action

$$A = \Phi_{k,\text{quant}}(E_{\text{reg}})$$

is a **continuous compression** of the emergent correlational constraints.

3. Quantum threshold

$$\hbar = \Phi_{k,\text{quant}}(C_f)$$

represents the **minimal granularity** induced by the fundamental binary structure.

Conclusion

The triple $(\mathcal{H}, A, \hbar)$ constitutes the **emergent quantum structure**, non-fundamental, devoid of primitive operators, non-commutativity, and primitive quantization.

Q2 — Emergent Quantum Stability

Statement

If the emergent structure is stable (Theorems 17–19), then the emergent quantum structure is stable:

$$\delta E_{\text{reg}} \rightarrow 0 \implies \delta \mathcal{H} \rightarrow 0, \delta A \rightarrow 0, \delta \hbar \rightarrow 0.$$

Conclusion

Quantum stability is a direct consequence of emergent stability and of the monotonicity of the quantum projection.

Q3 — Emergent Quantum Continuity

Statement

If $D_S^{(e)}$ is continuous and C_e is regular, then the quantum projection is continuous:

$$\Phi_{k,\text{quant}}: E_{\text{reg}} \rightarrow (\mathcal{H}, A, \hbar),$$

and satisfies:

$$\lim_{x \rightarrow y} C_e(x, y) = 1 \Rightarrow \psi_x \rightarrow \psi_y \in \mathcal{H}.$$

Conclusion

The continuity of the state space is a **compressed image** of emergent continuity.

Q4 — Functional Determination of the Quantum Structure

Statement

The emergent quantum structure is fully determined by the regularized emergent structure:

$$(\mathcal{H}, A, \hbar) = F_{\text{quant}}(E_{\text{reg}}) = F_{\text{quant}}(D_S^{(e)}, C_e).$$

In particular:

$$\mathcal{H} = f_{\mathcal{H}}(C_e), A = f_A(E_{\text{reg}}), \hbar = f_{\hbar}(C_f).$$

Conclusion

The quantum structure is not autonomous: it is an **effective representation** of the emergent structure.

12. Physical Corollaries

Corollary F1 — Existence of Emergent Curvature

Context

From Theorems 25–30 we know that a physical emergent metric exists, defined by:

$$g_{\mu\nu} = \Gamma(E_{\text{reg}})$$

where:

- $E_{\text{reg}} = R(E)$ is the regularized emergent domain,
- Γ is a continuous, isotropic, deterministic, compressive metric operator.

The emergent metric is:

- continuous,
- regular,
- stable,
- isotropic,
- fully determined by the regularized emergent structure.

We now show that **emergent physical curvature necessarily exists**.

Statement of the Corollary

The emergent metric $g_{\mu\nu}$ is sufficiently regular to guarantee the existence of:

- the Levi-Civita connection $\Gamma_{\mu\nu}^\rho$,
- the Riemann tensor $R_{\sigma\mu\nu}^\rho$,
- the Ricci tensor $R_{\mu\nu}$,
- the scalar curvature R ,
- the Einstein tensor $G_{\mu\nu}$.

In particular:

$$g_{\mu\nu} \in C^0 \Rightarrow \Gamma_{\mu\nu}^\rho \in C^{-1}, R_{\sigma\mu\nu}^\rho \in C^{-2}.$$

Proof

1. Regularity of the metric

From Theorem 26:

$$g_{\mu\nu} = \Gamma(E_{\text{reg}}) \in C^0.$$

The metric is continuous.

2. Existence of the Levi-Civita connection

The connection is defined by:

$$\Gamma_{\mu\nu}^{\rho} = \frac{1}{2} g^{\rho\sigma} (\partial_{\mu} g_{\sigma\nu} + \partial_{\nu} g_{\sigma\mu} - \partial_{\sigma} g_{\mu\nu}).$$

Since $g_{\mu\nu} \in C^0$, its weak derivatives ∂g exist in the distributional sense:

$$\Gamma_{\mu\nu}^{\rho} \in C^{-1}.$$

3. Existence of the Riemann tensor

$$R_{\sigma\mu\nu}^{\rho} = \partial_{\mu} \Gamma_{\sigma\nu}^{\rho} - \partial_{\nu} \Gamma_{\sigma\mu}^{\rho} + \Gamma_{\mu\nu}^{\lambda} \Gamma_{\sigma\lambda}^{\rho} - \Gamma_{\nu\mu}^{\lambda} \Gamma_{\sigma\lambda}^{\rho}.$$

Since:

- $\Gamma_{\mu\nu}^{\rho} \in C^{-1}$,
- $\partial \Gamma \in C^{-2}$,

we obtain:

$$R_{\sigma\mu\nu}^{\rho} \in C^{-2}.$$

4. Ricci tensor and scalar curvature

$$R_{\mu\nu} = R_{\mu\rho\nu}^{\rho}, \quad R = g^{\mu\nu} R_{\mu\nu}.$$

Since $g^{\mu\nu} \in C^0$ and $R_{\mu\nu} \in C^{-2}$:

$$R \in C^{-2}.$$

5. Einstein tensor

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R.$$

All quantities are well-defined:

$$G_{\mu\nu} \in C^{-2}.$$

6. Functional determination

From Theorem 30:

$$g_{\mu\nu} = \Gamma(E_{\text{reg}})$$

is unique and fully determined by the regularized emergent structure.

Since curvature is a functional of $g_{\mu\nu}$:

$$R_{\sigma\mu\nu}^{\rho}, R_{\mu\nu}, R, G_{\mu\nu}$$

are completely determined by E_{reg} .

7. Isotropy

From Theorem 22 (updated):

$$g_{\mu\nu}^{(\sigma)} = g_{\mu\nu}$$

for any renumbering σ of the emergent domain.

Since curvature is constructed from $g_{\mu\nu}$:

$$R_{\sigma\mu\nu}^{\rho}{}^{(\sigma)} = R_{\sigma\mu\nu}^{\rho}.$$

Curvature is isotropic.

Conclusion

The emergent physical curvature:

$$R_{\sigma\mu\nu}^{\rho}, R_{\mu\nu}, R, G_{\mu\nu}$$

is:

- well-defined,
- regular,
- stable,
- isotropic,
- deterministic,
- non-amplifying,
- fully determined by the regularized emergent structure E_{reg} .

Corollary F2 — Existence of Derived Tensor Fields

Context

From Theorems 25–30 we know that the physical domain contains:

1. **the emergent metric**

$$g_{\mu\nu} = \Gamma(E_{\text{reg}})$$

2. **the emergent physical fields**

$$\mathcal{F}_{\text{phys}} = \Xi(E_{\text{reg}})$$

3. **the emergent curvature**

$$R_{\sigma\mu\nu}^{\rho}, R_{\mu\nu}, R, G_{\mu\nu}$$

where:

- $E_{\text{reg}} = R(E)$ is the regularized emergent domain,
- Γ and Ξ are continuous, isotropic, deterministic, compressive operators.

We now show that the physical domain necessarily gives rise to **derived tensor fields**, obtained through geometric and differential operations on the metric and on the primary fields.

Statement of the Corollary

From the emergent physical domain:

$$P = (g_{\mu\nu}, \mathcal{F}_{\text{phys}})$$

the following derived tensor fields necessarily emerge:

- energy-momentum tensor $T_{\mu\nu}$,
- spin tensor $S_{\mu\nu\rho}$,
- correlation tensor $K_{\mu\nu}$,
- flux tensor J_μ ,
- potential tensors A_μ ,
- density tensors $\rho_{\mu\nu}$,
- transport tensors $\Theta_{\mu\nu}$,
- deformation tensors $\Sigma_{\mu\nu}$,
- field tensors $F_{\mu\nu}$,
- covariant derivative tensors $\nabla_\mu \phi$,
- second-order derivative tensors $\nabla_\mu \nabla_\nu \phi$.

In particular:

$$\mathcal{T}_{\text{der}} = \Lambda(P)$$

where Λ is a continuous, isotropic, deterministic tensorial operator.

Proof

1. Existence of the metric and the connection

From Theorems 25–26:

$$g_{\mu\nu} \in C^0, \Gamma_{\mu\nu}^\rho \in C^{-1}.$$

The Levi-Civita connection exists.

2. Existence of primary fields

From Theorem 27:

$$\mathcal{F}_{\text{phys}} = \Xi(E_{\text{reg}}) \in C^0.$$

Scalar, vector, and tensor primary fields exist.

3. Existence of covariant derivatives

Since:

- $g_{\mu\nu}$ is continuous,
- $\Gamma_{\mu\nu}^\rho$ is defined,
- physical fields are continuous,

the covariant derivatives:

$$\nabla_\mu \phi, \nabla_\mu A_\nu, \nabla_\mu T_{\nu\rho}$$

are well-defined in the distributional sense.

4. Existence of field tensors

For any vector field A_μ :

$$F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu$$

is well-defined.

5. Existence of energy-momentum tensors

For any physical field ϕ :

$$T_{\mu\nu} = \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{2} g_{\mu\nu} g^{\rho\sigma} \nabla_\rho \phi \nabla_\sigma \phi$$

is well-defined.

6. Existence of spin tensors

For fields with internal structure:

$$S_{\mu\nu\rho} = \phi_\mu \nabla_\nu \phi_\rho - \phi_\nu \nabla_\mu \phi_\rho$$

is well-defined.

7. Existence of correlation tensors

From the regularized emergent domain:

$$K_{\mu\nu} = \nabla_\mu \nabla_\nu C_e$$

is well-defined.

8. Existence of flux tensors

For any vector field A_μ :

$$J_\mu = \nabla_\nu F_{\mu\nu}$$

is well-defined.

9. Functional determination

Since:

- $g_{\mu\nu}$ and $\mathcal{F}_{\text{phys}}$ are fully determined by E_{reg} ,
- Λ is deterministic,

we obtain:

$$\mathcal{T}_{\text{der}} = \Lambda(P)$$

which is unique.

10. Isotropy

For any renumbering σ :

$$P^{(\sigma)} = P \Rightarrow \Lambda(P^{(\sigma)}) = \Lambda(P).$$

Derived tensors are isotropic.

Conclusion

The derived tensor fields:

$$\mathcal{T}_{\text{der}} = \Lambda(P)$$

are:

- well-defined,
- regular,
- stable,
- isotropic,
- deterministic,
- non-amplifying,
- fully determined by the regularized emergent structure E_{reg} .

Corollary F3 — Existence of Emergent Equations of Motion

Context

From the physical emergence theorems we already have:

- **emergent metric**

$$g_{\mu\nu} = \Gamma(E_{\text{reg}})$$

- **emergent physical fields**

$$\mathcal{F}_{\text{phys}} = \Xi(E_{\text{reg}})$$

- **emergent physical quantities**

$$Q = \Omega(P)$$

- **emergent dynamics**

$$\mathcal{D}_{\text{phys}} = \Delta(P)$$

where:

$$P = (g_{\mu\nu}, \mathcal{F}_{\text{phys}})$$

is the emergent physical domain.

We now show that **the physical equations of motion necessarily emerge** from the PSE physical domain.

Statement of the Corollary

From the emergent physical domain:

$$P = (g_{\mu\nu}, \mathcal{F}_{\text{phys}})$$

the following equations of motion necessarily emerge for every physical field:

- evolution equations for scalar fields,
- transport equations for vector fields,
- tensorial equations for higher-order fields,
- conservation equations,
- balance equations,
- geometric dynamical equations,
- propagation equations,
- flux equations,
- correlation equations.

Formally:

$$E_{\text{moto}} = \Delta(P)$$

where Δ is a continuous, isotropic, deterministic dynamical operator.

Proof

1. Existence of the metric and the connection

From Corollary F1:

$$g_{\mu\nu} \in C^0, \Gamma_{\mu\nu}^\rho \in C^{-1}.$$

The Levi-Civita connection exists and allows the definition of:

- covariant derivatives,
- dynamical operators,
- equations of motion.

2. Existence of physical fields

From Theorem 27:

$$\mathcal{F}_{\text{phys}} = \Xi(E_{\text{reg}}) \in C^0.$$

Scalar, vector, and tensor fields exist.

3. Existence of derived tensors

From Corollary F2:

$$\mathcal{T}_{\text{der}} = \Lambda(P)$$

the following tensors are well-defined:

- $T_{\mu\nu},$

- $F_{\mu\nu}$,
- J_μ ,
- $S_{\mu\nu\rho}$,
- $K_{\mu\nu}$,
- $\nabla_\mu \phi$,
- $\nabla_\mu \nabla_\nu \phi$.

These tensors are the **ingredients of the equations of motion**.

4. Existence of the dynamical operator

From Theorem 30:

$$\mathcal{D}_{\text{phys}} = \Delta(P)$$

with Δ continuous, isotropic, deterministic.

Thus a dynamical operator exists acting on:

- the metric,
- the fields,
- the derived tensors,
- the physical quantities.

5. Construction of the equations of motion

Since Δ is defined on P , the equations of motion emerge as:

$$E_{\text{moto}} = \Delta(P)$$

i.e., as dynamical relations among:

- $g_{\mu\nu}$,
- $\Gamma_{\mu\nu}^\rho$,
- $R_{\mu\nu}$,
- $T_{\mu\nu}$,
- $F_{\mu\nu}$,
- J_μ ,
- $\nabla_\mu \phi$,
- $\nabla_\mu \nabla_\nu \phi$.

6. Functional determination

Since:

- $g_{\mu\nu}$ and $\mathcal{F}_{\text{phys}}$ are fully determined by E_{reg} ,

- Δ is deterministic,

we obtain:

$$E_{\text{moto}} = \Delta(P)$$

which is unique.

7. Isotropy

For any renumbering σ :

$$P^{(\sigma)} = P \Rightarrow \Delta(P^{(\sigma)}) = \Delta(P).$$

The equations of motion are isotropic and invariant under permutations, renumberings, and relabelings.

Conclusion

The emergent equations of motion:

$$E_{\text{moto}} = \Delta(P)$$

are:

- well-defined,
- regular,
- stable,
- isotropic,
- deterministic,
- non-amplifying,
- fully determined by the regularized emergent structure E_{reg} .

Corollary F4 — Invariance of Emergent Physical Laws

Context

From the fundamental emergence theorems we have:

- **Theorem 22 — Invariance under isomorphisms of the fundamental domain**

$$F^{(\sigma)} = F \Rightarrow E^{(\sigma)} = E, P^{(\sigma)} = P.$$

- **Theorem 24 — Functional determination of the physical domain** The physical domain obtained from the emergence chain is **unique**.

From the physical emergence theorems:

- **Theorems 25–26:** the emergent metric is deterministic and isotropic.
- **Theorems 27–28:** emergent fields are deterministic and isotropic.
- **Theorem 29:** emergent physical quantities are deterministic and isotropic.
- **Theorem 30:** emergent dynamics is deterministic and isotropic.

We now show that **emergent physical laws are invariant under any isomorphism of the fundamental domain.**

Statement of the Corollary

For any isomorphism σ of the fundamental domain:

$$F^{(\sigma)} = F,$$

the emergent physical laws:

$$\mathcal{L}_{\text{phys}} = \Delta(P)$$

are invariant, i.e.:

$$\mathcal{L}_{\text{phys}}^{(\sigma)} = \mathcal{L}_{\text{phys}}.$$

In particular, the following remain unchanged:

- equations of motion,
- conservation equations,
- balance equations,
- propagation equations,
- geometric equations,
- correlation equations.

Proof

1. Invariance of the emergent domain

From Theorem 22:

$$F^{(\sigma)} = F \Rightarrow E^{(\sigma)} = E.$$

Thus the emergent domain is invariant.

2. Invariance of the physical domain

Again from Theorem 22:

$$E^{(\sigma)} = E \Rightarrow P^{(\sigma)} = P.$$

Therefore:

- the metric,
- the fields,
- the physical quantities

are invariant.

3. Invariance of the metric

From Theorems 25–26:

$$g_{\mu\nu}^{(\sigma)} = g_{\mu\nu}.$$

4. Invariance of physical fields

From Theorems 27–28:

$$\mathcal{F}_{\text{phys}}^{(\sigma)} = \mathcal{F}_{\text{phys}}.$$

5. Invariance of physical quantities

From Theorem 29:

$$Q^{(\sigma)} = Q.$$

6. Invariance of dynamics

From Theorem 30:

$$\mathcal{D}_{\text{phys}}^{(\sigma)} = \mathcal{D}_{\text{phys}}.$$

7. Invariance of physical laws

Since emergent physical laws are defined by:

$$\mathcal{L}_{\text{phys}} = \Delta(P),$$

and since:

$$P^{(\sigma)} = P,$$

we obtain:

$$\Delta(P^{(\sigma)}) = \Delta(P).$$

Thus:

$$\mathcal{L}_{\text{phys}}^{(\sigma)} = \mathcal{L}_{\text{phys}}.$$

Conclusion

The emergent physical laws:

$$\mathcal{L}_{\text{phys}} = \Delta(P)$$

are:

- invariant under any isomorphism of the fundamental domain,
- independent of renumberings,
- independent of permutations,
- independent of relabelings,
- objective,
- deterministic,
- stable,

- fully determined by the fundamental structure F .

Corollary F5 — Stability of Physical Observables (updated version)

Context

From the physical emergence theorems we know:

- **Theorem 26:** the emergent metric is continuous and stable.
- **Theorems 27–28:** emergent physical fields are continuous and stable.
- **Theorem 29:** emergent physical quantities $Q = \Omega(P)$ are deterministic and stable.
- **Theorem 30:** emergent dynamics is continuous and stable.

We now show that **all emergent physical observables**—that is, all measurable physical quantities in the emergent physical domain—are stable with respect to microscopic variations of the fundamental structure F .

Statement of the Corollary

Every emergent physical observable:

$$Q = \Omega(P)$$

is stable, i.e.:

$$\|Q - Q'\|_{\mathcal{Q}_{\text{phys}}} \leq K_{\Omega} K_{\Phi} K_{\Psi} \|F - F'\|_{\mathcal{F}},$$

where:

- K_{Ψ} is the stability coefficient of emergence (Theorem 19),
- K_{Φ} is the stability coefficient of the physical projection (Theorem 23),
- K_{Ω} is the compressivity coefficient of the physical-quantity operator (Theorem 29).

In particular, **no physical observable amplifies microscopic perturbations** of the fundamental domain.

Thus, observables are:

- continuous,
- regular,
- Lipschitz-stable,
- non-amplifying,
- isotropic,
- deterministic,
- fully determined by F .

Proof

1. Stability of the emergent domain

From Theorem 19:

$$\| E - E' \|_E \leq K_\Psi \| F - F' \|_{\mathcal{F}}.$$

2. Stability of the physical domain

From Theorem 23:

$$\| P - P' \|_{\mathcal{P}} \leq K_\Phi K_\Psi \| F - F' \|_{\mathcal{F}}.$$

Therefore:

- the metric,
- the fields,
- the physical quantities,

are stable with respect to microscopic variations of F .

3. Stability of physical quantities

From Theorem 29:

$$\| \Omega(P) - \Omega(P') \|_{\mathcal{Q}_{\text{phys}}} \leq K_\Omega \| P - P' \|_{\mathcal{P}}.$$

Composing:

$$\| \Omega(P) - \Omega(P') \|_{\mathcal{Q}_{\text{phys}}} \leq K_\Omega K_\Phi K_\Psi \| F - F' \|_{\mathcal{F}}.$$

Thus, physical observables are **Lipschitz-stable** with respect to the fundamental structure.

4. Isotropy

From Theorem 22:

$$P^{(\sigma)} = P \Rightarrow \Omega(P^{(\sigma)}) = \Omega(P).$$

Observables are invariant under renumberings, permutations, and relabelings.

5. Functional determination

From Theorem 24 (updated):

the emergent physical domain P is unique.

Since Ω is deterministic:

$$Q = \Omega(P)$$

is unique.

Conclusion

Every emergent physical observable:

$$Q = \Omega(P)$$

is:

- stable,

- regular,
- isotropic,
- deterministic,
- non-amplifying,
- fully determined by the fundamental structure F .

Corollary F6 — Complete Functional Determination of Emergent Physics

Context

From the fundamental and physical emergence theorems we have:

- **Theorem 24 — Functional determination of the physical domain** The physical domain obtained from the emergence chain is unique.
- **Theorems 25–26 — Emergent metric**

$$g_{\mu\nu} = \Gamma(E_{\text{reg}})$$

- **Theorems 27–28 — Emergent fields**

$$\mathcal{F}_{\text{phys}} = \Xi(E_{\text{reg}})$$

- **Theorem 29 — Emergent physical quantities**

$$Q = \Omega(P)$$

- **Theorem 30 — Emergent dynamics**

$$\mathcal{D}_{\text{phys}} = \Delta(P)$$

Corollaries F1–F5 established that:

- emergent curvature exists,
- derived tensors exist,
- equations of motion exist,
- physical laws are invariant,
- observables are stable.

We now show that **the entire emergent physics is a deterministic function of the fundamental structure F** .

Statement of the Corollary

For every fundamental structure F , the entire emergent physics:

$$\mathcal{P}_{\text{phys}} = (g_{\mu\nu}, \mathcal{F}_{\text{phys}}, \mathcal{T}_{\text{der}}, Q, E_{\text{moto}}, \mathcal{L}_{\text{phys}})$$

is completely determined by F .

Formally:

$$\mathcal{P}_{\text{phys}} = \Lambda_{\text{tot}}(F)$$

where Λ_{tot} is a map that is:

- continuous,
- monotone,
- isotropic,
- compressive,
- deterministic,
- non-amplifying.

In particular:

$$\| \mathcal{P}_{\text{phys}} - \mathcal{P}'_{\text{phys}} \| \leq K_{\Delta} K_{\Omega} K_{\Phi} K_{\Psi} \| F - F' \|.$$

Proof

1. Determination of the emergent domain

From Theorem 18:

$$E = \Psi(F)$$

is unique.

2. Determination of the metric

From Theorem 25:

$$g_{\mu\nu} = \Gamma(E_{\text{reg}})$$

is unique and deterministic.

3. Determination of physical fields

From Theorem 27:

$$\mathcal{F}_{\text{phys}} = \Xi(E_{\text{reg}})$$

is unique and deterministic.

4. Determination of derived tensors

From Corollary F2:

$$\mathcal{T}_{\text{der}} = \Lambda(P)$$

is unique and deterministic.

5. Determination of physical quantities

From Theorem 29:

$$Q = \Omega(P)$$

is unique and deterministic.

6. Determination of dynamics

From Theorem 30:

$$\mathcal{D}_{\text{phys}} = \Delta(P)$$

is unique and deterministic.

7. Determination of physical laws

From Corollary F4:

$$\mathcal{L}_{\text{phys}}^{(\sigma)} = \mathcal{L}_{\text{phys}}$$

so physical laws are invariant and deterministic.

8. Composition of determinations

Composing all maps:

$$F \rightarrow \Psi \rightarrow E \rightarrow \Gamma, \Xi \rightarrow (g_{\mu\nu}, \mathcal{F}_{\text{phys}}) \rightarrow \Lambda, \Omega \rightarrow (\mathcal{T}_{\text{der}}, Q) \rightarrow \Delta \rightarrow \mathcal{D}_{\text{phys}} \rightarrow \mathcal{L}_{\text{phys}}$$

we obtain the global map:

$$\Lambda_{\text{tot}}: F \mapsto \mathcal{P}_{\text{phys}}.$$

Since each map is deterministic, the composition is deterministic.

9. Global stability

Composing stability constants:

$$\| \mathcal{P}_{\text{phys}} - \mathcal{P}'_{\text{phys}} \| \leq K_{\Delta} K_{\Omega} K_{\Phi} K_{\Psi} \| F - F' \|.$$

Thus the entire emergent physics is Lipschitz-stable.

10. Global isotropy

Since each map is isotropic:

$$F^{(\sigma)} = F \Rightarrow \mathcal{P}_{\text{phys}}^{(\sigma)} = \mathcal{P}_{\text{phys}}.$$

Thus the entire emergent physics is invariant under isomorphisms.

Conclusion

The entire emergent physics:

$$\mathcal{P}_{\text{phys}} = \Lambda_{\text{tot}}(F)$$

is:

- deterministic,
- stable,
- isotropic,
- non-amplifying,
- regular,
- fully determined by the fundamental structure F .

13. Final Synthesis of the Mathematical Formalization of the PSE

1. Complete Ontological Chain

The fundamental structure $F \in \mathcal{F}$ generates the entire emergent physics through a chain of deterministic, continuous, stable, and isotropic maps:

$$F \xrightarrow{\Psi} E \xrightarrow{R} E_{\text{reg}} \xrightarrow{\Theta_k} E_{\text{reg}} \xrightarrow{(\Gamma, \Xi)} P = (g_{\mu\nu}, \mathcal{F}_{\text{phys}}) \xrightarrow{(\Lambda, \Omega, \Delta)} \mathcal{P}_{\text{phys}}$$

Each level is unique, non-amplifying, Lipschitz-stable, invariant under isomorphisms, and fully determined by F .

2. Emergent Domain

The emergent domain is:

$$E = \Psi(F),$$

and the regularized emergent domain is:

$$E_{\text{reg}} = R(E).$$

The fundamental emergent structures:

- emergent distance $D_S^{(e)}$,
- emergent correlation $\mathcal{C}_e$,

are continuous, stable, unique, and isotropic.

Emergence stability:

$$\| E - E' \|_E \leq K_\Psi \| F - F' \|_{\mathcal{F}}.$$

3. Physical Projection

Physical projection is performed through the metric and field operators:

$$g_{\mu\nu} = \Gamma(E_{\text{reg}}), \quad \mathcal{F}_{\text{phys}} = \Xi(E_{\text{reg}}).$$

Stability:

$$\| P - P' \|_{\mathcal{P}} \leq K_\Phi K_\Psi \| F - F' \|_{\mathcal{F}}.$$

4. Emergent Physical Metric

An emergent metric exists:

$$g_{\mu\nu} = \Gamma(E_{\text{reg}}),$$

with:

- continuity,
- regularity,
- isotropy,

- stability:

$$\|g - g'\|_{\mathcal{G}} \leq K_{\Gamma} K_{\Psi} \|F - F'\|_{\mathcal{F}}.$$

Emergent curvature is well-defined:

$$\Gamma_{\mu\nu}^{\rho} \in C^{-1}, R_{\sigma\mu\nu}^{\rho} \in C^{-2}.$$

5. Emergent Physical Fields

Physical fields exist:

$$\mathcal{F}_{\text{phys}} = \Xi(E_{\text{reg}}),$$

with:

- continuity,
- isotropy,
- stability:

$$\|\Xi(E) - \Xi(E')\| \leq K_{\Xi} K_{\Psi} \|F - F'\|.$$

6. Derived Tensors

Derived tensors emerge from the physical domain:

$$\mathcal{T}_{\text{der}} = \Lambda(P),$$

including:

- $T_{\mu\nu},$
- $F_{\mu\nu},$
- $J_{\mu},$
- $S_{\mu\nu\rho},$
- $K_{\mu\nu},$
- $\nabla_{\mu}\phi,$
- $\nabla_{\mu}\nabla_{\nu}\phi.$

All deterministic and isotropic.

7. Emergent Physical Quantities

Observable physical quantities:

$$Q = \Omega(P)$$

are:

- continuous,
- stable,
- isotropic,

- deterministic.

Stability:

$$\| Q - Q' \| \leq K_{\Omega} K_{\Phi} K_{\Psi} \| F - F' \|.$$

8. Emergent Dynamics

Emergent dynamics:

$$\mathcal{D}_{\text{phys}} = \Delta(P)$$

is:

- continuous,
- isotropic,
- deterministic,
- non-amplifying.

Stability:

$$\| \Delta(P) - \Delta(P') \| \leq K_{\Delta} K_{\Phi} K_{\Psi} \| F - F' \|.$$

9. Emergent Physical Laws

Physical laws:

$$\mathcal{L}_{\text{phys}} = \Delta(P)$$

are invariant under any isomorphism of the fundamental domain:

$$F^{(\sigma)} = F \Rightarrow \mathcal{L}_{\text{phys}}^{(\sigma)} = \mathcal{L}_{\text{phys}}.$$

10. Complete Functional Determination

By composing all maps:

$$\Lambda_{\text{tot}}: F \mapsto \mathcal{P}_{\text{phys}} = (g_{\mu\nu}, \mathcal{F}_{\text{phys}}, \mathcal{T}_{\text{der}}, Q, E_{\text{moto}}, \mathcal{L}_{\text{phys}}),$$

the entire emergent physics is a deterministic, stable, isotropic, non-amplifying function of the fundamental structure.

Global stability:

$$\| \mathcal{P}_{\text{phys}} - \mathcal{P}'_{\text{phys}} \| \leq K_{\Delta} K_{\Omega} K_{\Phi} K_{\Psi} \| F - F' \|.$$

Final Conclusion

The mathematical formalization of the PSE shows that:

- emergent physics is completely determined by the fundamental structure;
- every level is continuous, stable, isotropic, non-amplifying;
- the ontological chain is rigorous, closed, and coherent;
- the regularized emergent domain E_{reg} is the key link between pregeometry and observable physics.

